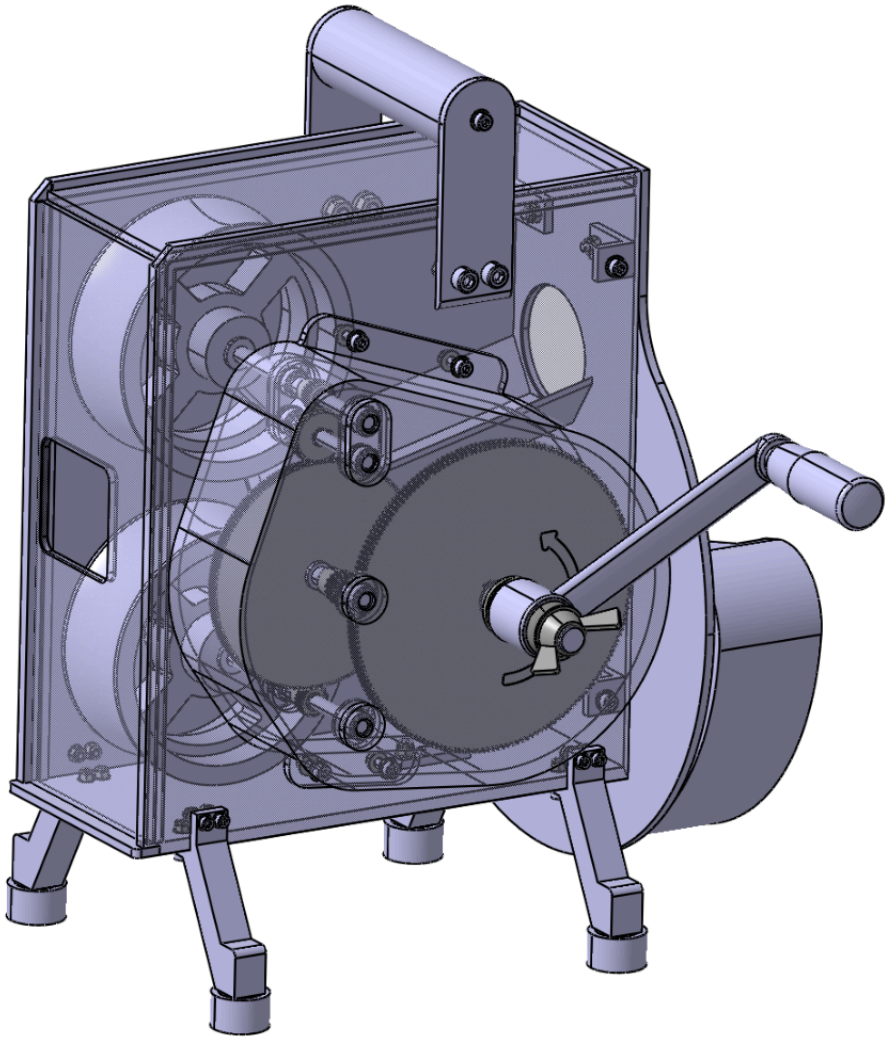
EPFL — Project ME-102/107

Manual Squash Ball Launcher

Microengineering Section (MT) — Group 07

Iyad Abid · Amine Ben Ayed · Mohamed Kamil Berrada · Ali Boudih

Submitted on 1 June 2025



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8. Safety checks
9. Choice of parts
10. Fasteners and guiding systems
11. Critical review of the project
12. Useful calculations
13. Assembly
14. Operating instructions

1 Drawings

1.1 Bill of materials

1.1.1 Machined parts

Name	Material	No.	Qty
Shaft 1	Steel	01	1
Shaft 2	Steel	02	1
Shaft 3	Steel	03	1
Shaft 4	Steel	04	2
Shaft 5	Steel	05	1
Rear frame panel	Aluminium	06	1
Front frame panel	Polycarbonate	07	1
Top frame panel	Polycarbonate	08	1
Bottom frame panel	Aluminium	09	1
Right frame panel	Aluminium	10	1
Left frame panel	Aluminium	11	1
Gear housing	Aluminium	12	1
Launch wheels	POM-H	13	2
Ball track	POM-H	14	1
Legs	Aluminium	15	2
Shaft support	Aluminium	16	1
Ball magazine	POM-H	17	1
Handle (mounts)	Aluminium	18	2
Handle (bar)	Aluminium	19	1
Long bracket	Hardened steel	20	1
Short bracket	Hardened steel	21	2

Table 1 – Bill of materials, machined parts

1.1.2 Modified commercial parts

Name	Supplier	Reference	No.	Qty
Gear	HPC Europe	G0.75-180	22	1
Gear	HPC Europe	G0.75-17	23	1
Gear	HPC Europe	G0.5-200	24	1
Gear	HPC Europe	G0.5-45	25	1
Gear	HPC Europe	G0.5-25	26	1
Gear	HPC Europe	M1.5-22	27	1
Crank	McMaster	6545N15	28	1

Table 2 – Bill of materials, modified parts

1.1.3 Commercial parts

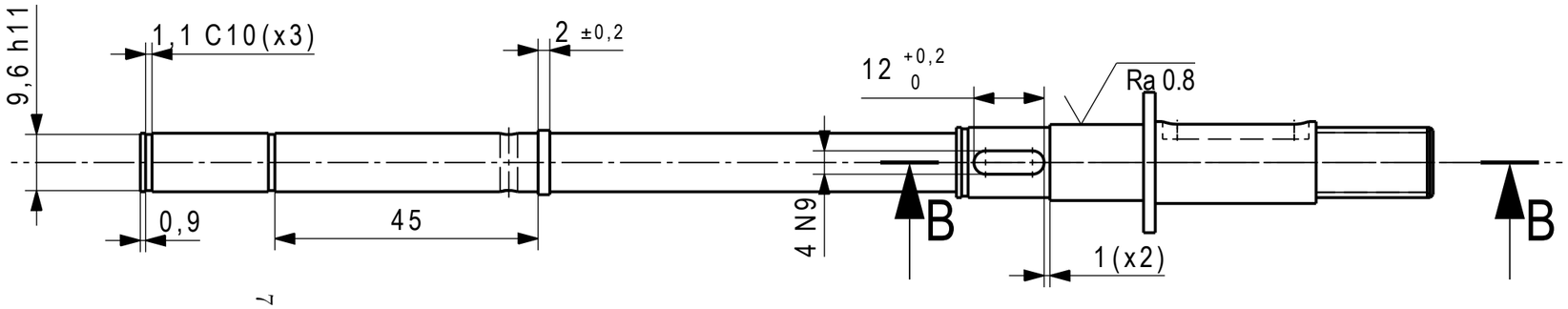
Name	Supplier	Reference	No.	Qty
Plain bearing	Misumi	SHFZ10-12	29	1
Plain bearing	Misumi	SHFZ13-12	30	1
Ball bearing	Misumi	MR126ZZ	31	10
Ball bearing	Misumi	B6700ZZ	32	1
Ball bearing	Misumi	B6800ZZ	33	1
External circlip	HPC Europe	CX-6S/B	34	1
External circlip	HPC Europe	CX-10S/B	35	1
External circlip	HPC Europe	CX-12S/B	36	1
Internal circlip	HPC Europe	CY-12S/B	37	4
Internal circlip	HPC Europe	CY-18S/B	38	4
Internal circlip	HPC Europe	CY-19S/B	39	1
Key	HPC Europe	KK2-10	40	2
Key	HPC Europe	KK3-10	41	2
Key	HPC Europe	KK3-20	42	1
Key	HPC Europe	KK4-12	43	1
Key	HPC Europe	KK5-25	44	1
Key	Misumi	KESF4-40	45	2
Key	Misumi	KESF2-12	46	1
Key	Misumi	KESF4-45	47	1
Dowel pin	HPC Europe	SRP3-24/B	48	1
Worm	HPC Europe	W1.5-1	49	1
Screw M3-14	Bossard	1003801	50	16
Screw M4-16	Bossard	1003941	51	11
Screw M4-8	Bossard	1003909	52	2
Screw M4-22	Bossard	1301705	53	2
Screw M6-16	Bossard	1004301	54	7
Self-locking nut M4	HPC Europe	NIN-4/B	55	15
Self-locking nut M3	HPC Europe	SHNA-3/B	56	4
Nut M6	HPC Europe	SHNA-6/B	57	7
Washer M3	HPC Europe	SHW-3/B	58	16
Washer M4	HPC Europe	SHW-4/B	59	15
Washer M6	HPC Europe	RCX-6/SS	60	7
Crank washer	HPC Europe	RCX-12/SS	61	2
Wing nut	Misumi	CHON12	62	1
Anti-slip foot	Misumi	93115K911	63	4

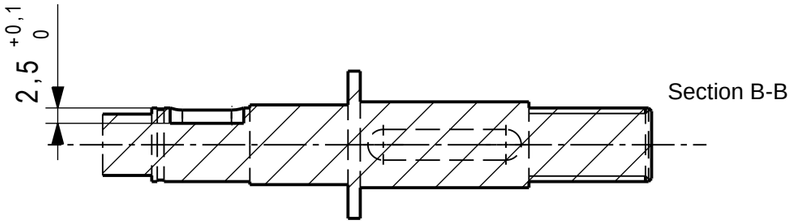
Table 3 – Bill of materials, purchased parts

1.2 Drawings of machined or modified parts

The detail and assembly drawings follow on the next pages (parts 01 to 28, assemblies A to F, and the general assembly).

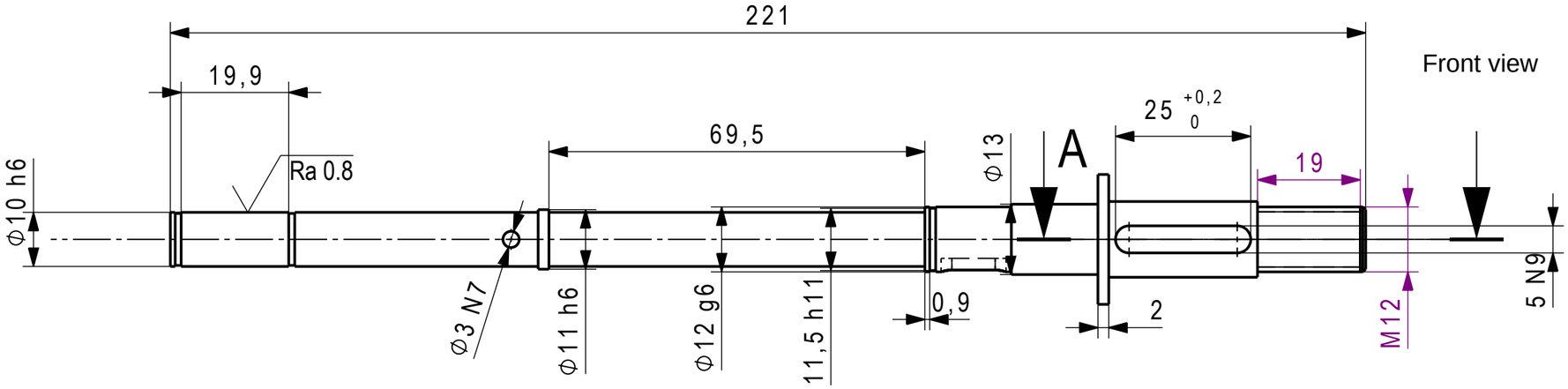
Bottom view



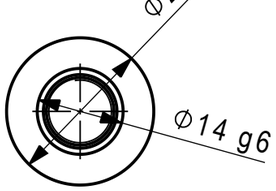


Section B-B

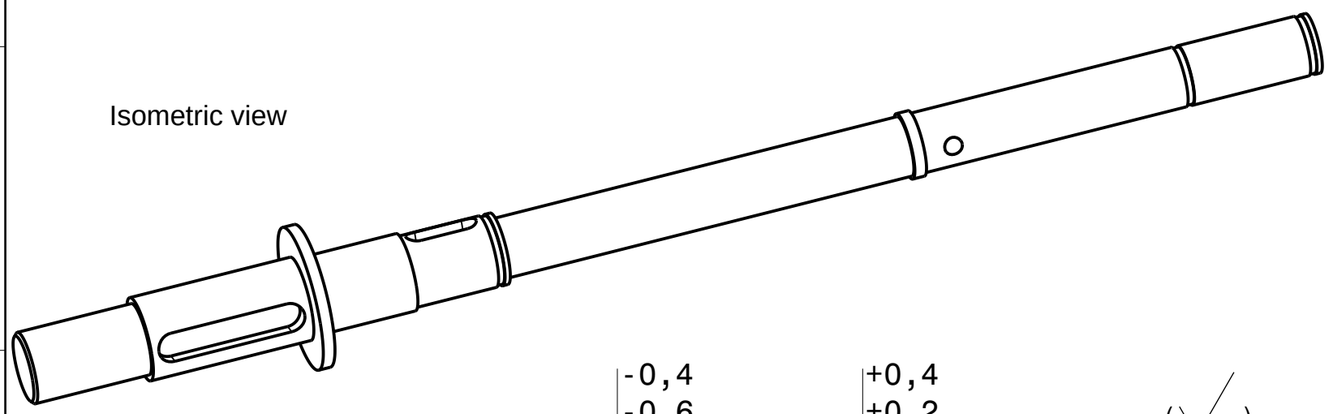
Front view

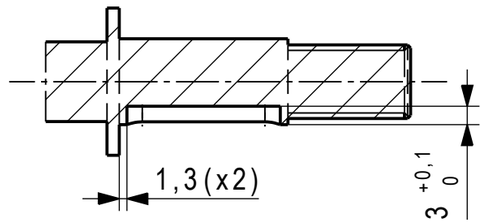


Right view



Isometric view





Section A-A

-0,4
-0,6

+0,4
+0,2

(√)

Ra 3.2

General tolerances:

Rev.		Rev.		Drawn 01/06/2025	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	42 Cr Mo 4 V <= ø 100	Origin	
Sep. parts list, diff. no. ☐		Mass	0.171 kg	Replaces	

Scale 1:1

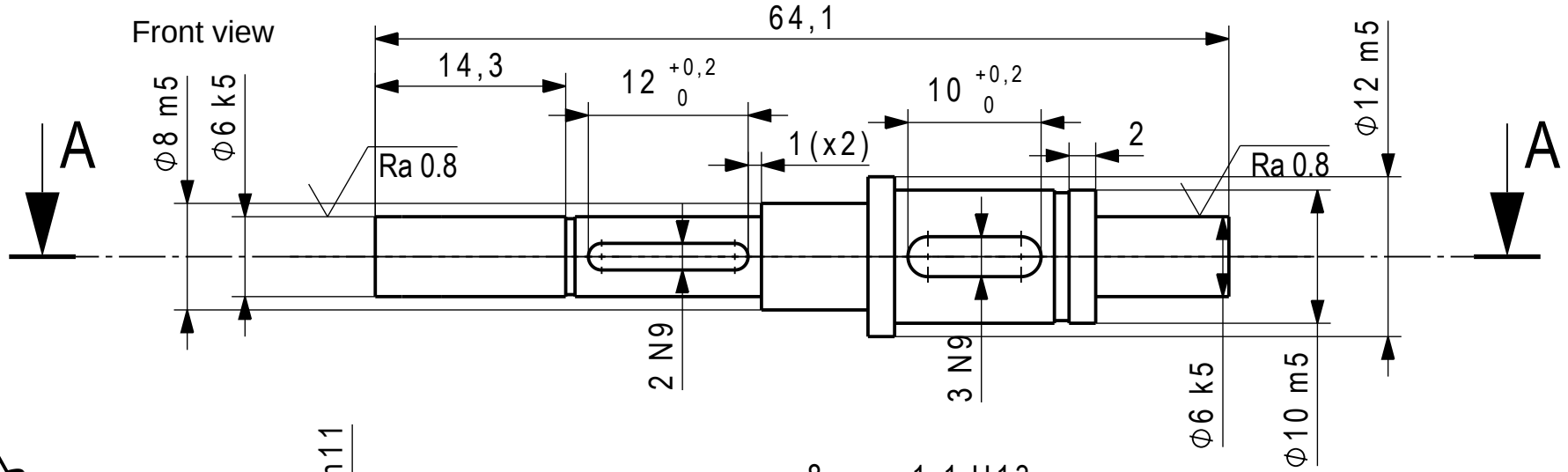
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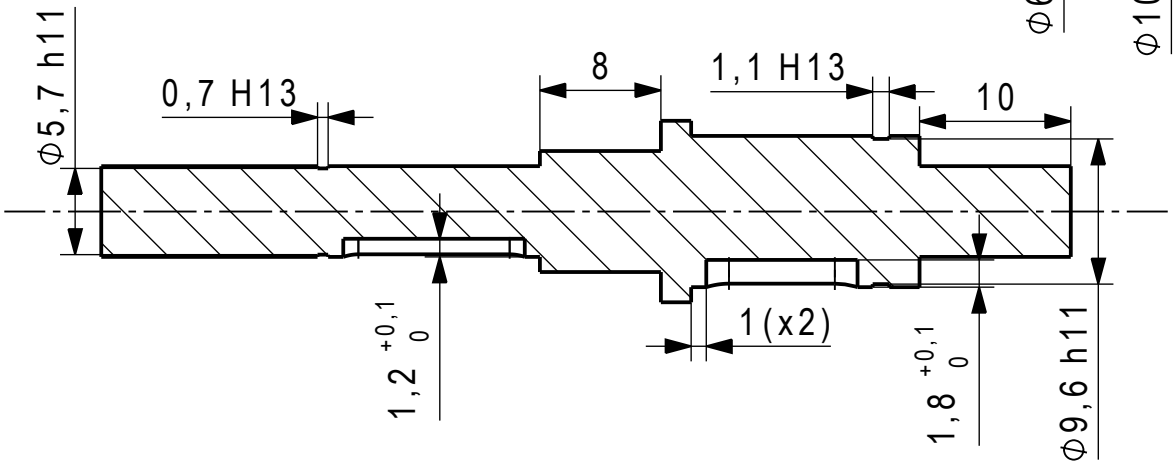
EPFL

Designation: Shaft 1

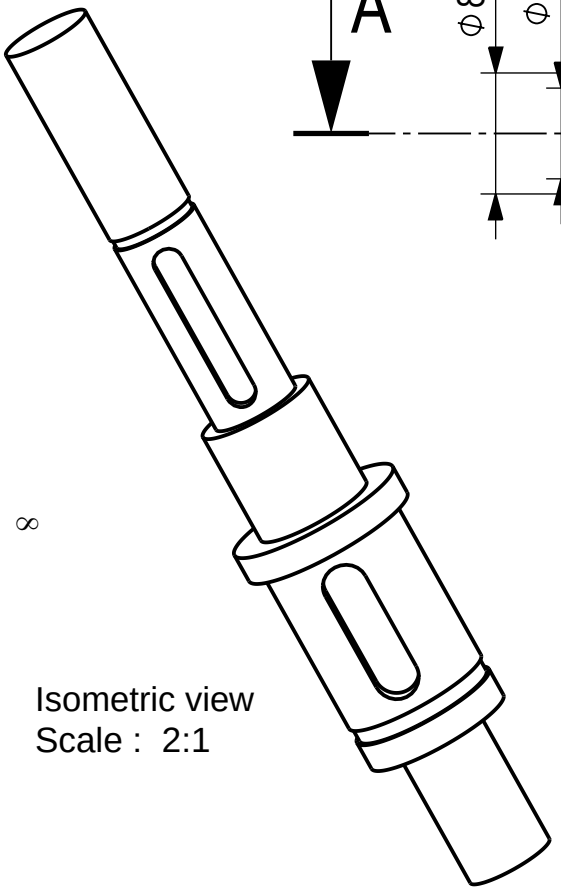
Drawing no.: 01

Front view





Section A-A



Isometric view
Scale : 2:1

+0,4
+0,2

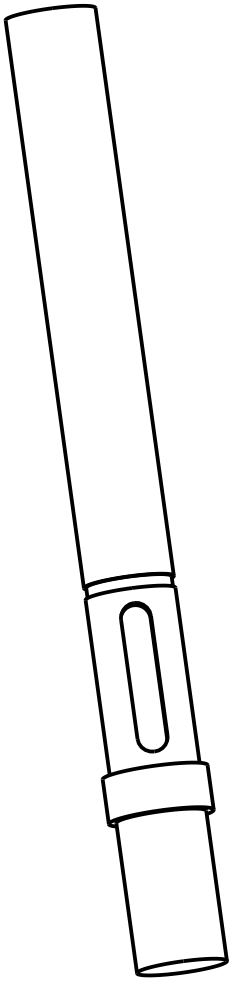
-0,4
-0,6

(√) General tolerances : ISO 2768-mK

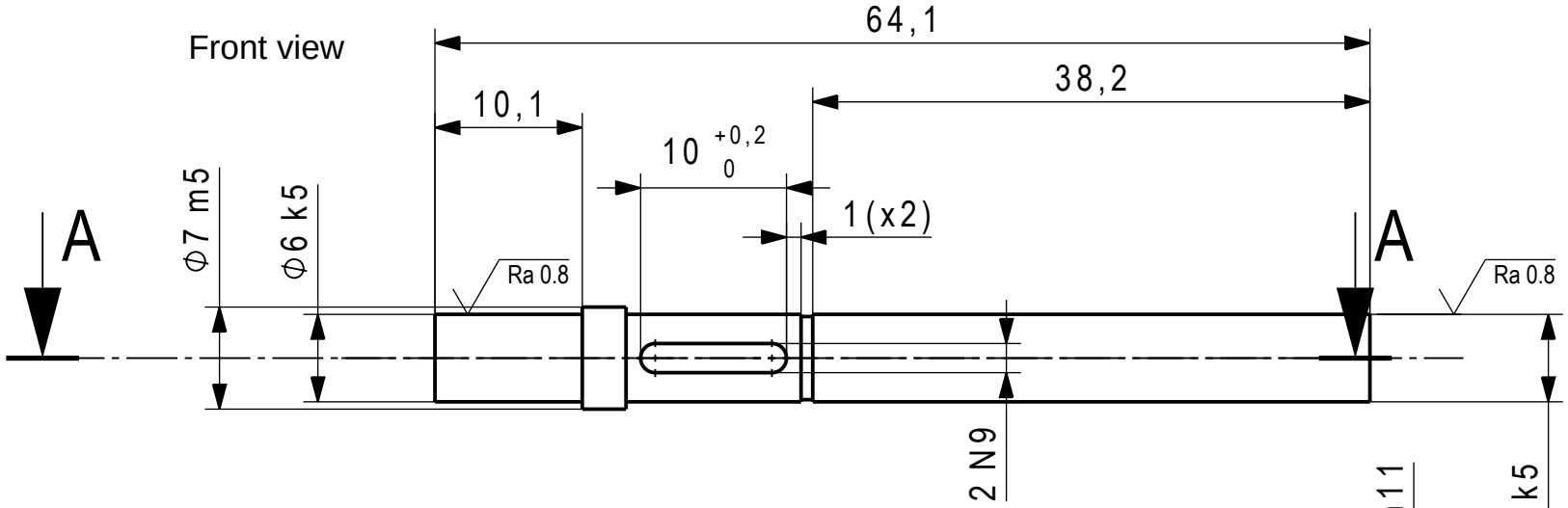
Ra 3.2

Rev.		Rev.		Drawn	30/05/2025	MT_07	Scale 2:1
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				Std. compliant			
				Approved for prod.			
No separate parts list ☐				Order no.			
Sep. parts list, same no. ☐		Material	42 Cr Mo 4 V <= ø 100	Origin		Format	No. of sheets / Sheet no.
Sep. parts list, diff. no. ☐		Mass	0.022 kg	Replaces			
EPFL		Designation	Shaft 2			Drawing no.	02

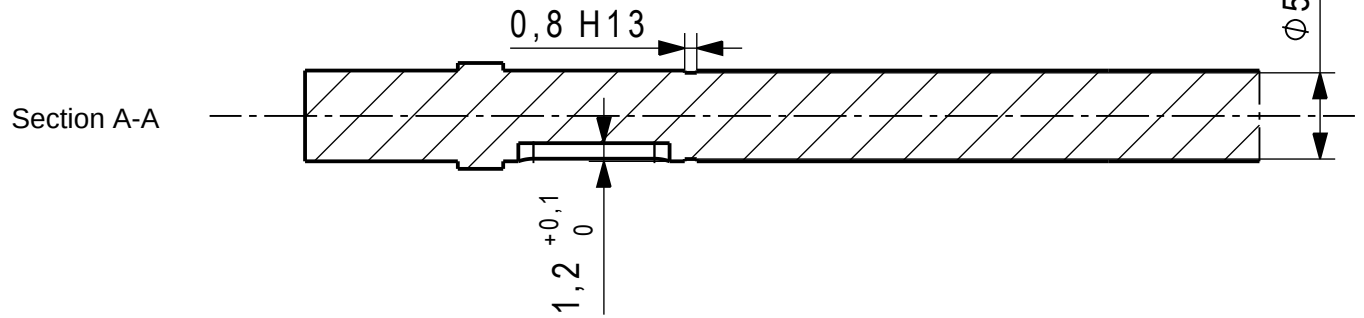
Isometric view
Scale : 2:1



Front view



Section A-A



(√) General tolerances : ISO 2768-mK √ Ra 3.2

+0,4 +0,2

-0,4 -0,6

Rev.		Rev.	Drawn	31/05/2025	MT_07
			Checked		
			Std. compliant		
			Approved for prod.		
No separate parts list ☐			Order no.		
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Sep. parts list, diff. no. ☐	Mass	0.014 kg	Replaces		

Scale 2:1

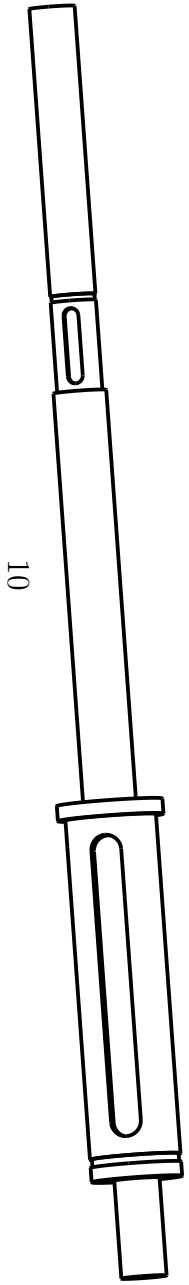
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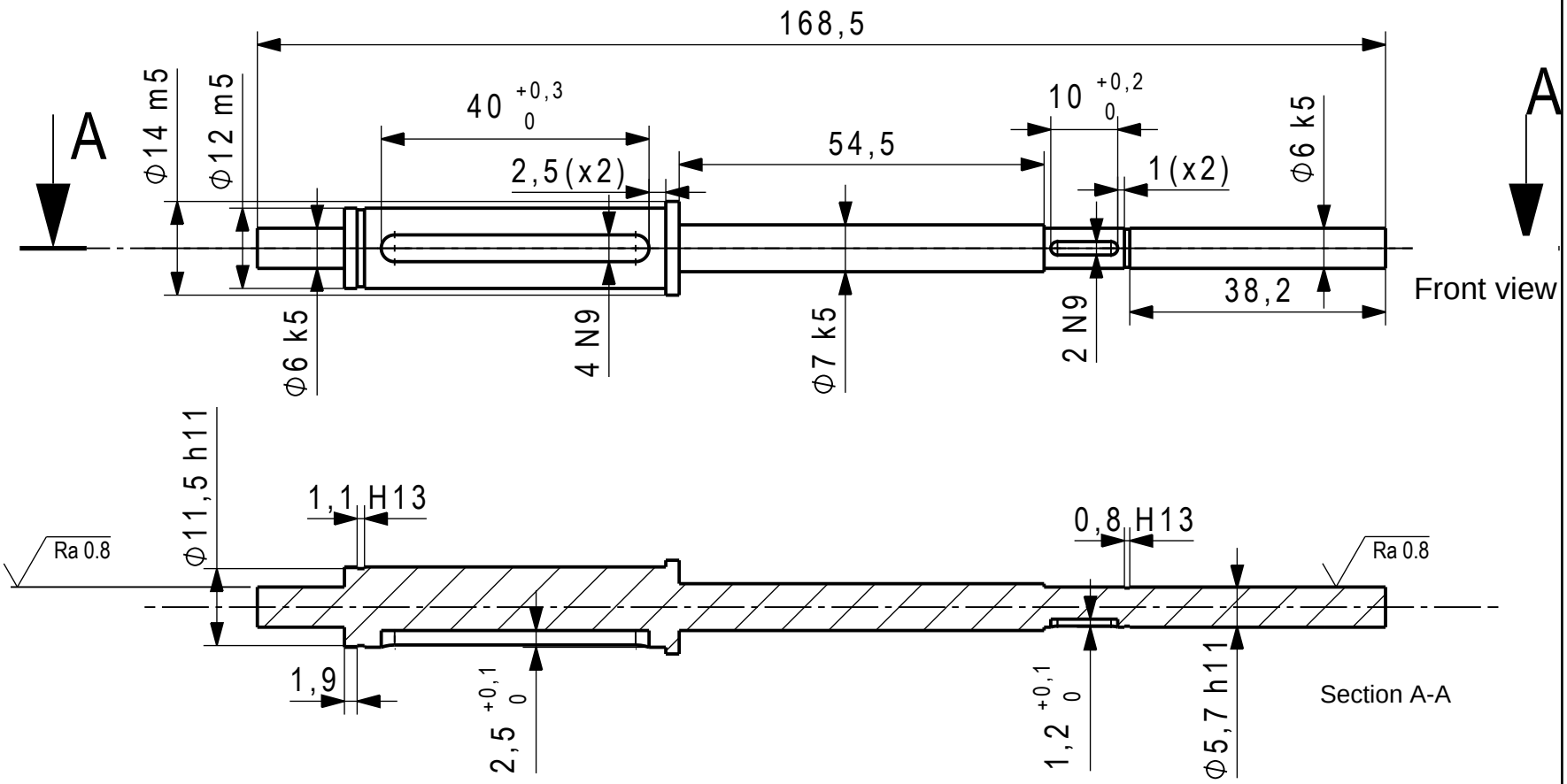
EPFL

Designation: Shaft 3

Drawing no.: 03

Isometric view
Scale : 1:1





(√) General tolerances : ISO 2768-mK

Ra 3.2

-0,4
-0,6

+0,4
+0,2

Rev.		Rev.		Drawn	31/05/2025 MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
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Sep. parts list, diff. no. ☐		Mass	0.072 kg	Replaces	

Scale 1:1

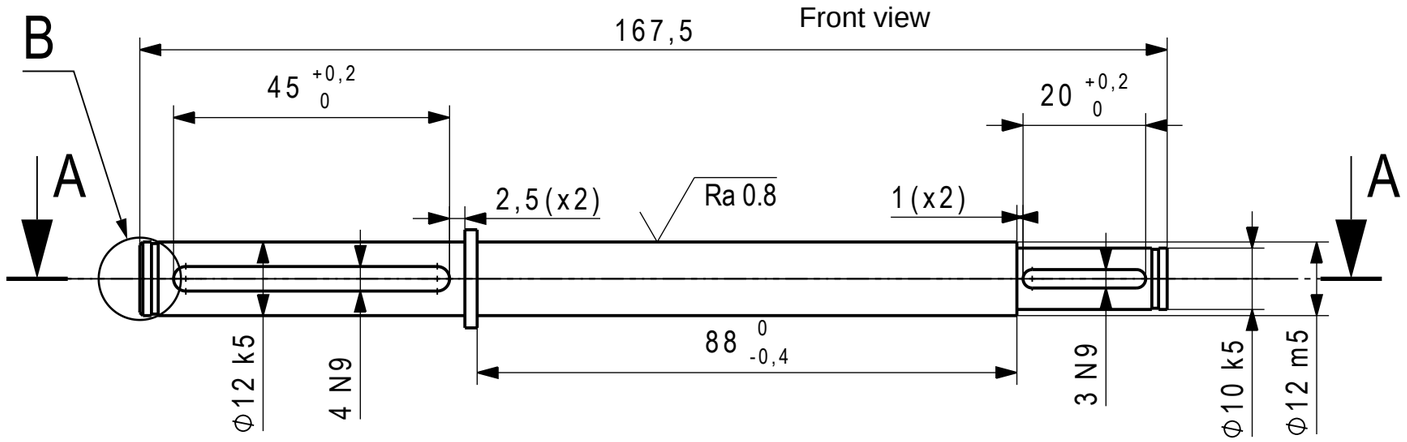
Format | No. of sheets | Sheet no.

EPFL

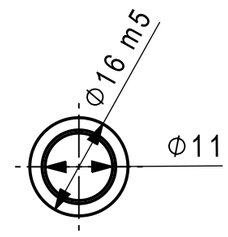
Designation: Shaft 4

Drawing no. 04

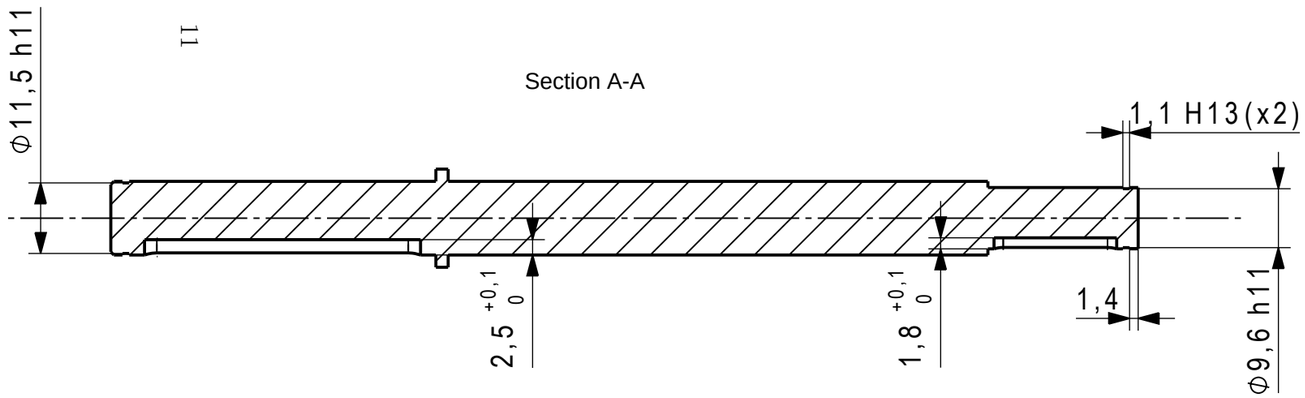
Front view

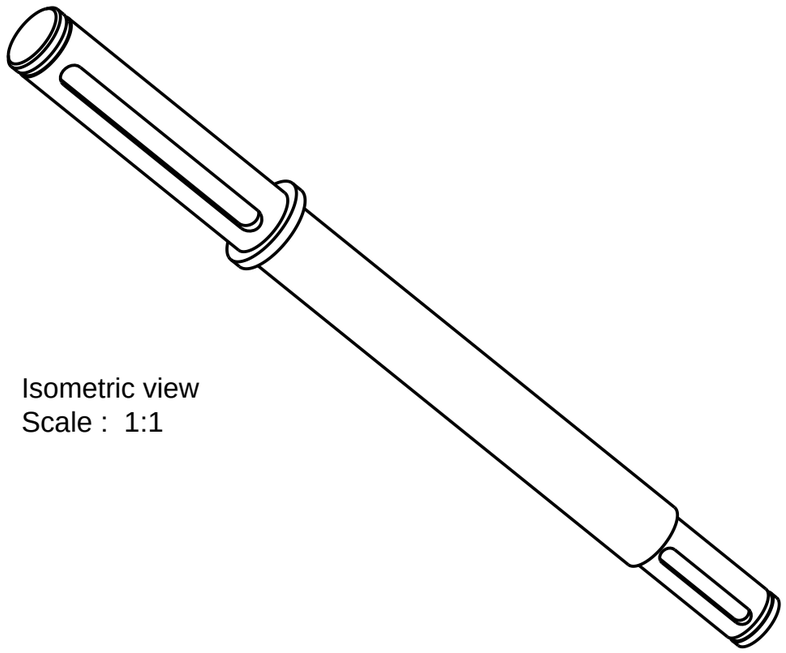


Left view

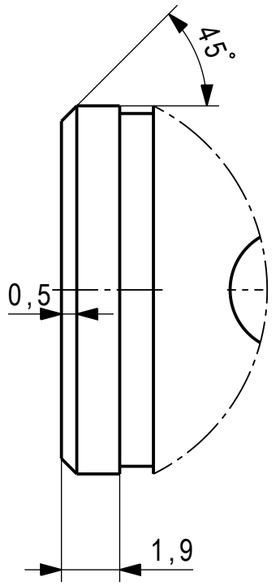


Section A-A

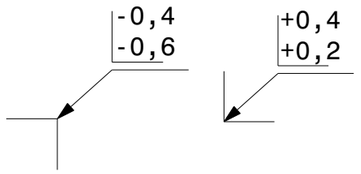




Isometric view
Scale : 1:1



Detail B
Scale : 5:1



(√) General tolerances : ISO 2768-mK

Ra 3.2

Rev.		Rev.		Drawn	31/05/2025	MT_07
				Checked		
				Std. compliant		
				Approved for prod.		

Scale 1:1

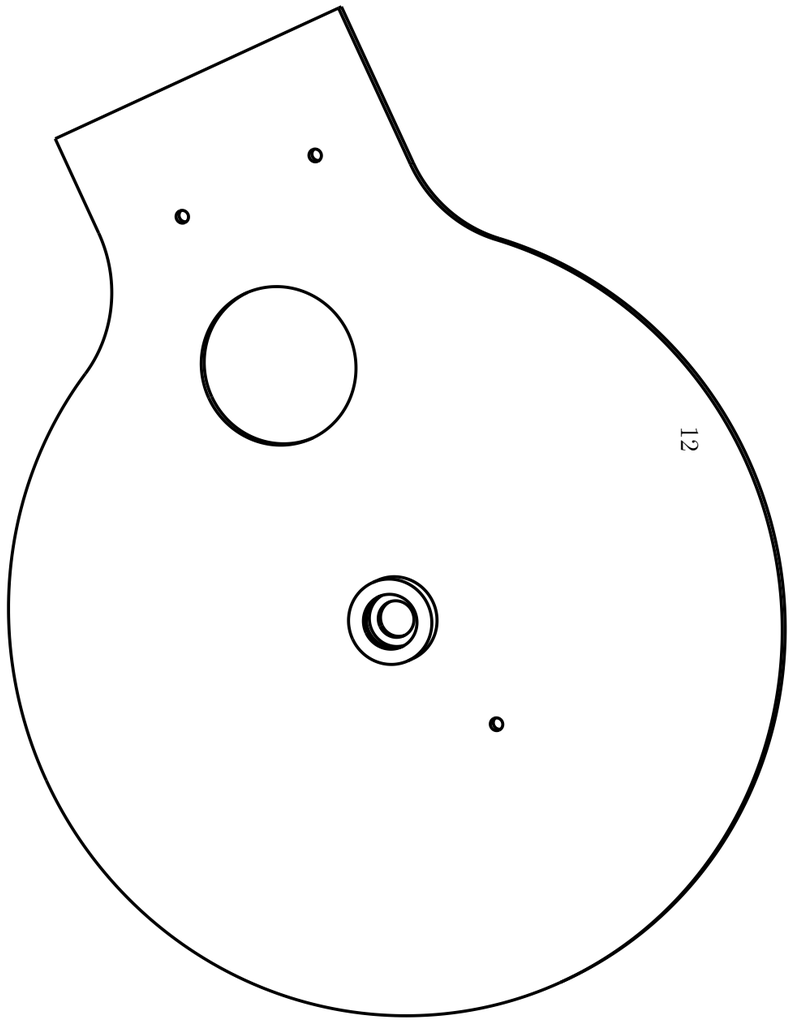
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Sep. parts list, diff. no. ☐	Mass	Replaces			

EPFL

Designation: Shaft 5

Drawing no.: 05

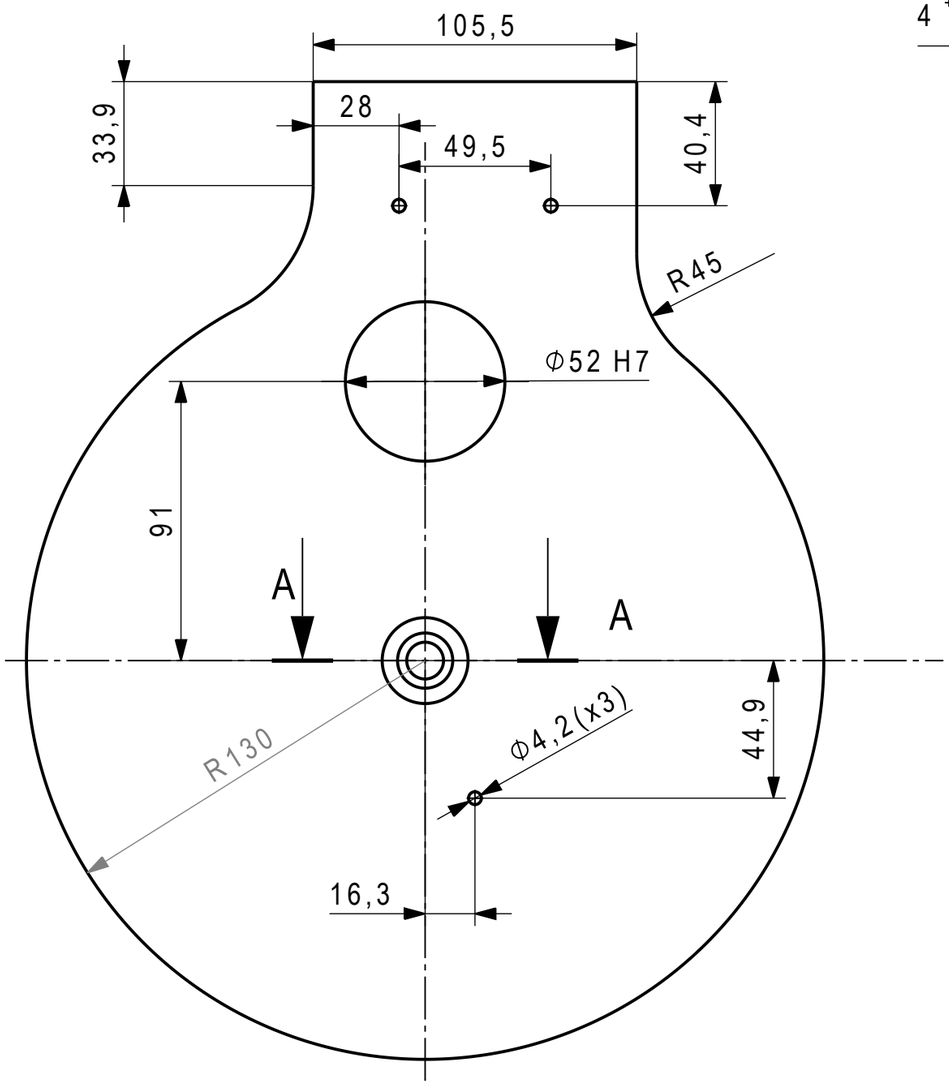
Isometric view
Scale : 1:2

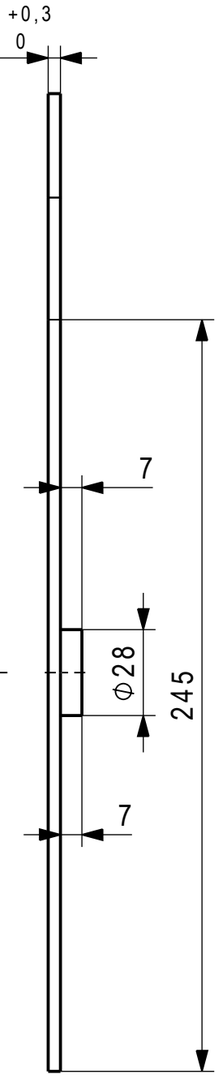


Right view

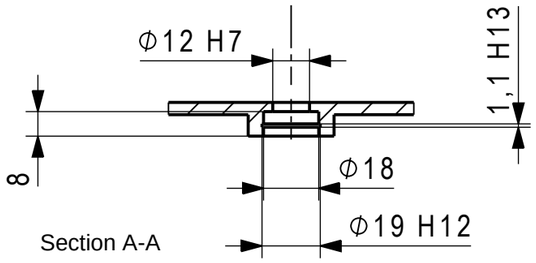


Front view

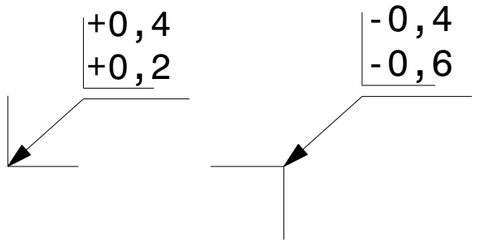




Left view



Section A-A

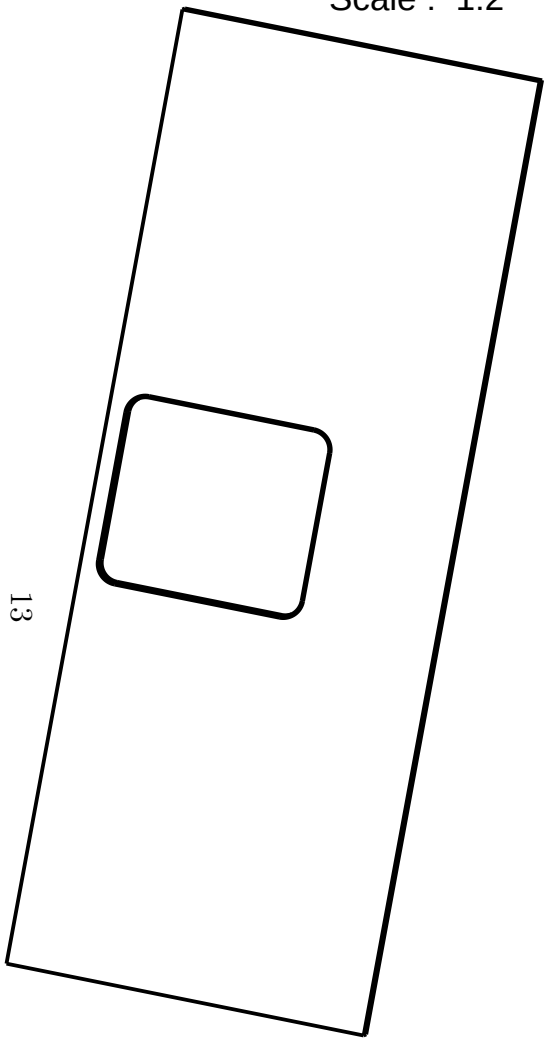


General tolerances :ISO 2768-mK

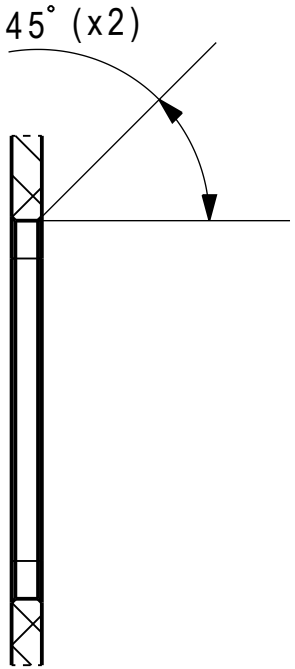
Ra 3.2

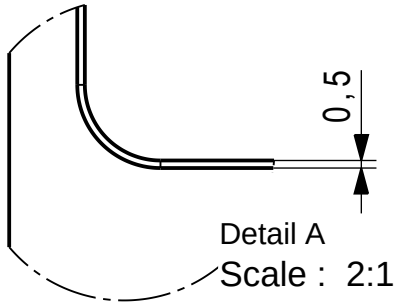
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		Std. compliant			
		Approved for prod.			
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Sep. parts list, same no. ☐	Material: EN AW-Al MgSi T6	Origin		Format	No. of sheets / Sheet no.
Sep. parts list, diff. no. ☐	Mass: 0.63 kg	Replaces			
EPFL	Designation: Rear frame panel			Drawing no.	06

Isometric view
Scale : 1:2

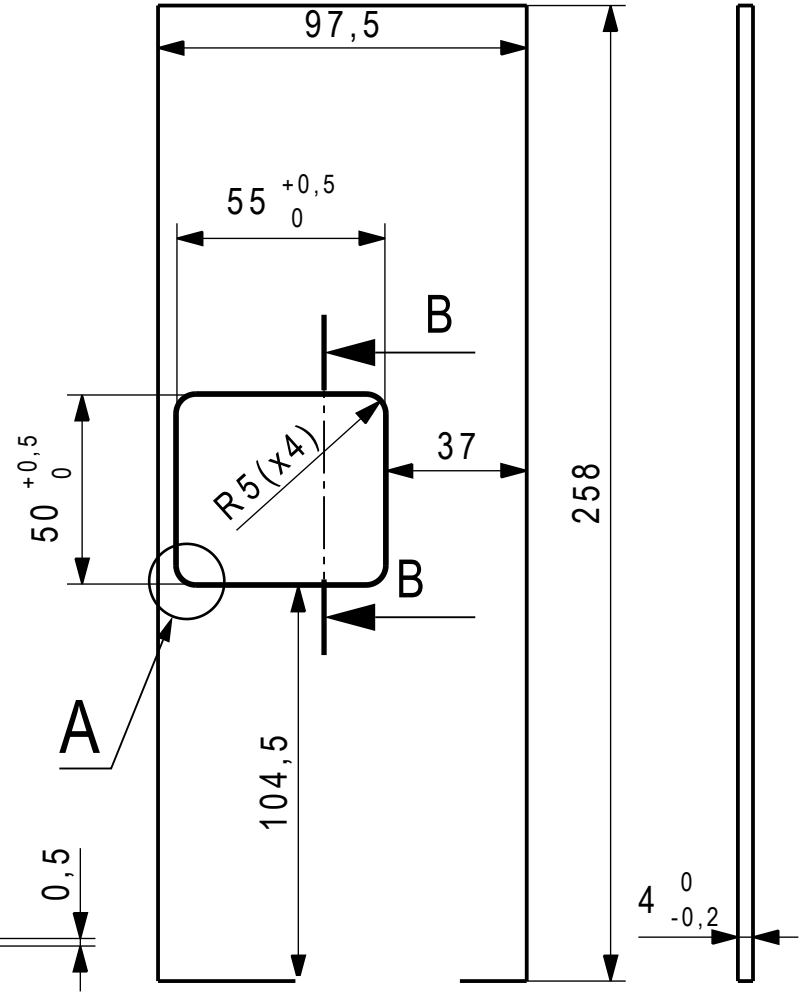


Section B-B
Scale : 1:1





Detail A
Scale : 2:1

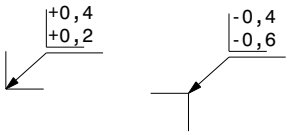


Front view

Left view

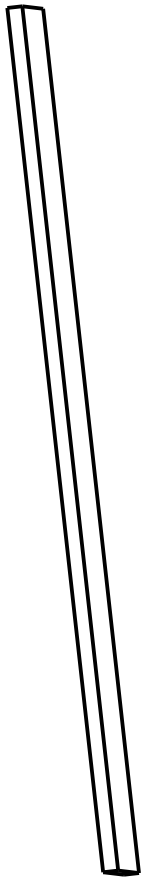
Ra 3.2

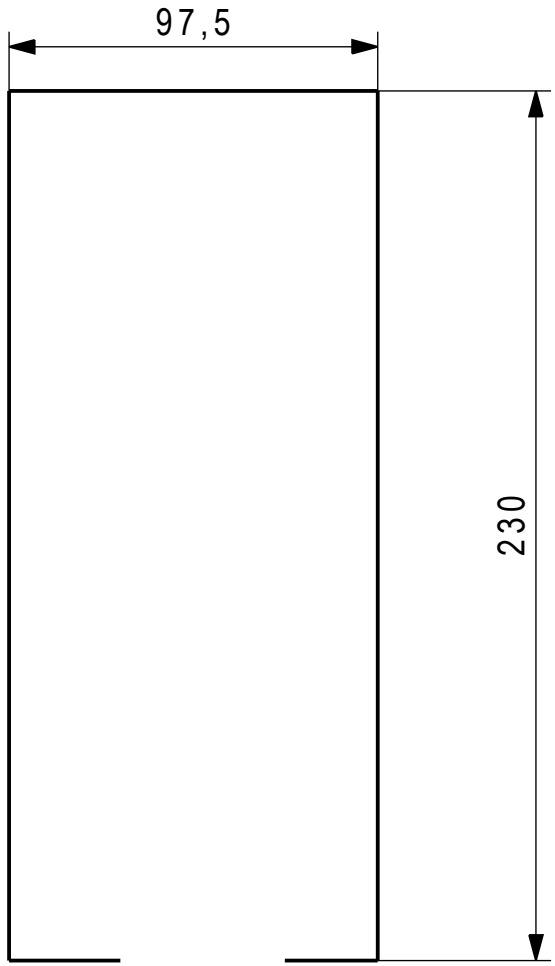
General tolerances 2768-mK



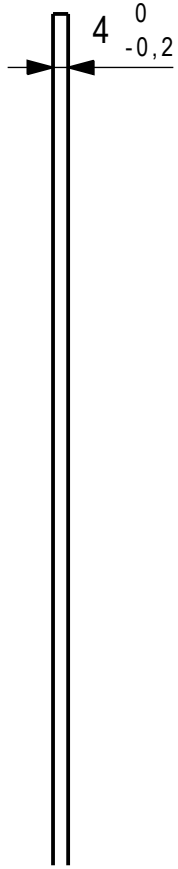
Rev.		Rev.		Drawn	31/05/2025	MT-07	Scale 1:2
				Checked			
				Std. compliant			
				Approved for prod.			
No separate parts list ☐				Order no.			
Sep. parts list, same no. ☐		Material	PC	Origin		Format	No. of sheets / Sheet no.
Sep. parts list, diff. no. ☐		Mass	0.108 kg	Replaces			
EPFL		Designation	Front frame panel			Drawing no.	07

Isometric view
Scale : 1:2





Front view



Left view

14

General tolerances ISO 2768-mK

Ra 3.2

+0,4 +0,2

-0,4 -0,6

Rev.			Rev.		Drawn	31/05/2025	MT_07
					Checked		
					Std. compliant		
					Approved for prod.		
No separate parts list	☐				Order no.		
Sep. parts list, same no.	☐	Material	PC		Origin		
Sep. parts list, diff. no.	☐	Mass	0.108 kg		Replaces		

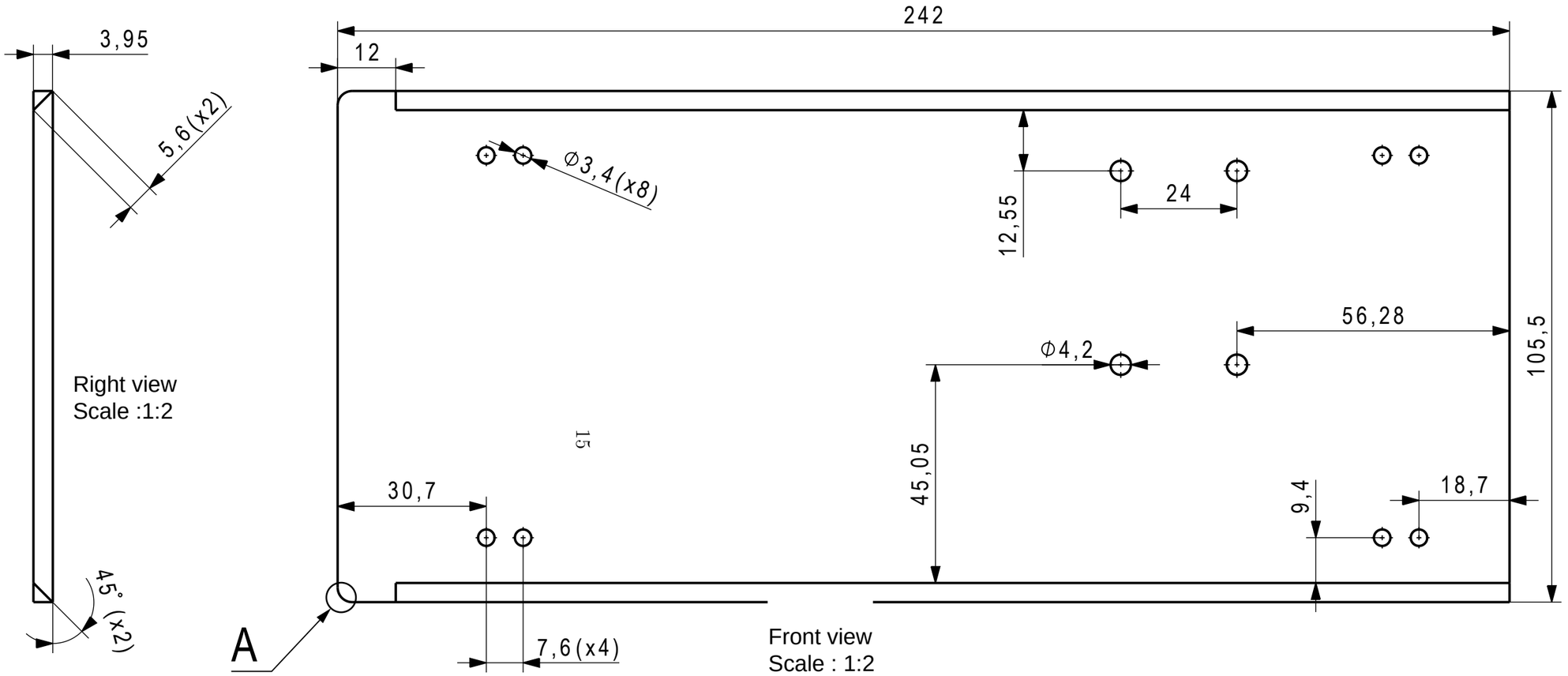
Scale 1:2

Format | No. of sheets | Sheet no.

EPFL

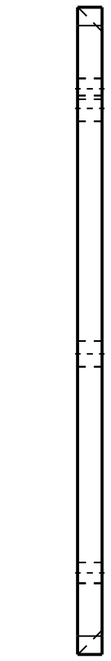
Designation: Top frame panel

Drawing no. 08

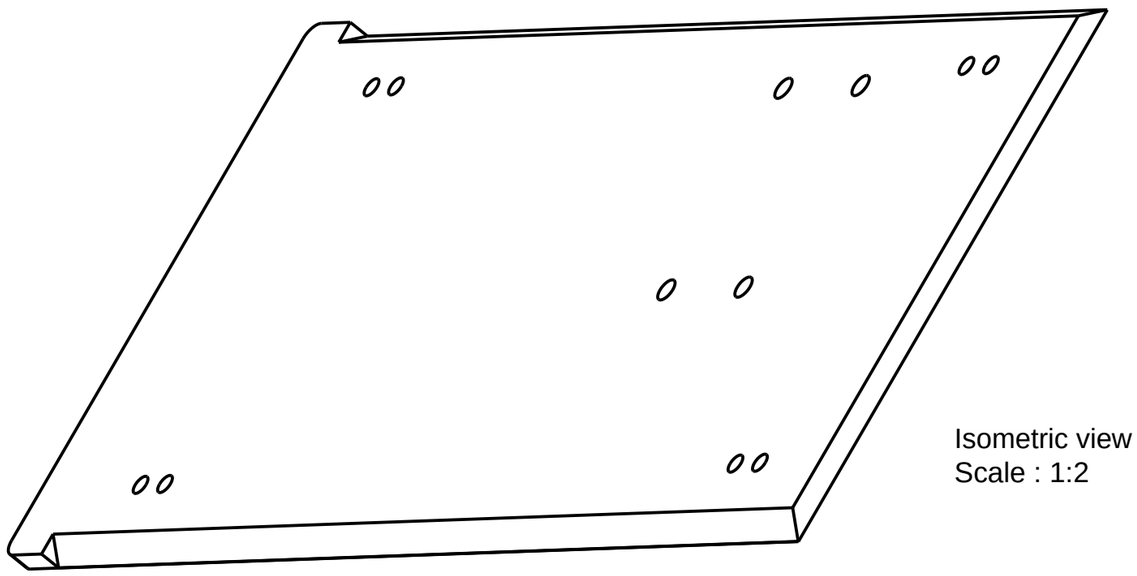


Right view
Scale :1:2

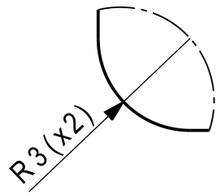
Front view
Scale : 1:2



Left view
Scale :1:2



Isometric view
Scale : 1:2



Detail A
Scale : 5:1

+0,4
+0,2

-0,4
-0,6

Ra 3.2

General tolerances : ISO 2768-mK

Rev.			Rev.		Drawn	31/05/2025	MT_07
					Checked		
					Std. compliant		
					Approved for prod.		
No separate parts list ☐					Order no.		
Sep. parts list, same no. ☐			Material	POM-H	Origin		
Sep. parts list, diff. no. ☐			Mass	0.14 kg	Replaces		

Scale 1:2

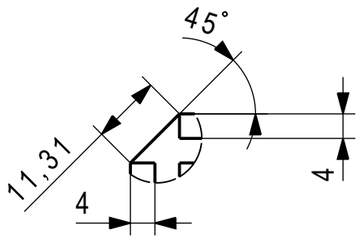
Format | No. of sheets | Sheet no.

EPFL

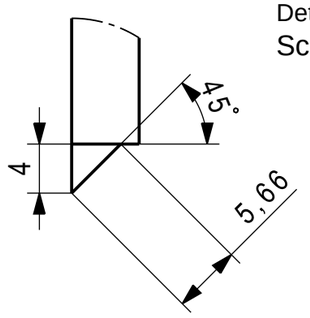
Designation: Bottom frame panel

Drawing no. 09

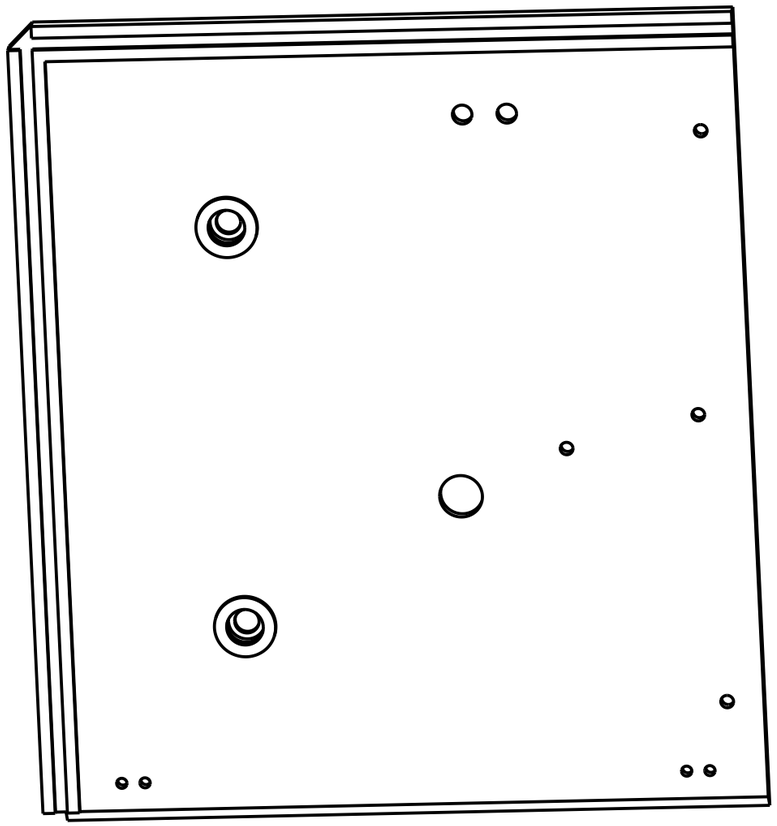
Detail D
Scale : 2:1

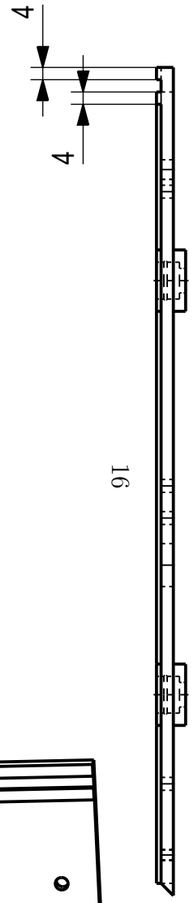


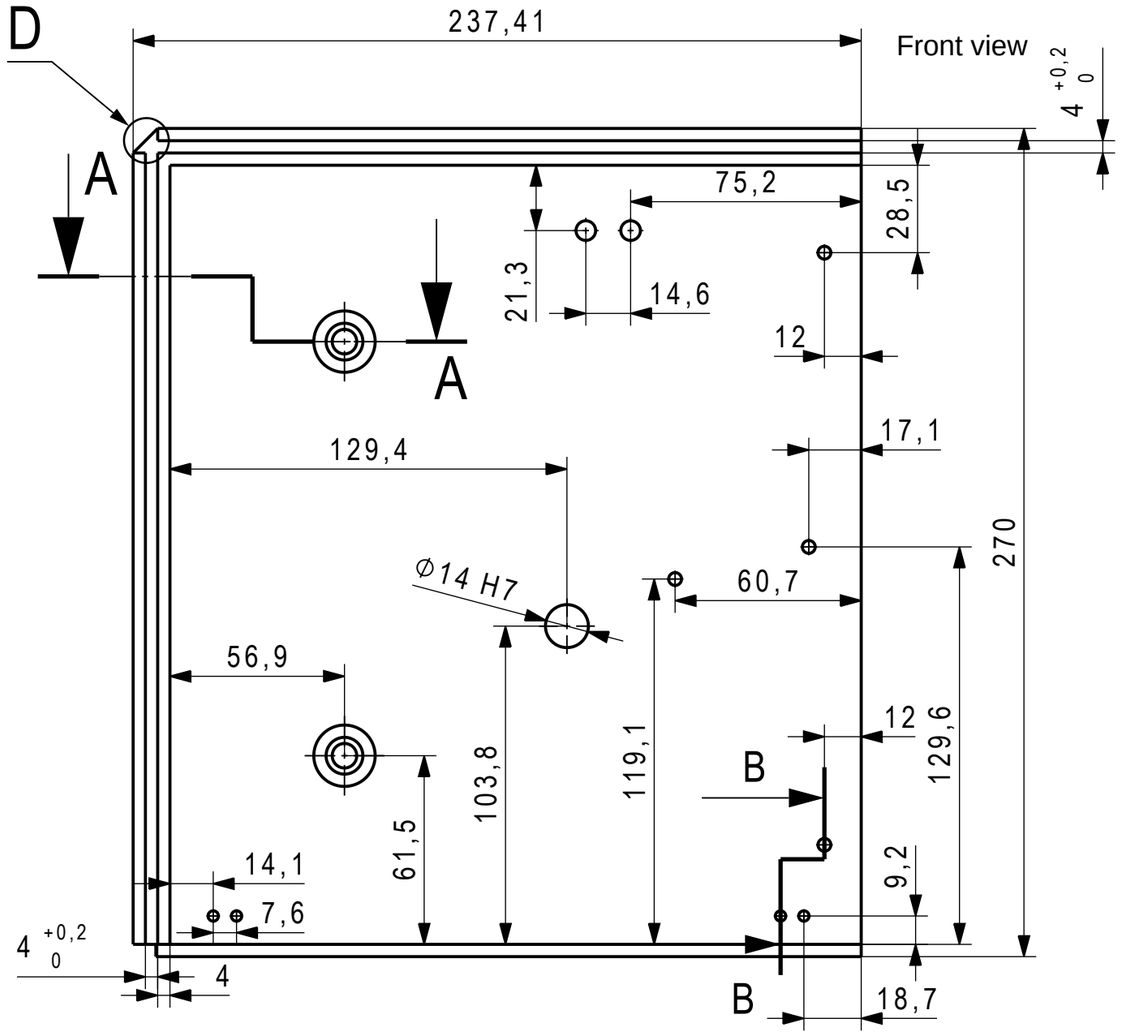
Detail C
Scale : 2:1



Isometric view
Scale : 1:2

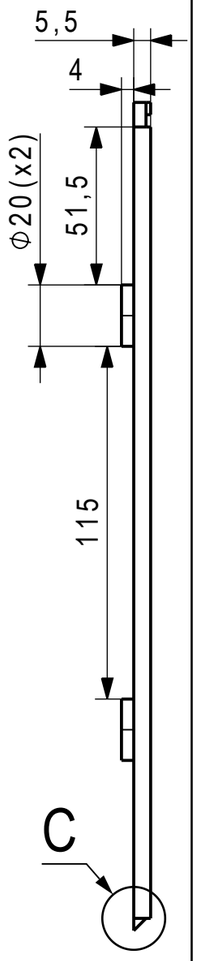




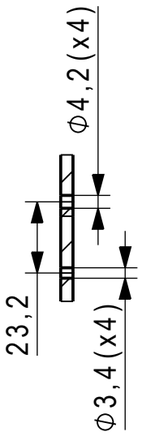


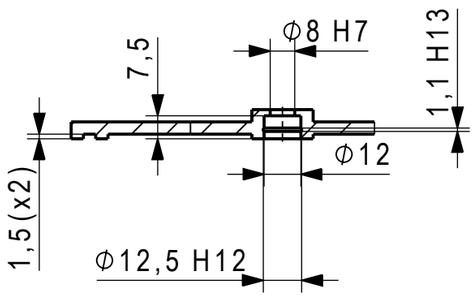
Front view

Left view

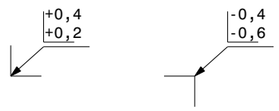


Section B-B





Section A-A

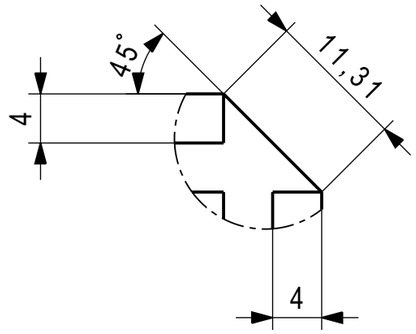


(√) General tolerances : ISO 2768-mK

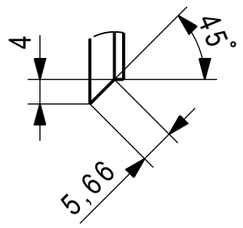
√ Ra 0.8

Rev.		Drawn	31/05/2025	MT_07	Scale 1:2
		Checked			
		Std. compliant			
		Approved for prod.			
No separate parts list ☐		Order no.			
Sep. parts list, same no. ☐	Material	EN AW-Al MgSi T6	Origin		Format / No. of sheets / Sheet no.
Sep. parts list, diff. no. ☐	Mass	0.703 kg	Replaces		
EPFL	Designation	Right frame panel		Drawing no.	10

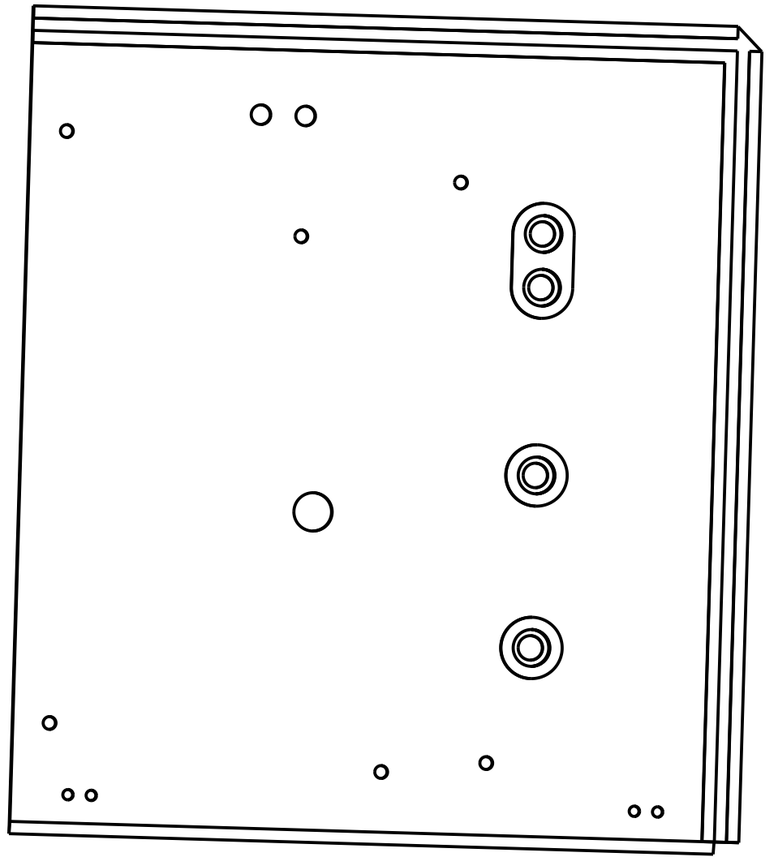
Detail C
Scale : 2:1



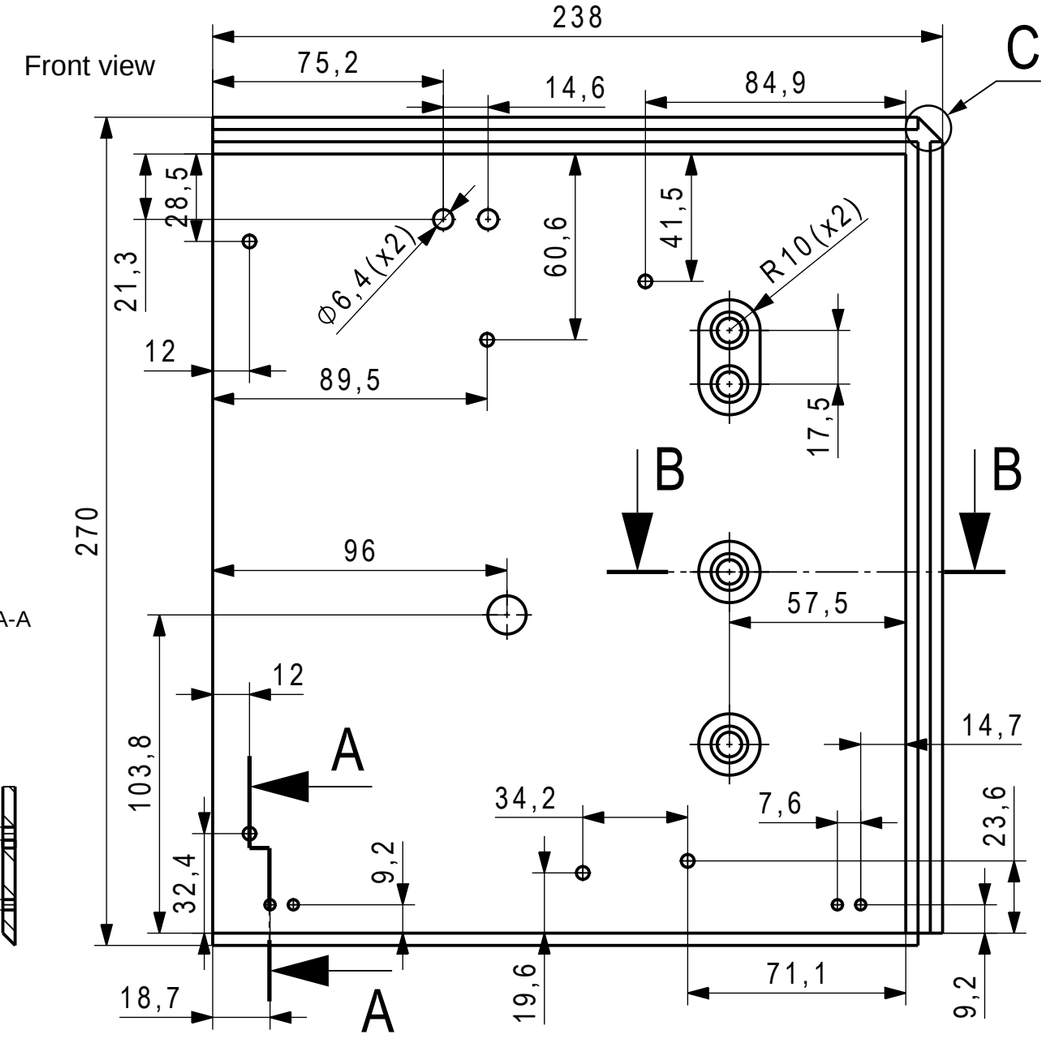
Detail D
Scale : 1:1



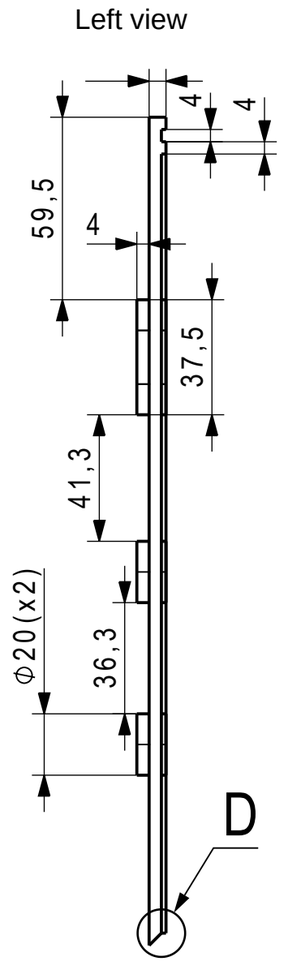
Isometric view
Scale : 1:2



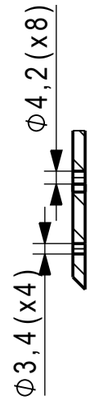
Front view



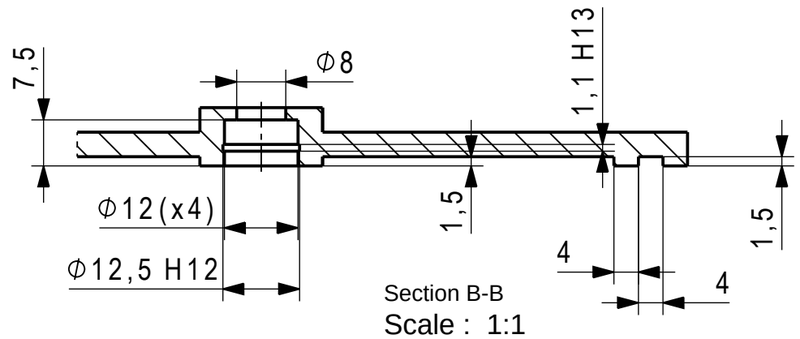
Left view



Section A-A

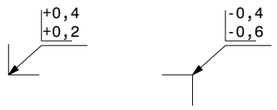


Section B-B
Scale : 1:1

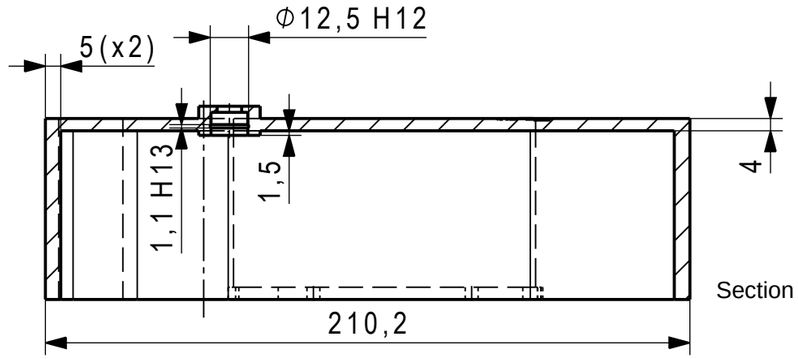


General tolerances : ISO 2768-mK

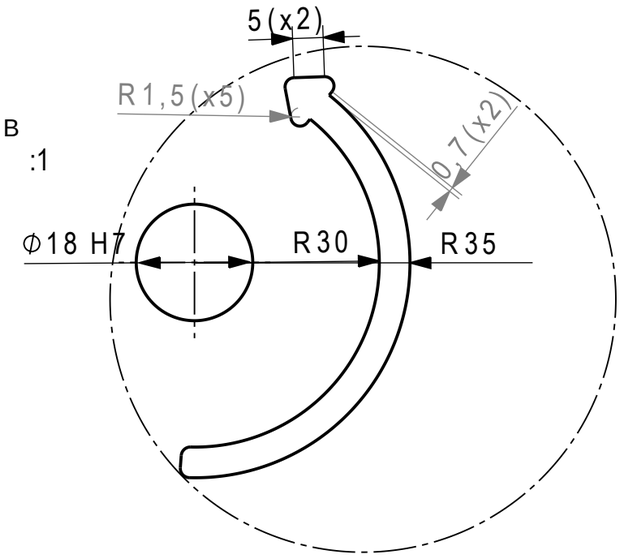
Ra 3.2



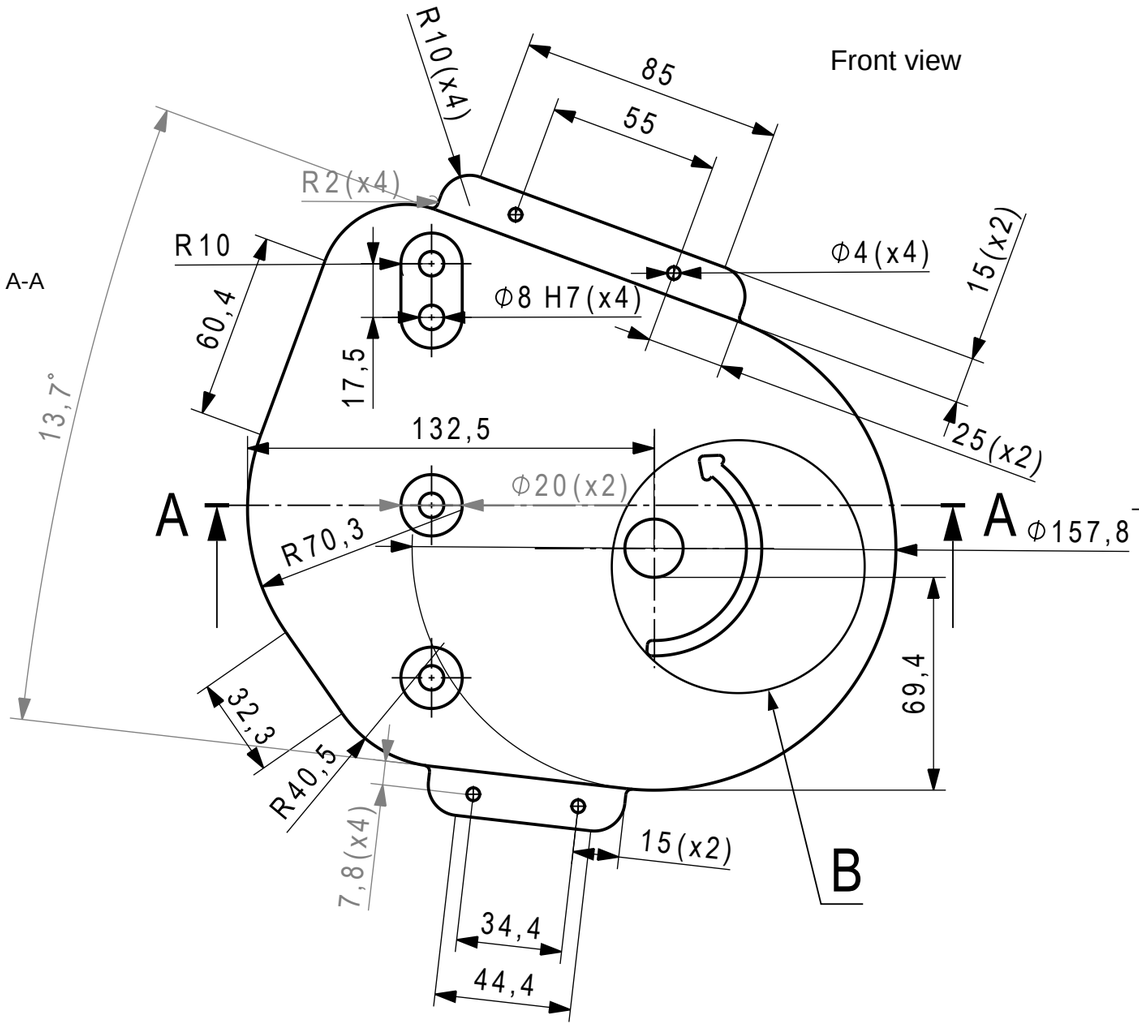
Rev.		Drawn	13/05/2025	MT_07	Scale 1:2
Rev.		Checked			
		Std. compliant			
		Approved for prod.			
No separate parts list ☐		Order no.			
Sep. parts list, same no. ☐	Material: EN AW-Al MgSi T6	Origin		Format	Sheet no.
Sep. parts list, diff. no. ☐	Mass: 0.71 kg	Replaces		No. of sheets	
EPFL	Designation: Left frame panel			Drawing no. 11	



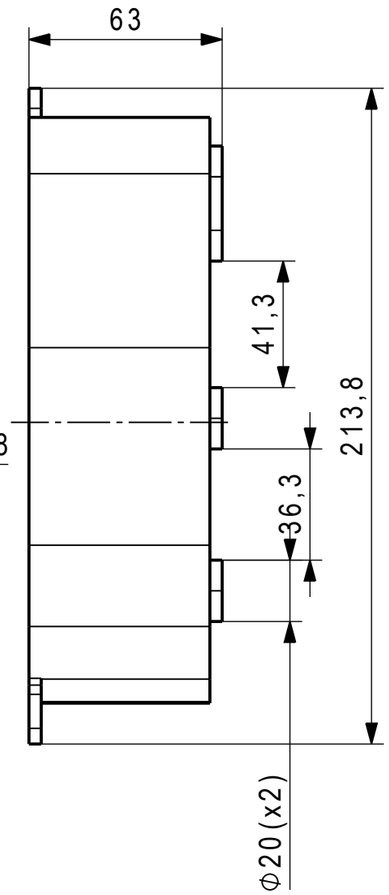
Section view A-A



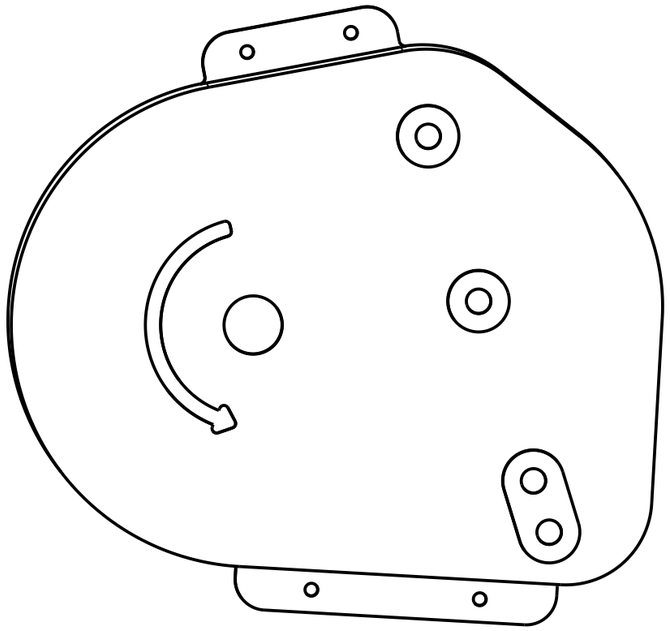
Detail B
:1



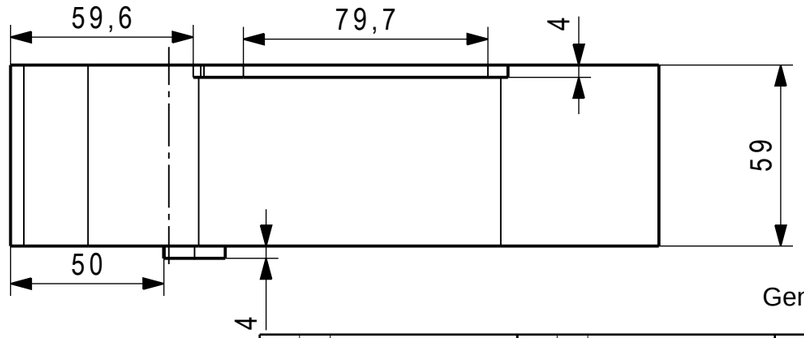
Front view



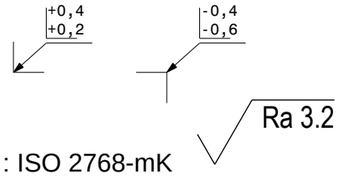
Left view



Isometric view
Scale 1:2

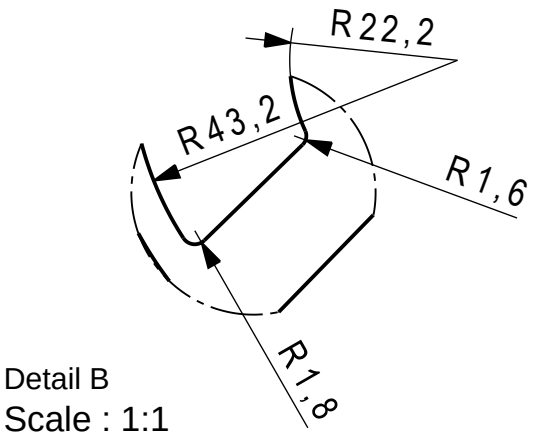


Top view



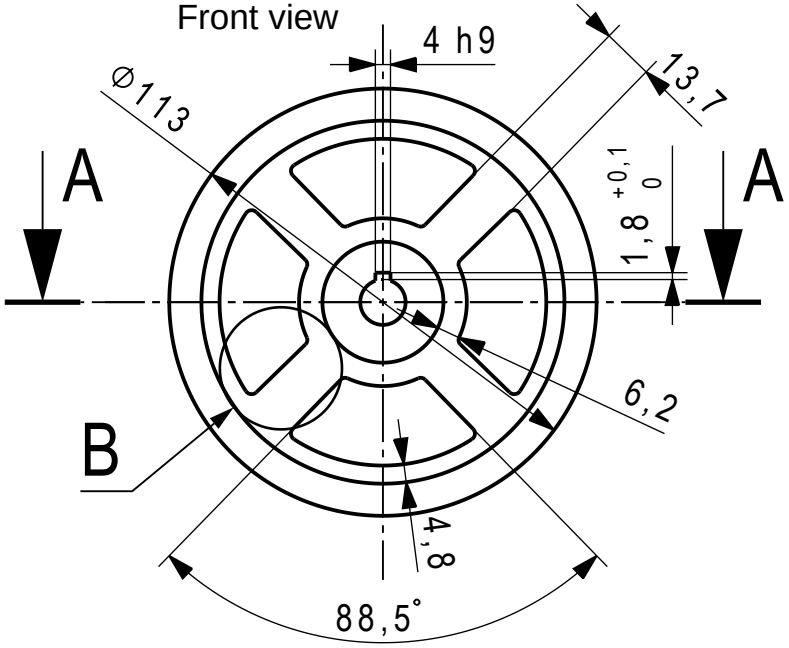
General tolerance : ISO 2768-mK

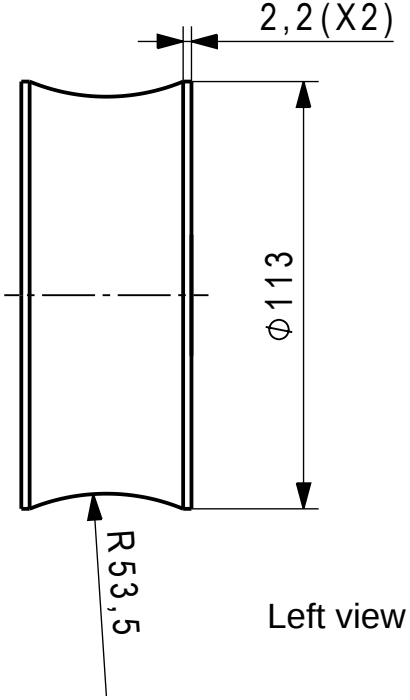
Rev.		Drawn	01/06/2025	MT-07	Scale 1:2
		Checked			
		Std. compliant			
		Approved for prod.			
No separate parts list ☐		Order no.			
Sep. parts list, same no. ☐	Material: EN AW-Al MgSi T6	Origin		Format	Sheet no.
Sep. parts list, diff. no. ☐	Mass: 0.821 kg	Replaces		No. of sheets	
EPFL	Designation: Gear housing			Drawing no. 12	



Detail B
Scale : 1:1

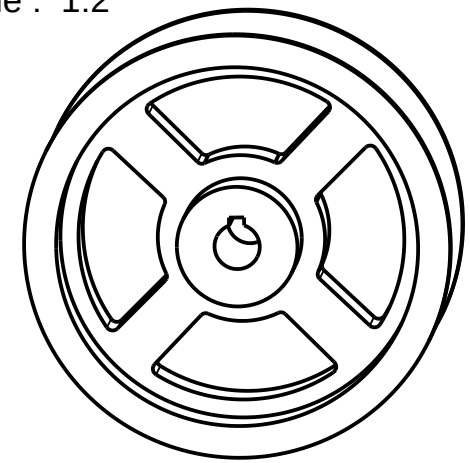
Front view

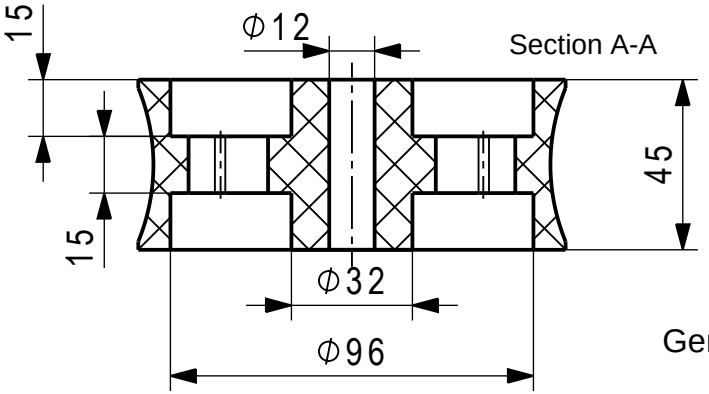




Left view

Isometric view
Scale : 1:2





General tolerances : ISO 2768-mK

Ra 3.2

+0,4
+0,2

-0,4
-0,6

Rev.		Rev.		Drawn	31/05/2025 MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	POM-H	Origin	
Sep. parts list, diff. no. ☐		Mass	0.24 kg	Replaces	

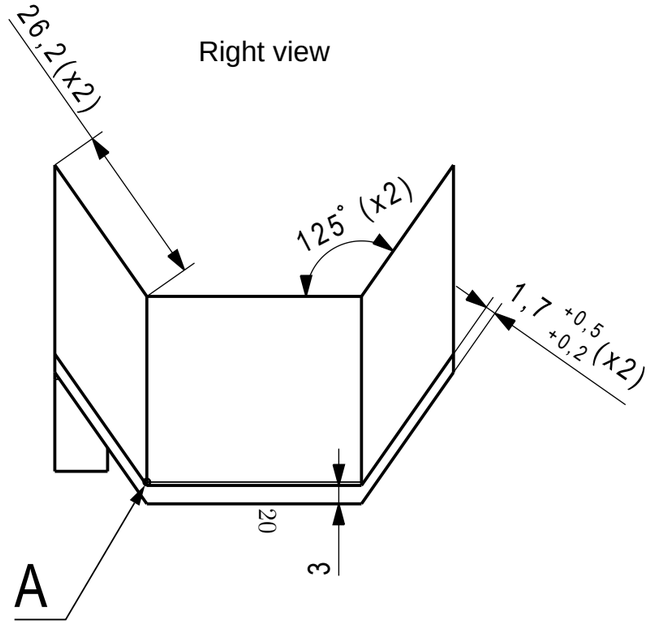
Scale 1:1

Format | No. of sheets | Sheet no.

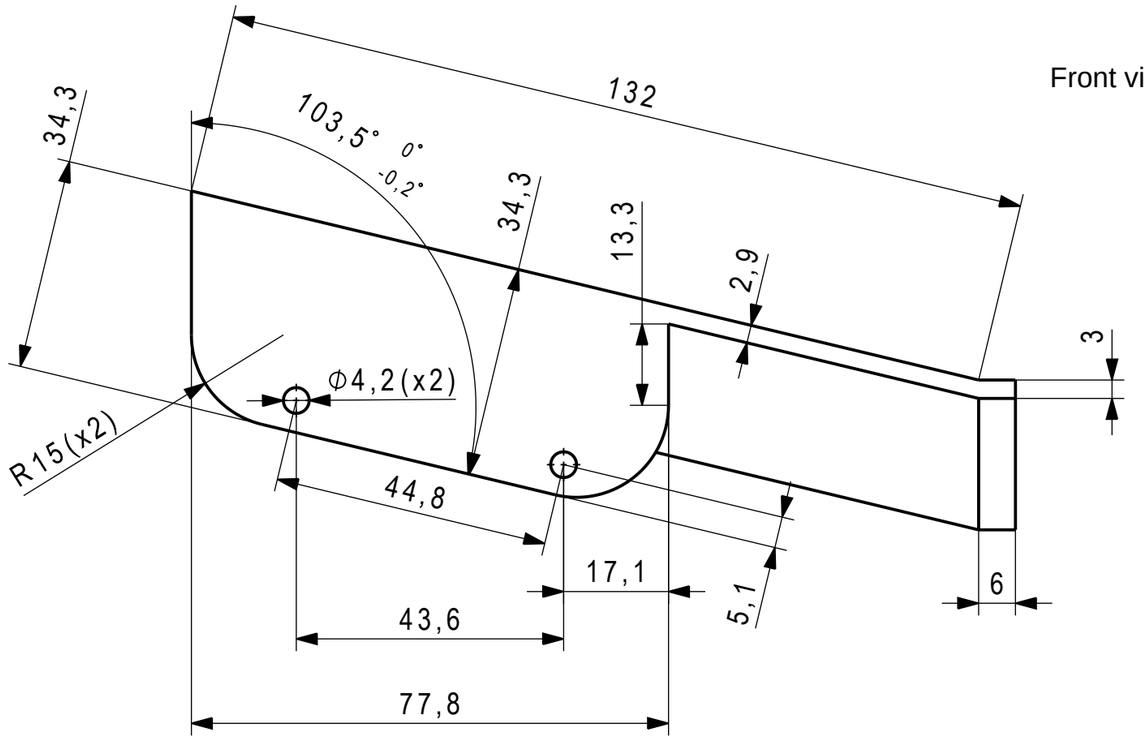
EPFL

Designation: Launch wheels

Drawing no. 13



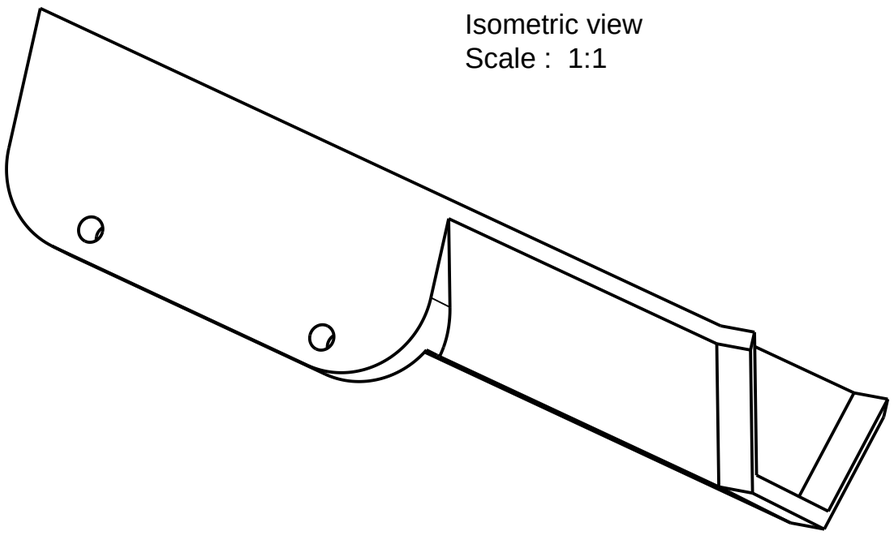
Right view



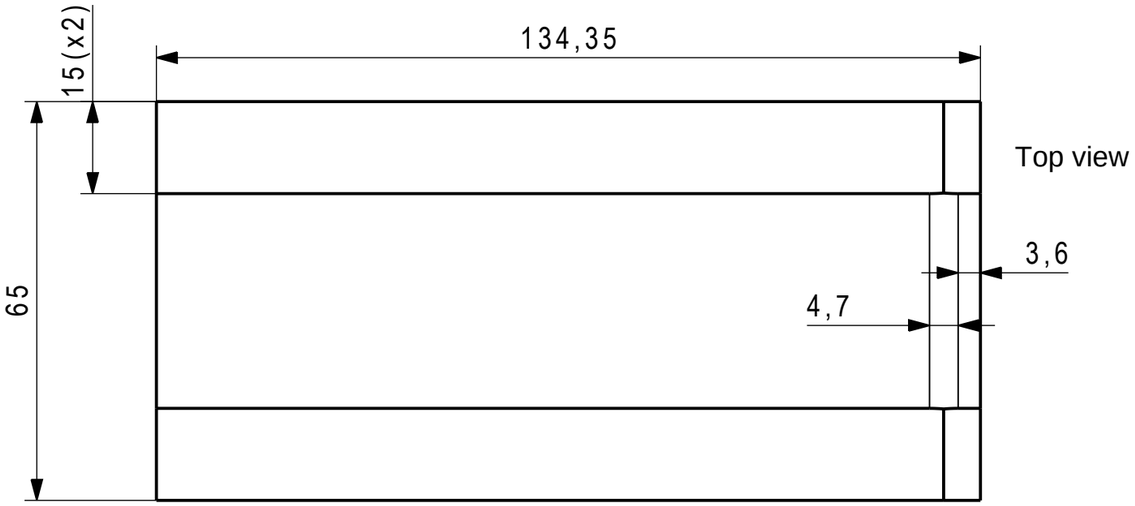
Front view

Detail A
Scale : 10:1





Isometric view
Scale : 1:1



Top view

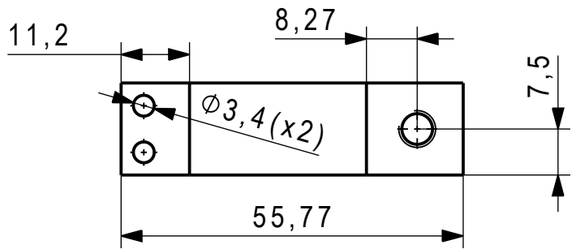
-0,4 -0,6 +0,4 +0,2

General tolerance : ISO 2768-mK

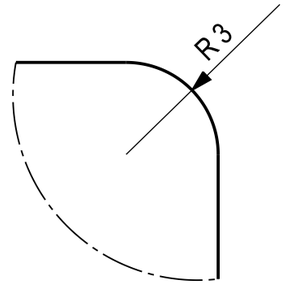
Ra 3.2

Rev.		Rev.		Drawn	01/06/2025	MT_07	Scale 1:1
				Checked			
				Std. compliant			
				Approved for prod.			
No separate parts list ☐				Order no.			
Sep. parts list, same no. ☐		Material	POM-H	Origin		Format / No. of sheets	Sheet no.
Sep. parts list, diff. no. ☐		Mass	0.064 kg	Replaces			
EPFL		Designation	Ball track			Drawing no.	14

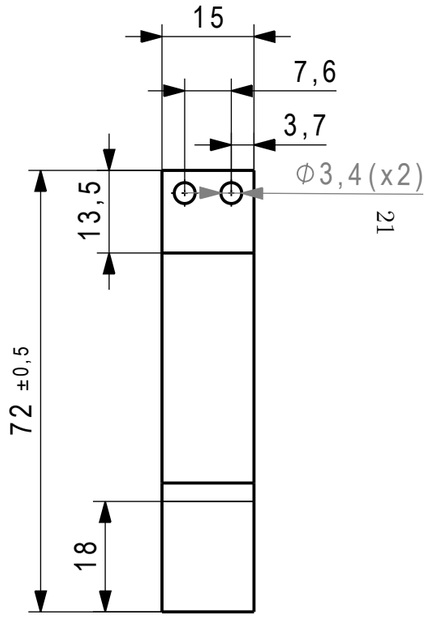
Bottom view



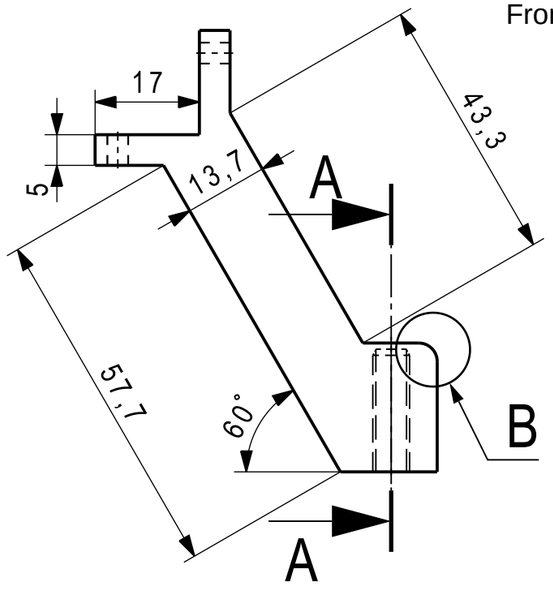
Detail B
Scale : 5:1

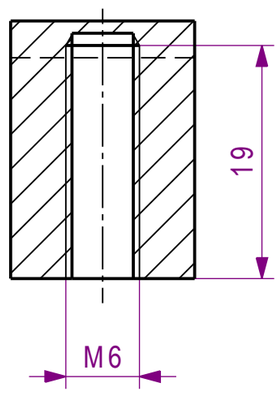


Right view



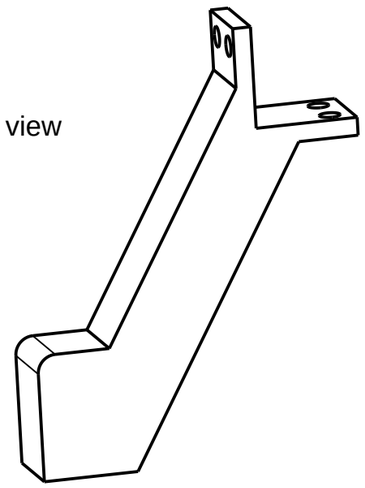
Front view





Section A-A
Scale : 2:1

Isometric view



General tolerances : ISO 2768-mK

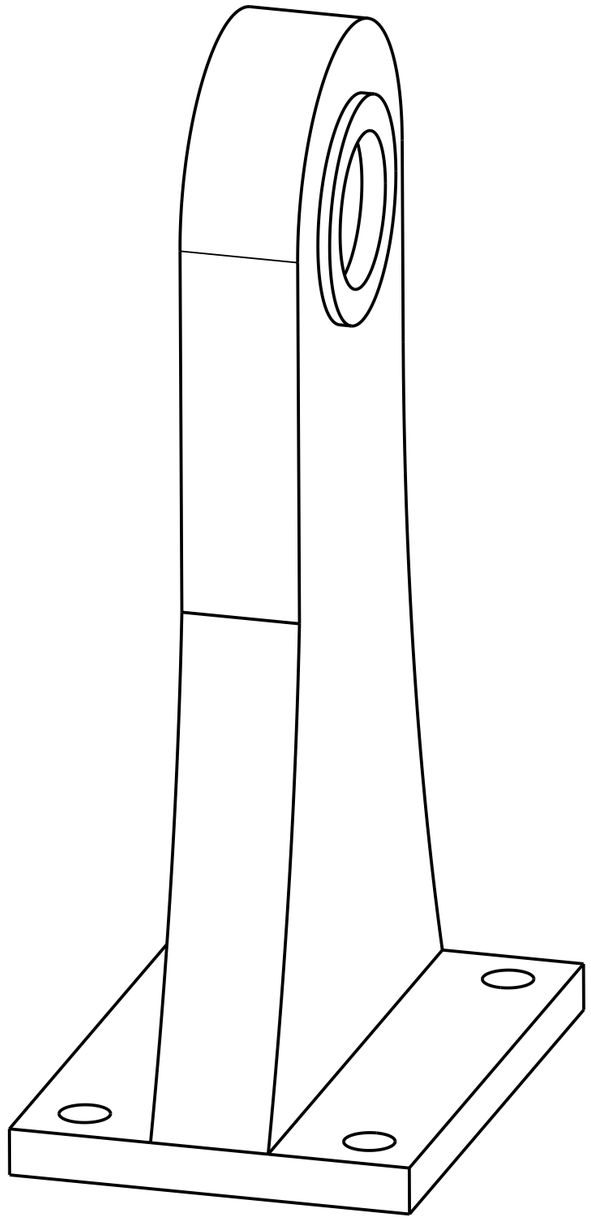
Ra 3.2

+0,4
+0,2

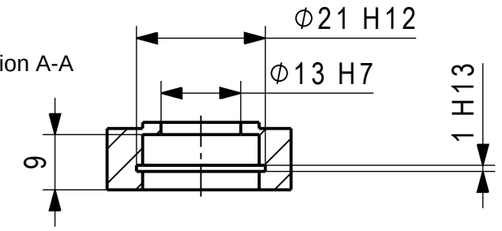
-0,4
-0,6

Rev.		Rev.		Drawn	30/05/2025	MT-07	Scale 1:1
				Checked			
				Std. compliant			
				Approved for prod.			
No separate parts list ☐				Order no.			
Sep. parts list, same no. ☐		Material	EN AW-Al MgSi T6	Origin		Format / No. of sheets	Sheet no.
Sep. parts list, diff. no. ☐		Mass	0.044 kg	Replaces			
EPFL		Designation	Legs			Drawing no.	15

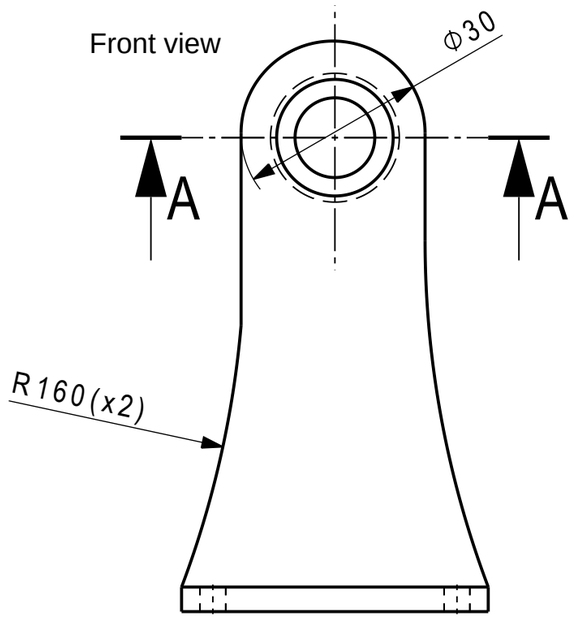
Isometric view
Scale : 2:1

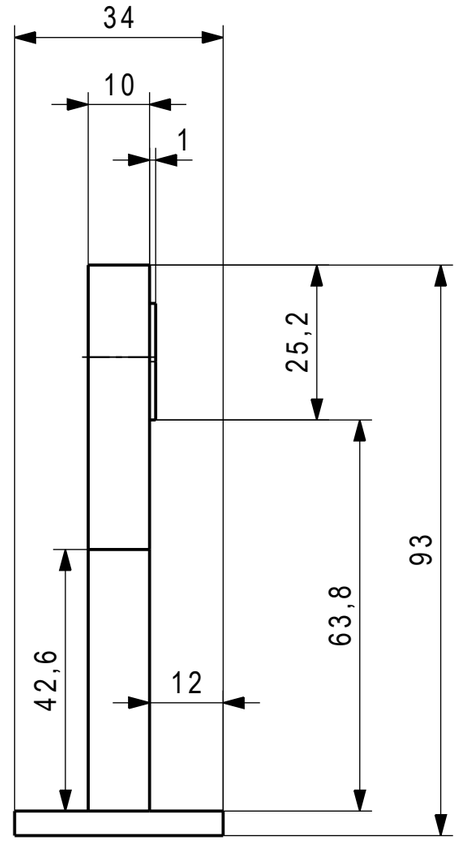


Section A-A

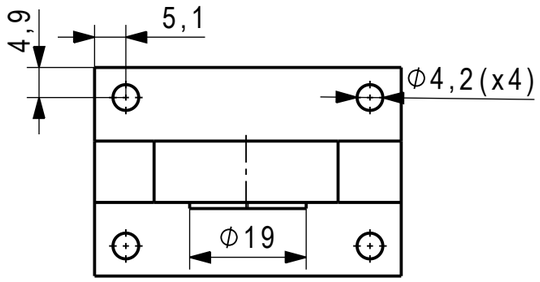


Front view





Top view



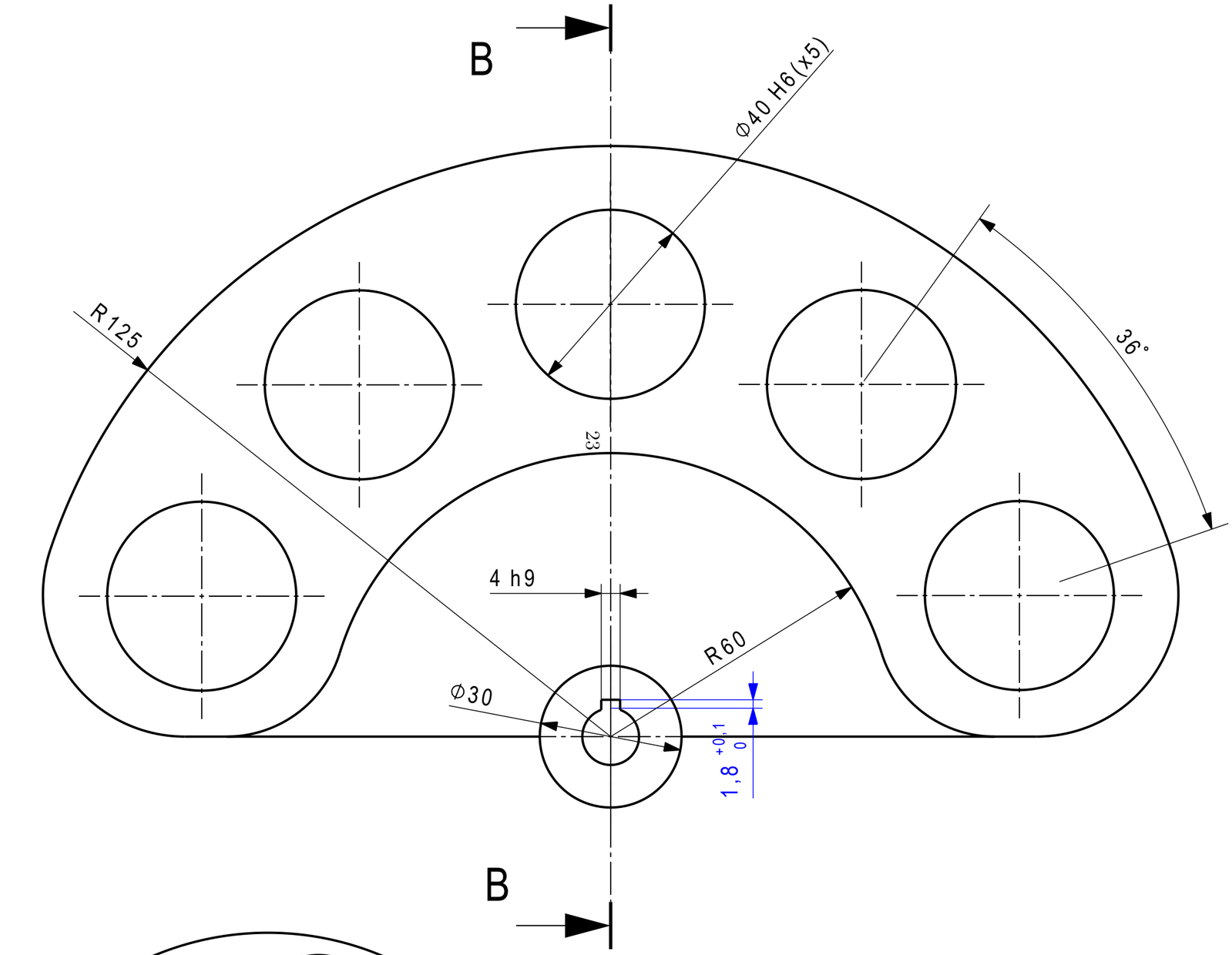
General tolerances ISO : 2768-f

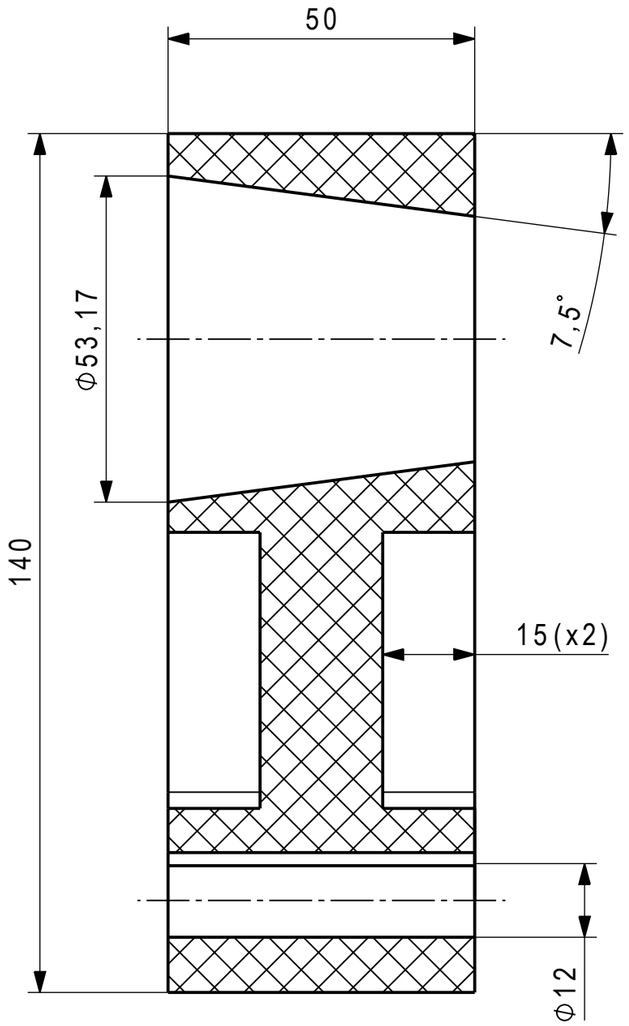
Ra 3.2

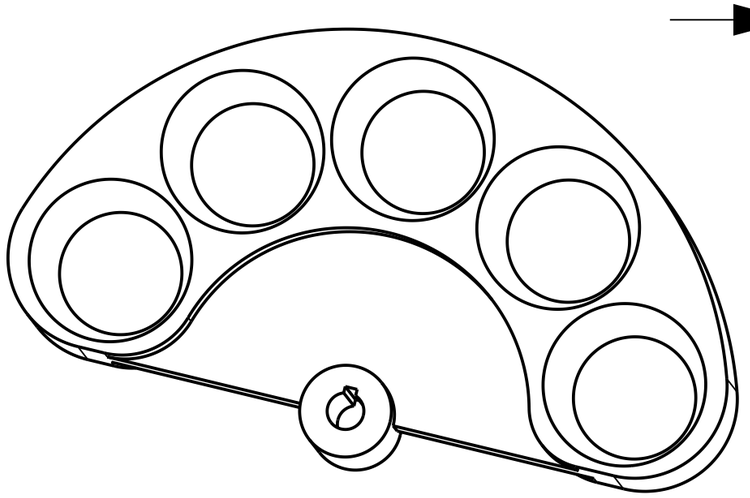
+0,4
+0,2

-0,4
-0,6

Rev.	Rev.		Drawn	MT_07	Scale 1:1
			Checked		
			Std. compliant		
			Approved for prod.		
No separate parts list ☐			Order no.		
Sep. parts list, same no. ☐	Material	EN AW-Al MgSi T6	Origin		Format / No. of sheets / Sheet no.
Sep. parts list, diff. no. ☐	Mass	0.09 kg	Replaces		
EPFL	Designation	Shaft support			Drawing no. 16







+0,4 +0,2

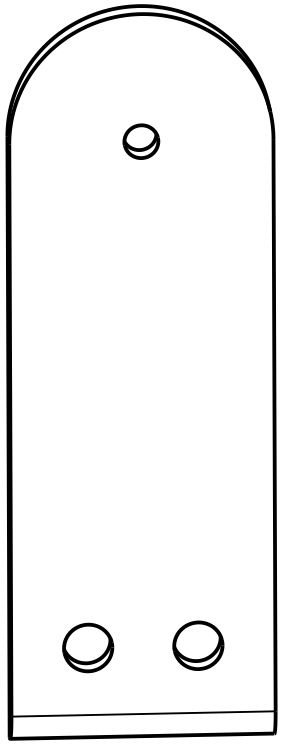
-0,4 -0,6

Ra 3.2

General tolerances:

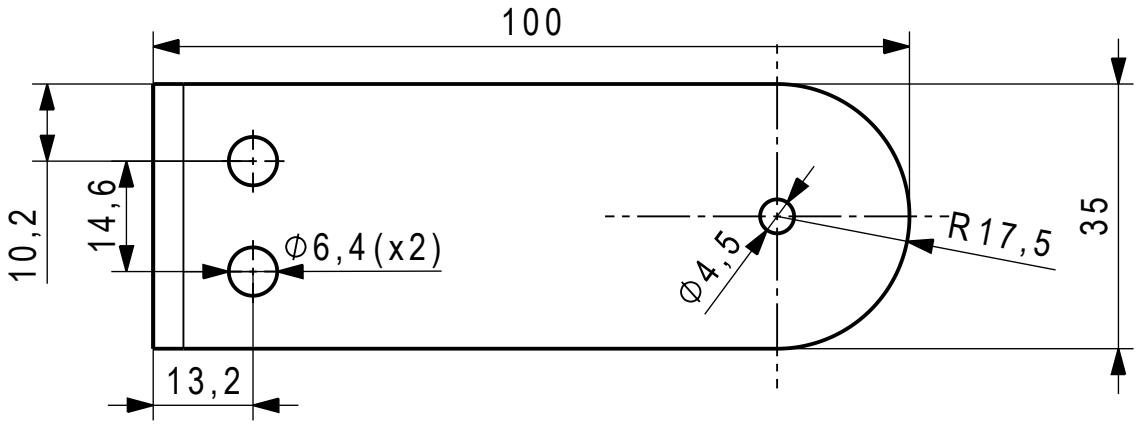
Rev.		Rev.		Drawn		MT-07	Scale 1:1
				Checked			
				Std. compliant			
				Approved for prod.			
No separate parts list ☐				Order no.			
Sep. parts list, same no. ☐		Material	POM-H	Origin		Format / No. of sheets	Sheet no.
Sep. parts list, diff. no. ☐		Mass	0.939 kg	Replaces			
EPFL		Designation	Ball magazine			Drawing no.	17

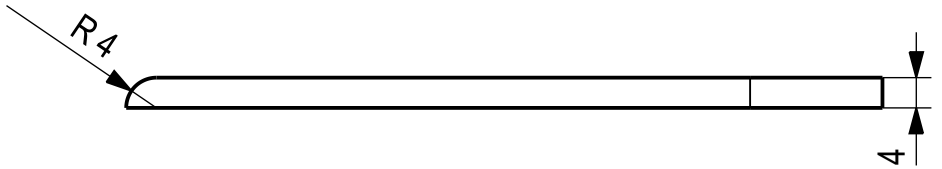
Isometric view
Scale : 1:1



24

Front view





Top view

Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	EN AW-Al MgSi T6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.07 kg	Replaces	

Scale: 1:1

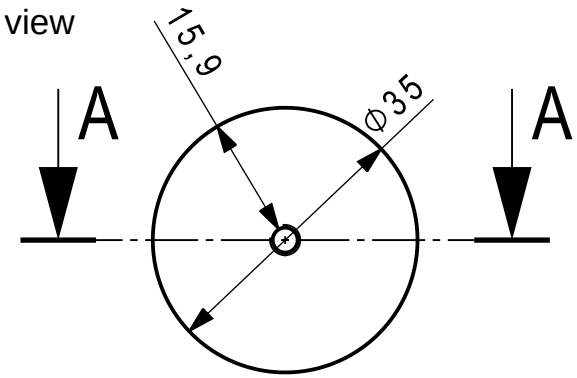
Format | No. of sheets | Sheet no.

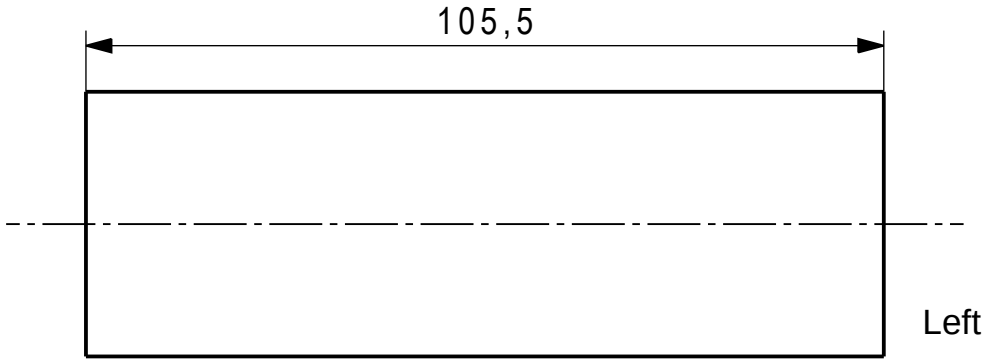
EPFL

Designation: Handle (mounts)

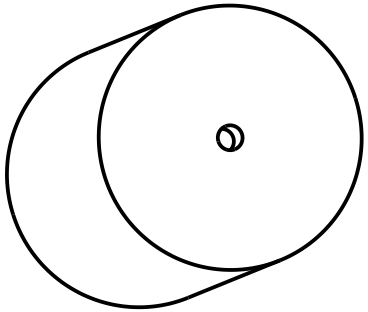
Drawing no.: 18

Front view





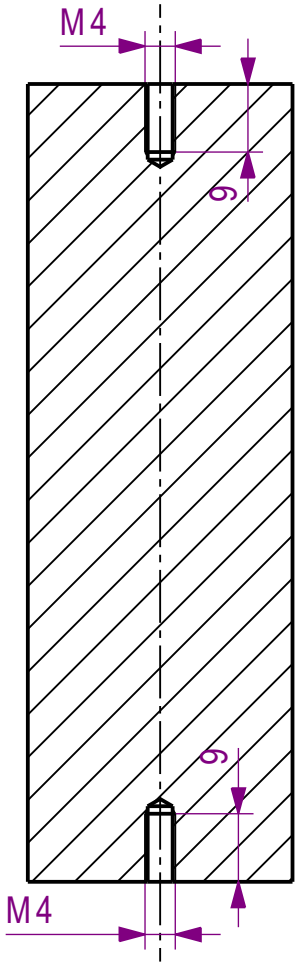
Left view

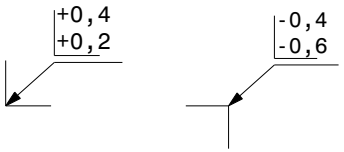


Isometric view
Scale : 1:1

25

Section A-A





General tolerances : ISO 2768-mK

Ra 3.2

Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	EN AW-Al MgSi T6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.274 kg	Replaces	

Scale 1:1

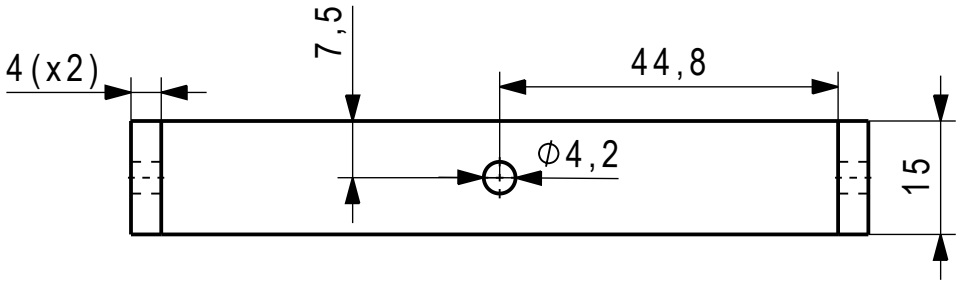
Format | No. of sheets | Sheet no.

EPFL

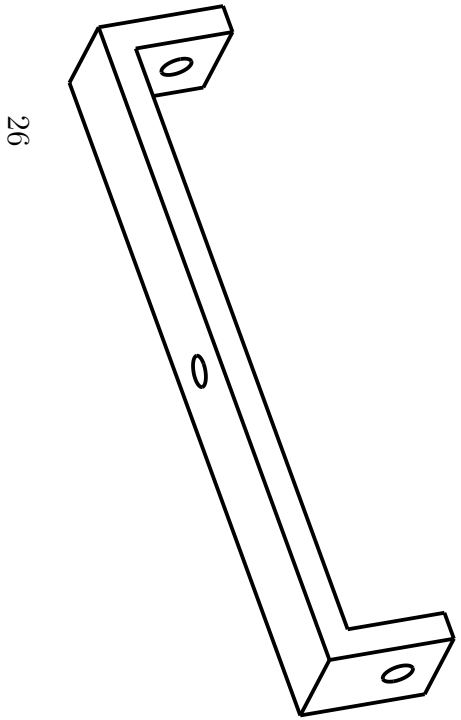
Designation: Handle (bar)

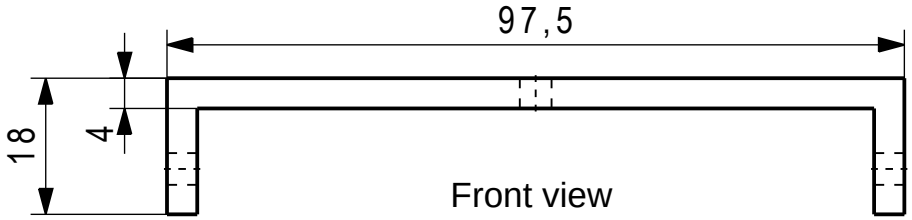
Drawing no. 19

Bottom view



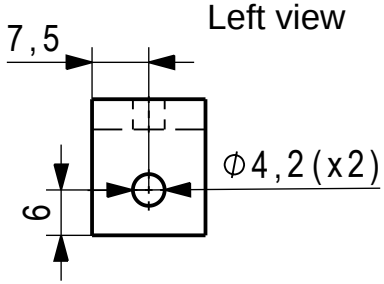
Isometric view
Scale : 1:1

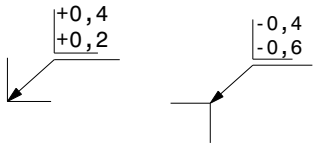




Front view

Left view





General tolerances : ISO 2768-mK

Ra 3.2

Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	42 Cr Mo 4 V <= ø 100	Origin	
Sep. parts list, diff. no. ☐		Mass	0.058 kg	Replaces	

Scale 1:1

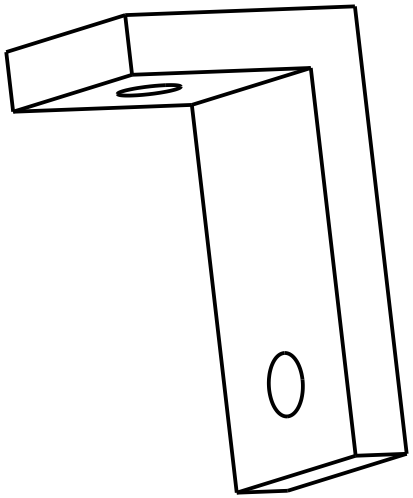
Format | No. of sheets | Sheet no.

EPFL

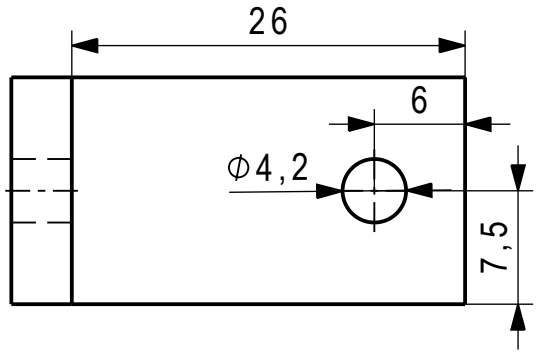
Designation: Long bracket

Drawing no. 20

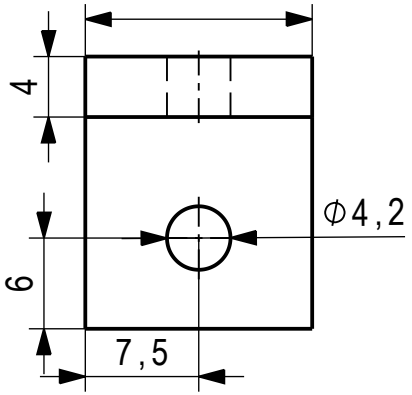
Isometric view
Scale : 2:1

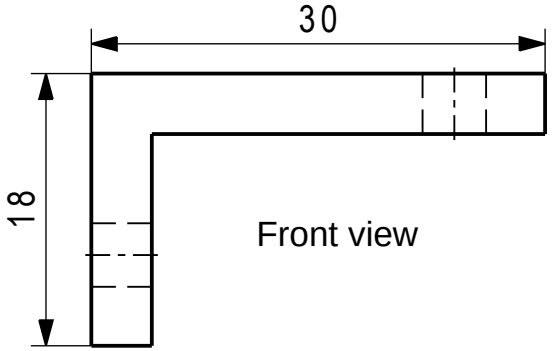


Bottom view

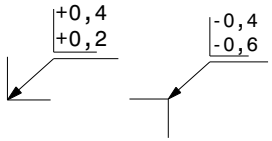


Right view





Front view



General tolerance : 2768-mK

Ra 3.2

27

Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	42 Cr Mo 4 V <= ø 100	Origin	
Sep. parts list, diff. no. ☐		Mass	0.02 kg	Replaces	

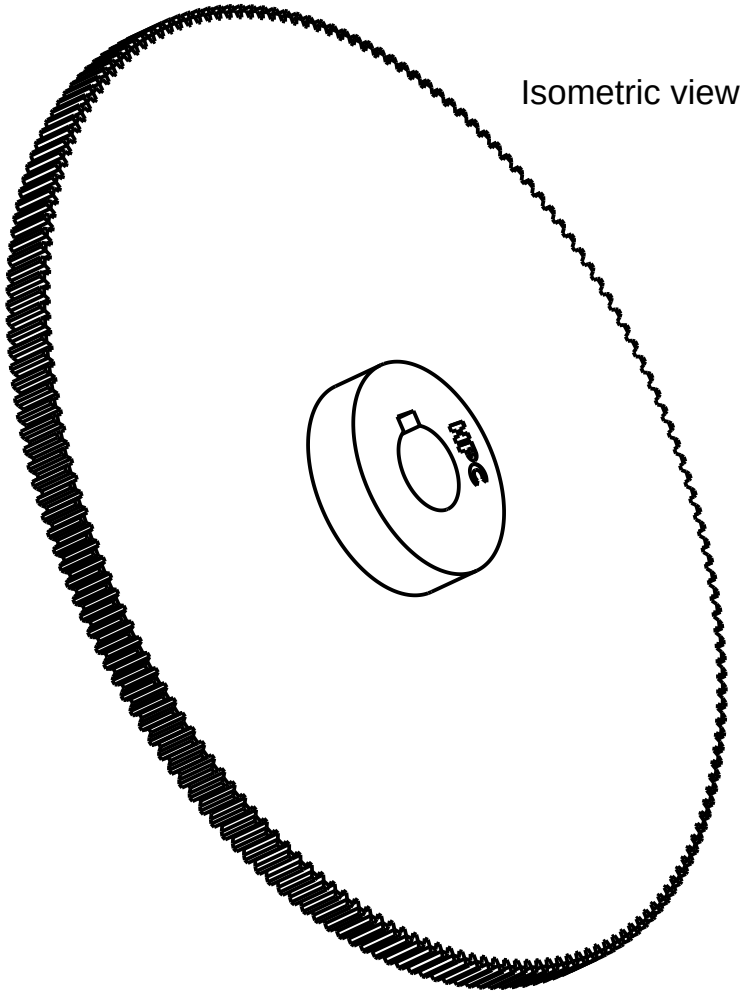
Scale 2:1

Format | No. of sheets | Sheet no.

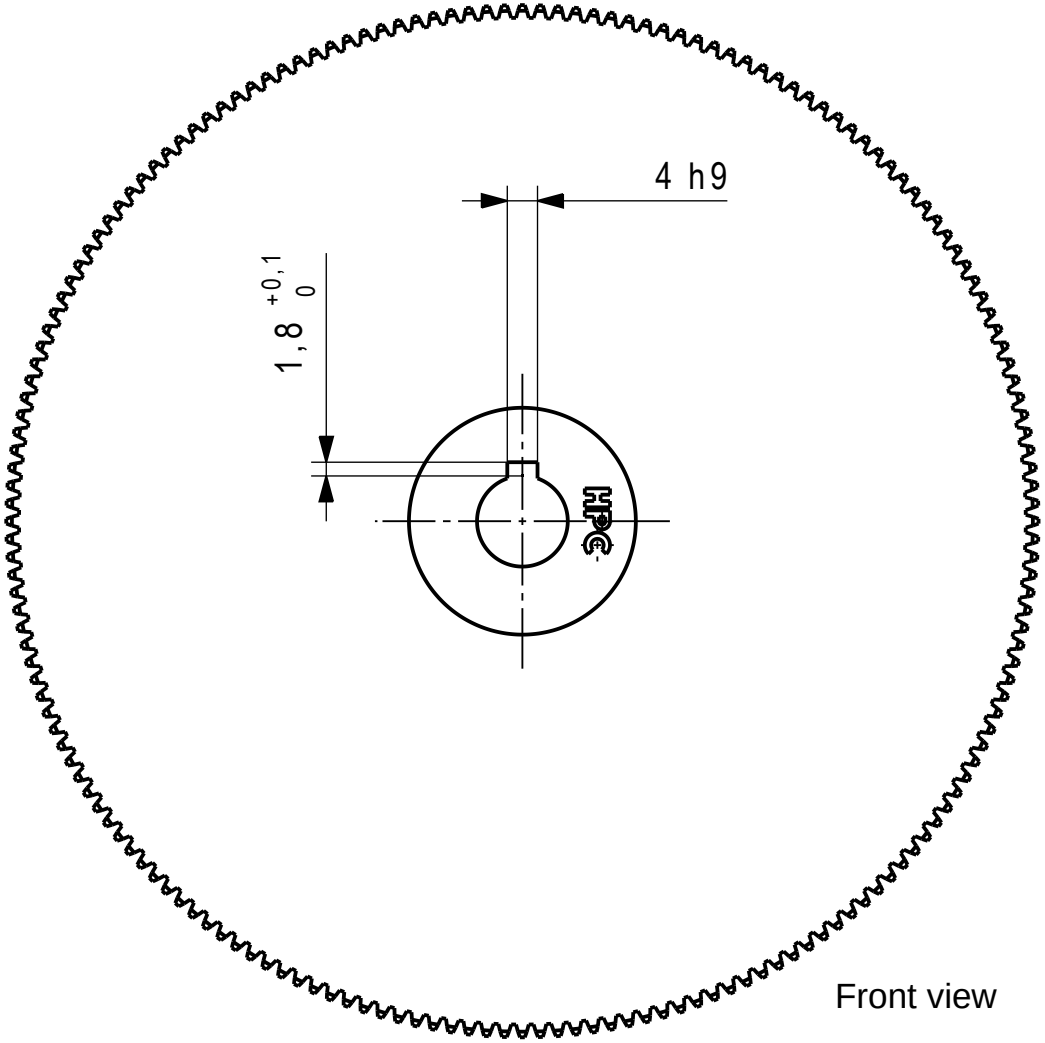
EPFL

Designation: Short bracket

Drawing no. 21



Isometric view



Front view

28

General tolerances : ISO 2768-mK

Ra 3.2

-0,4
-0,6

+0,4
+0,2

Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	34 Cr Ni Mo 6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.704 kg	Replaces	

Scale 1:1

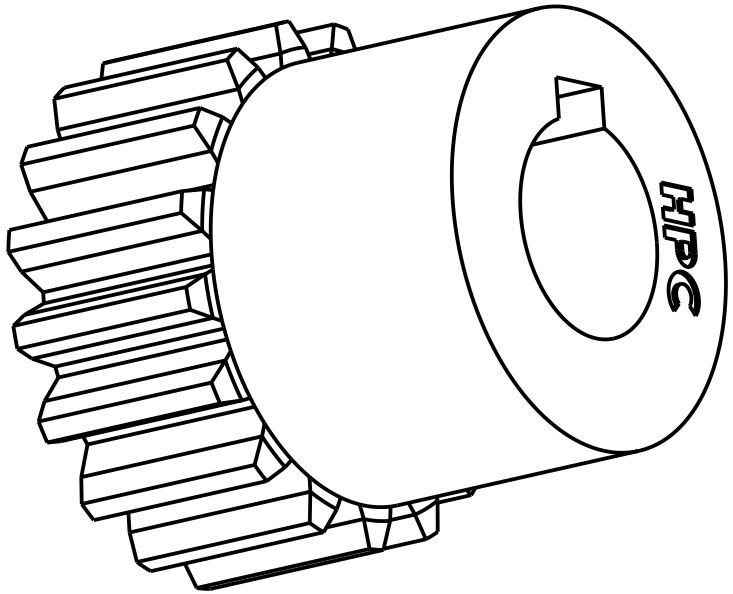
Format | No. of sheets | Sheet no.

EPFL

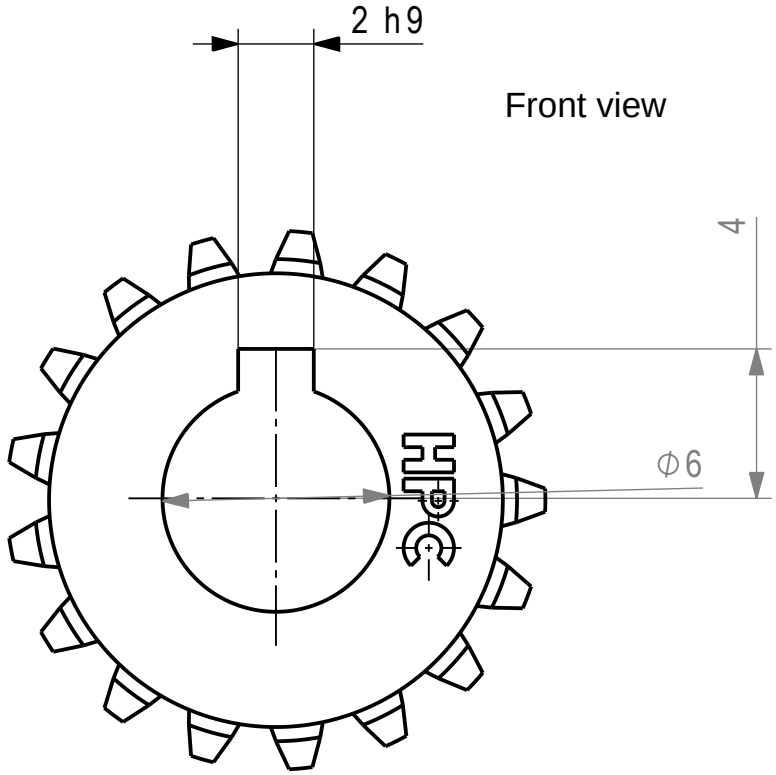
Designation: Gear G0.75-180

Drawing no. 22

Isometric view
Scale: 5:1



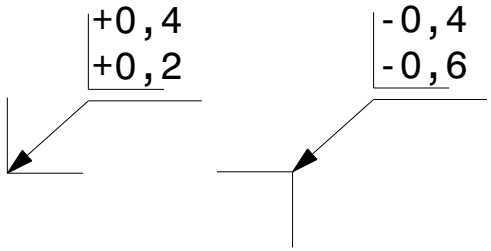
Front view



29

General tolerances:

Ra 3.2



Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	34 Cr Ni Mo 6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.01 kg	Replaces	

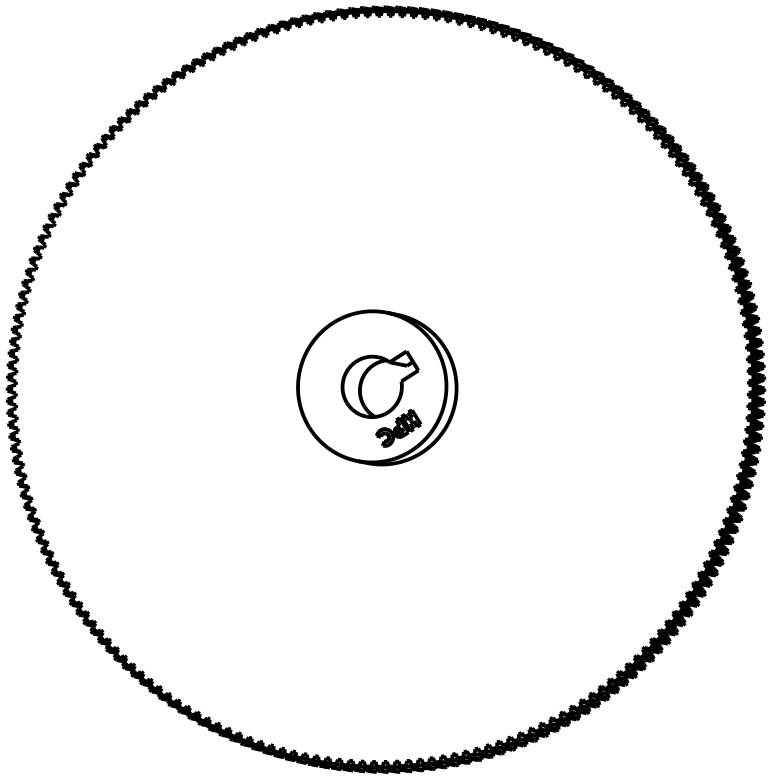
Scale: 5:1

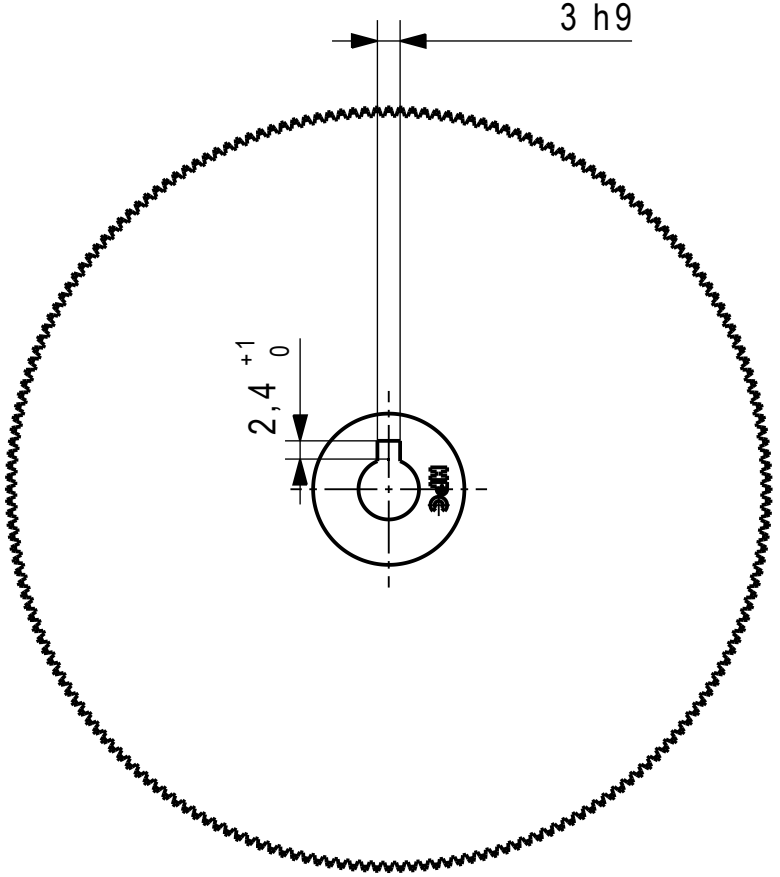
Format | No. of sheets | Sheet no.

EPFL

Designation: Gear G0.75-17

Drawing no.: 23





30

General tolerances: Ra 3.2

+0,4
+0,2

-0,4
-0,6

Rev.		Rev.		Drawn	MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	34 Cr Ni Mo 6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.319 kg	Replaces	

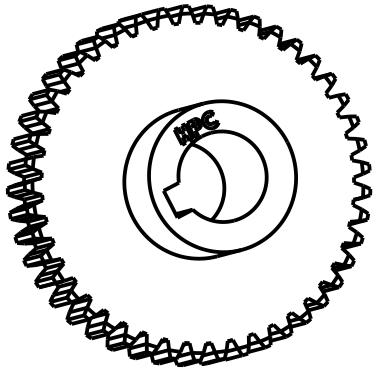
Scale 1:1

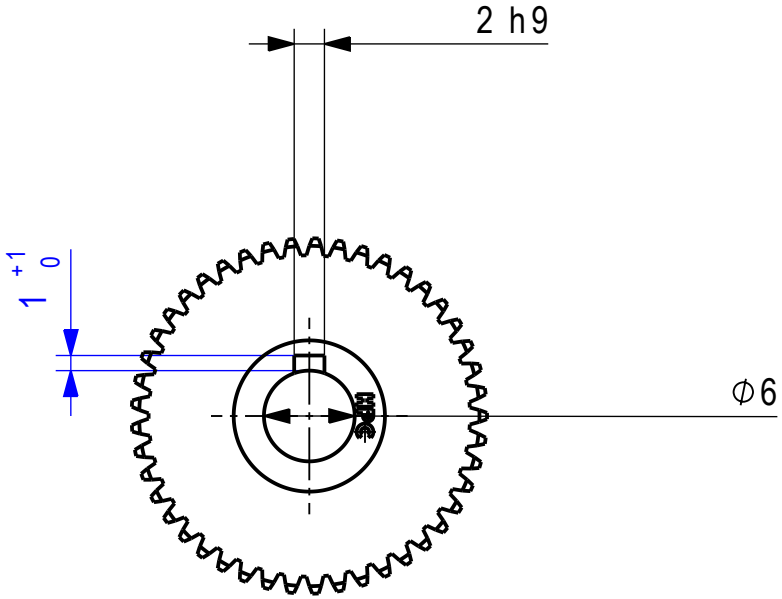
Format | No. of sheets | Sheet no.

EPFL

Designation: Gear G0.5-200

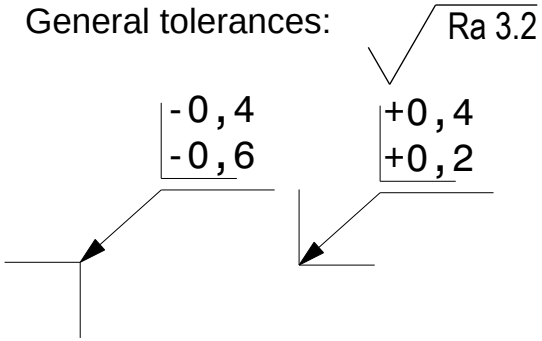
Drawing no. 24





31

General tolerances:



Rev.		Rev.		Drawn	01/06/2025 MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	34 Cr Ni Mo 6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.017 kg	Replaces	

Scale 2:1

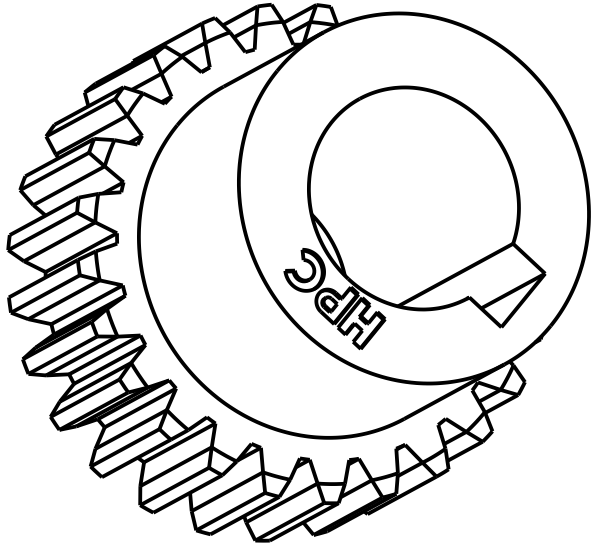
Format | No. of sheets | Sheet no.

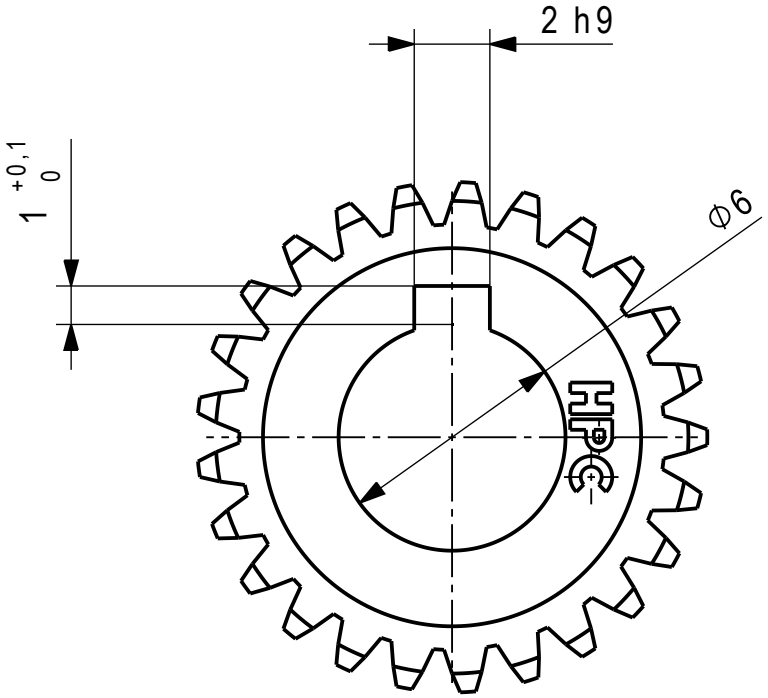
EPFL

Designation: Gear G0.5-45

Drawing no. 25

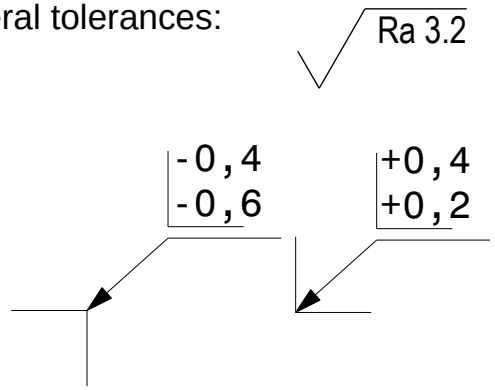
Isometric view





32

General tolerances:



Rev.		Rev.		Drawn	31/05/2025 MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	34 Cr Ni Mo 6	Origin	
Sep. parts list, diff. no. ☐		Mass	0.006 kg	Replaces	

Scale 5:1

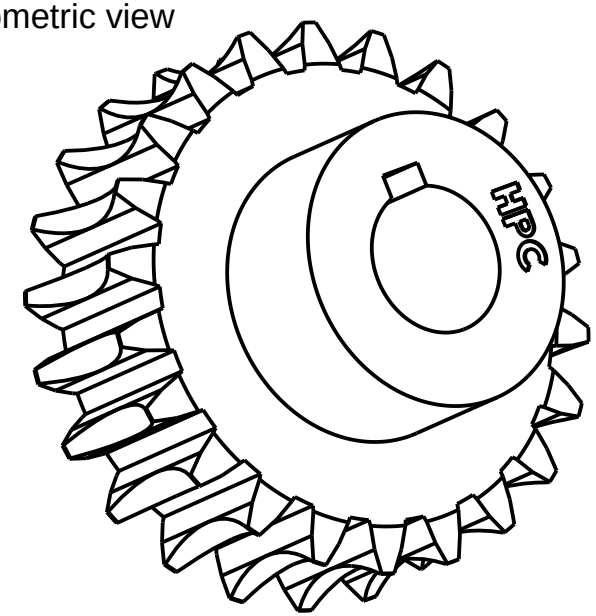
Format | No. of sheets | Sheet no.

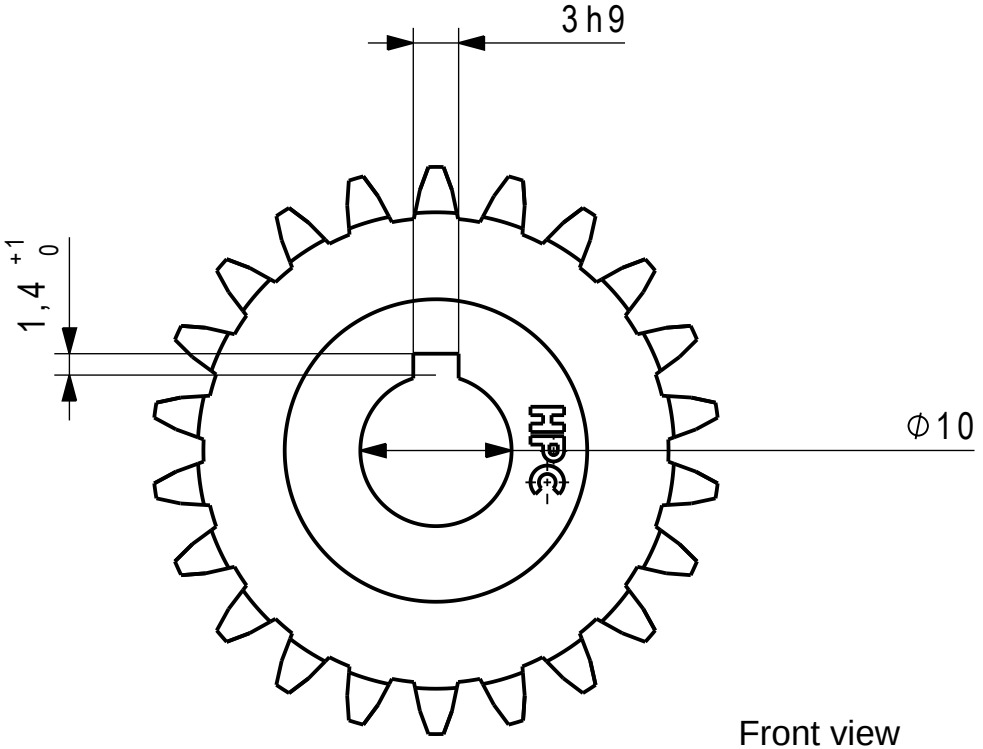
EPFL

Designation: Gear M0.5-25

Drawing no. 26

Isometric view

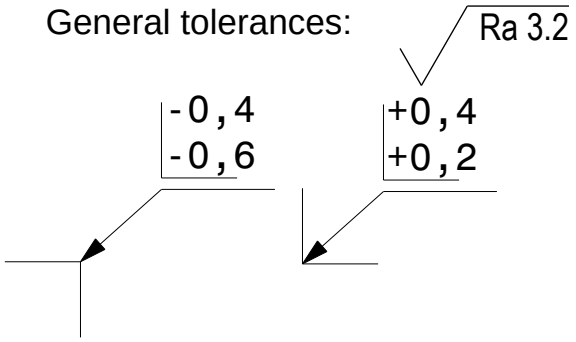




Front view

33

General tolerances:



Rev.			Rev.		Drawn	01/06/2025	MT_07
					Checked		
					Std. compliant		
					Approved for prod.		
No separate parts list ☐					Order no.		
Sep. parts list, same no. ☐		Material	Cu Al 10 Ni		Origin		
Sep. parts list, diff. no. ☐		Mass	0.089 kg		Replaces		

Scale 2:1

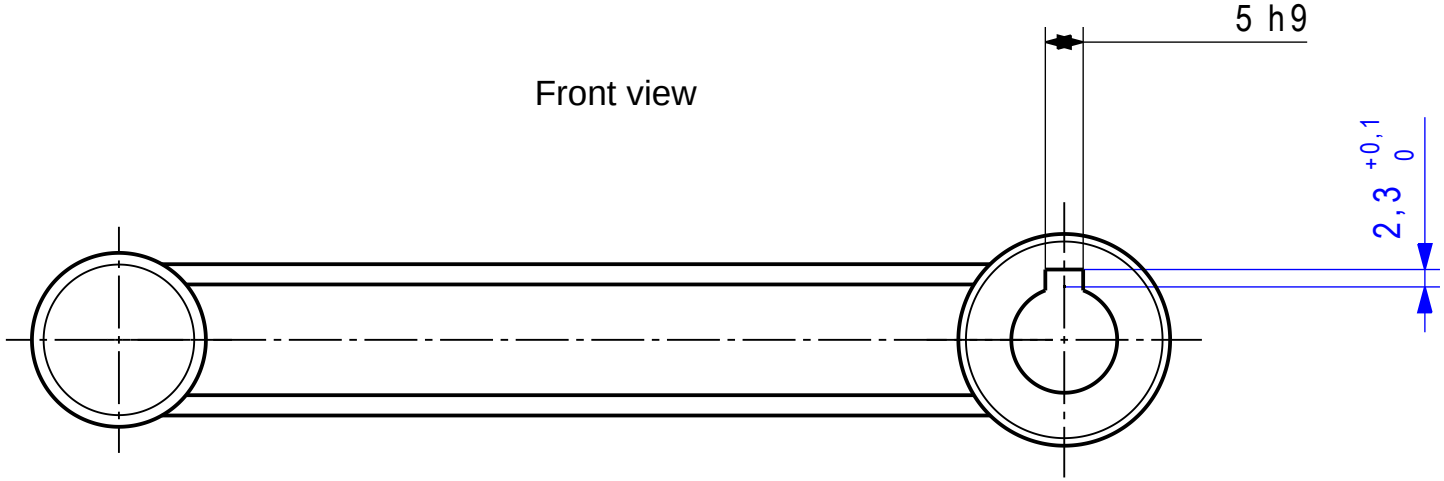
Format | No. of sheets | Sheet no.

EPFL

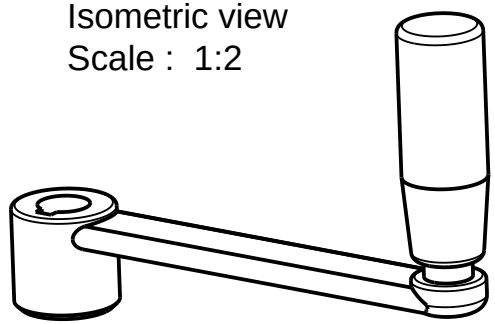
Designation: Gear M1.5-22

Drawing no. 27

Front view

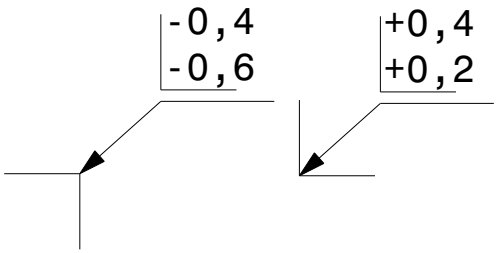


Isometric view
Scale : 1:2



General tolerances:

Ra 3.2



Rev.		Rev.		Drawn	29/05/2025 MT_07
				Checked	
				Std. compliant	
				Approved for prod.	
No separate parts list ☐				Order no.	
Sep. parts list, same no. ☐		Material	X 10 Cr Ni S 18 10	Origin	
Sep. parts list, diff. no. ☐		Mass	0.433 kg	Replaces	

Scale 1:1

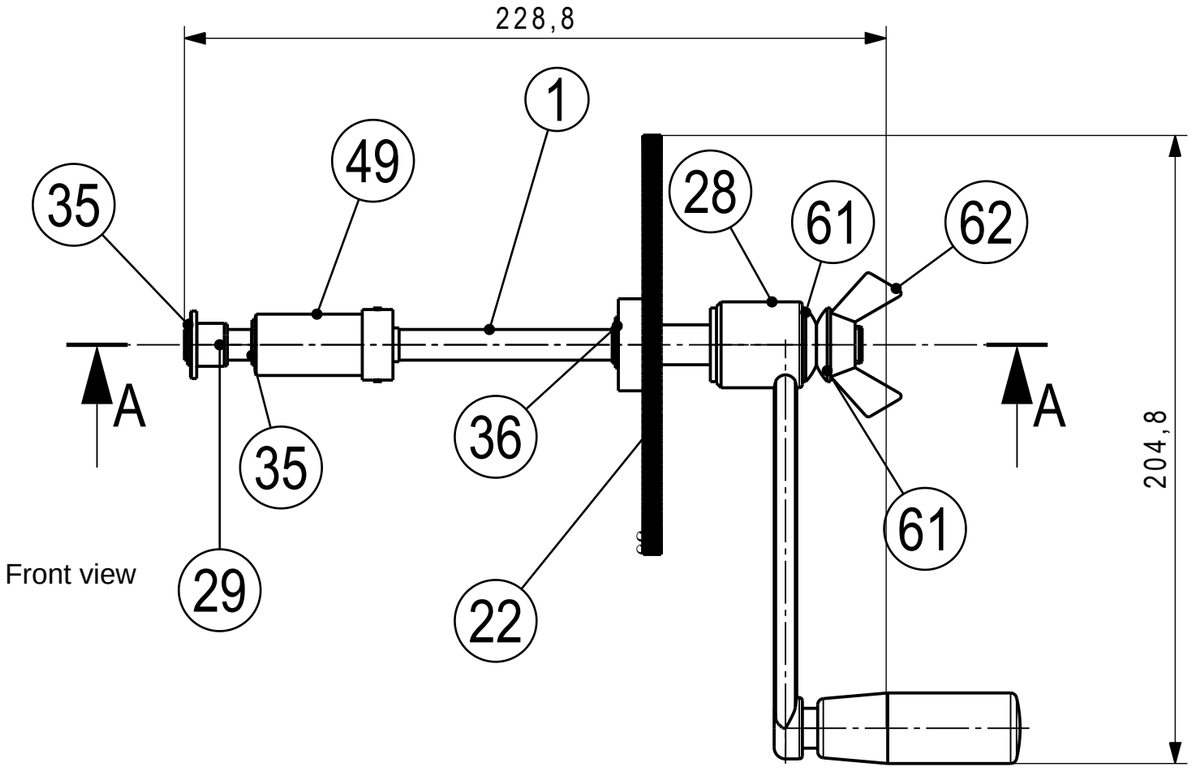
Format | No. of sheets | Sheet no.

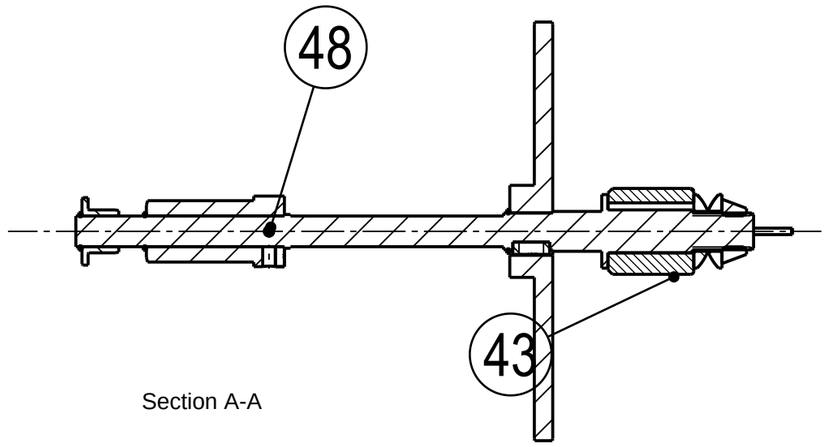
EPFL

Designation: Crank

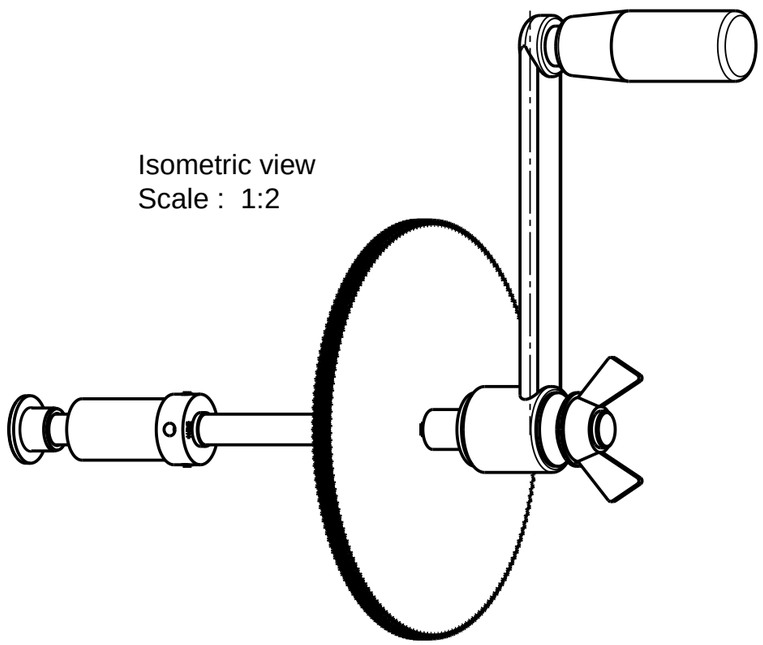
Drawing no. 28

34





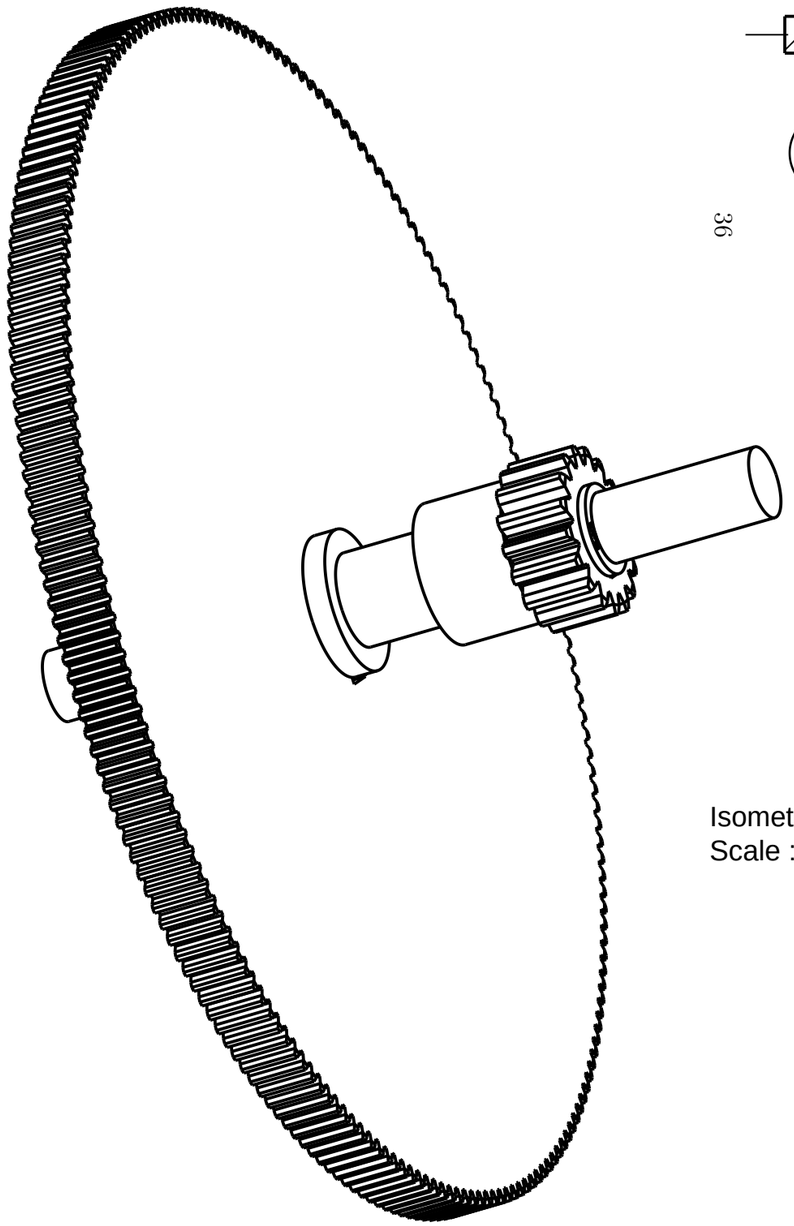
Section A-A



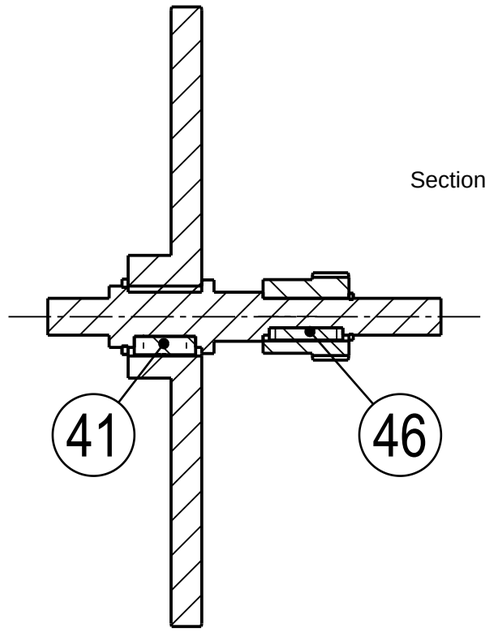
Isometric view
Scale : 1:2

Part number	Quantity	Parts list	Stainless steel
		Shaft 1	Stainless steel
22	1	Gear G0.75-180	Stainless steel
28	1	Crank	Stainless steel
29	1	Plain bearing SHFZ10-12	
35	2	External circlip CX-10S/B	Stainless steel
36	1	External circlip CX-12S/B	Stainless steel
43	1	Key KK4-12	Stainless steel
48	1	Dowel pin	Stainless steel
49	1	Worm	Stainless steel
61	2	Crank washer	Stainless steel
62	1	Wing nut	Stainless steel

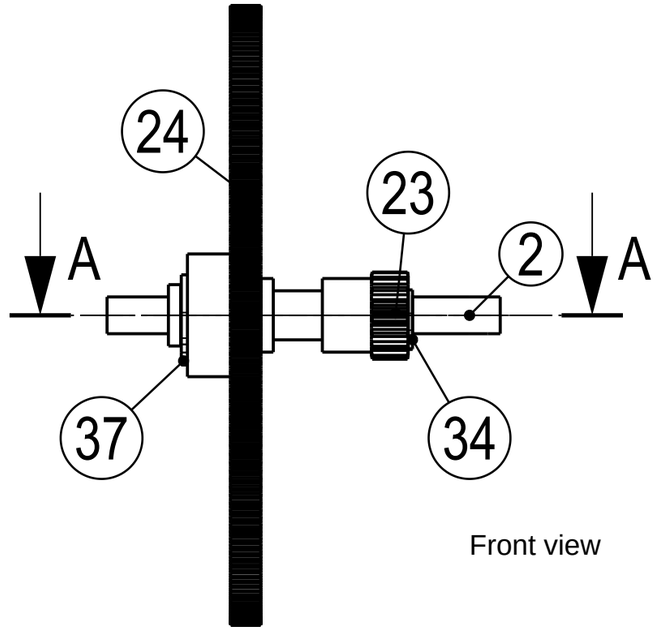
Rev.	Rev.	Drawn	01/06/2025	MT_07	Scale 1:2
		Checked			
		Std. compliant			
		Approved for prod.			
No separate parts list ☐		Order no.			
Sep. parts list, same no. ☐	Material	Origin		Format	No. of sheets / Sheet no.
Sep. parts list, diff. no. ☐	Mass	Replaces			
EPFL	Designation: Assembly A			Drawing no.: A	



Isometric view
Scale : 2:1



Section A-A

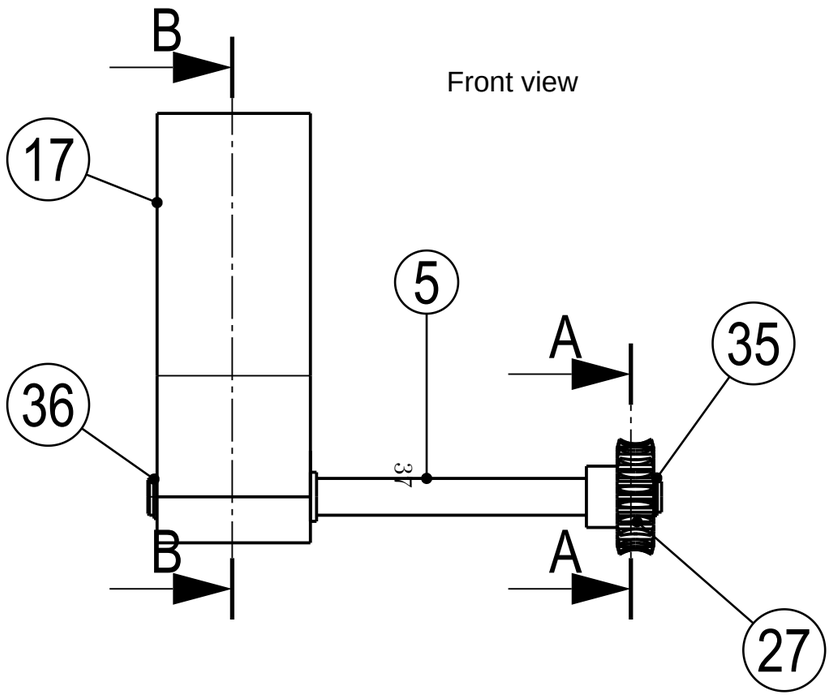


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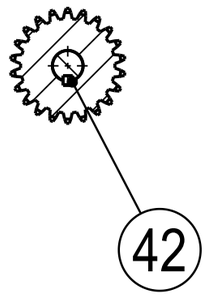
36

Part number	Quantity	Parts list	Material
		Shaft 2	Steel
23	1	Gear G0.75-17	Stainless steel
24	1	Gear G0.5-200	Stainless steel
34	1		Stainless steel
37	1		Stainless steel
41	1	Key KK3-10	Alloy steel
46	1	Key KESF2-12	Alloy steel

Rev.	Rev.	Drawn: MT_07	Scale: 1:1
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No separate parts list ☐		Order no.	
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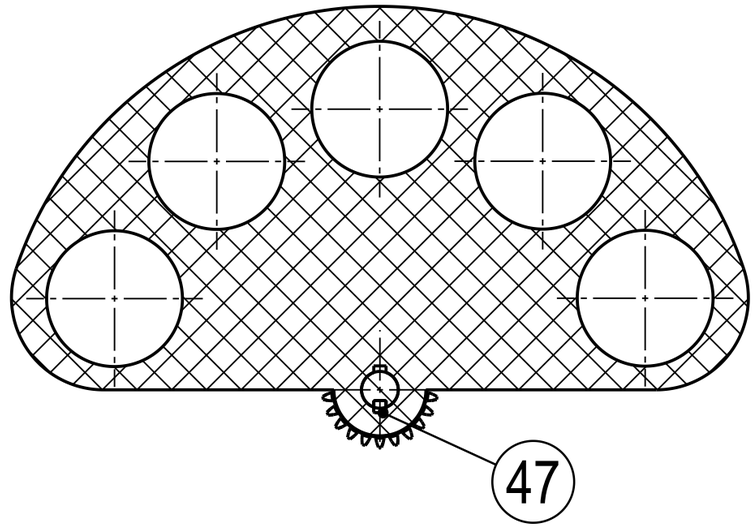


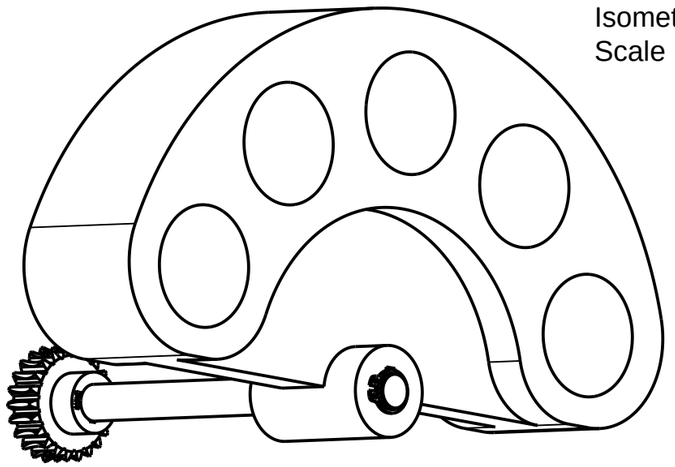
Front view



Section A-A

Section B-B



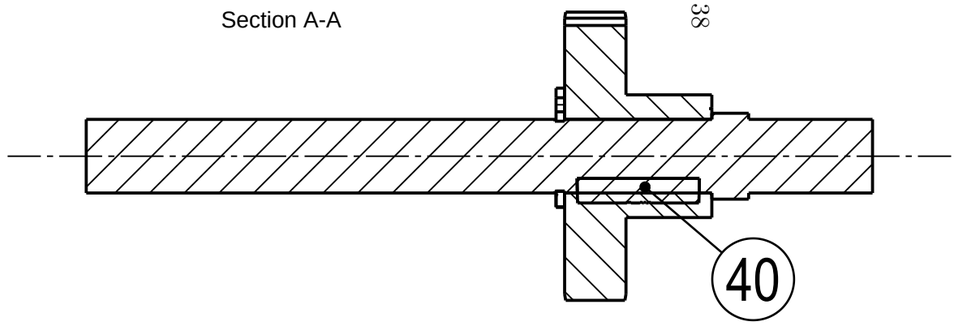


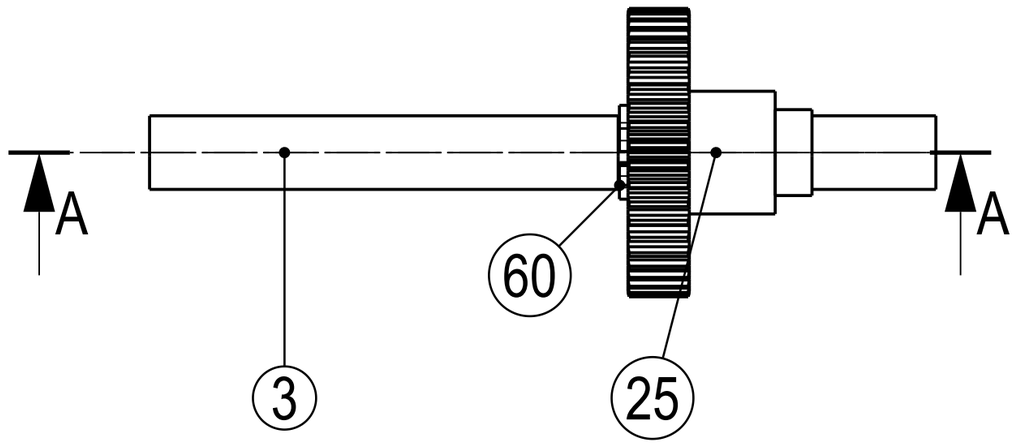
Isometric view
Scale : 1:2

Part number	Quantity	Parts list	Material
		Shaft 5	Stainless steel
17	1	Ball magazine	
27	1	Gear M1.5-22	Stainless steel
35	1	External circlip CX-12S/B	Stainless steel
36	1	External circlip CX-10S/B	Stainless steel
42	1	Key KK3-20	Stainless steel
47	1	Key KESF4-45	Stainless steel

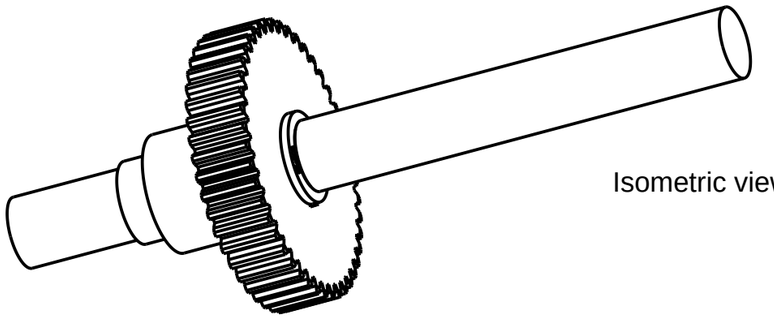
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Sep. parts list, diff. no. ☐	Mass	Replaces			
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Section A-A





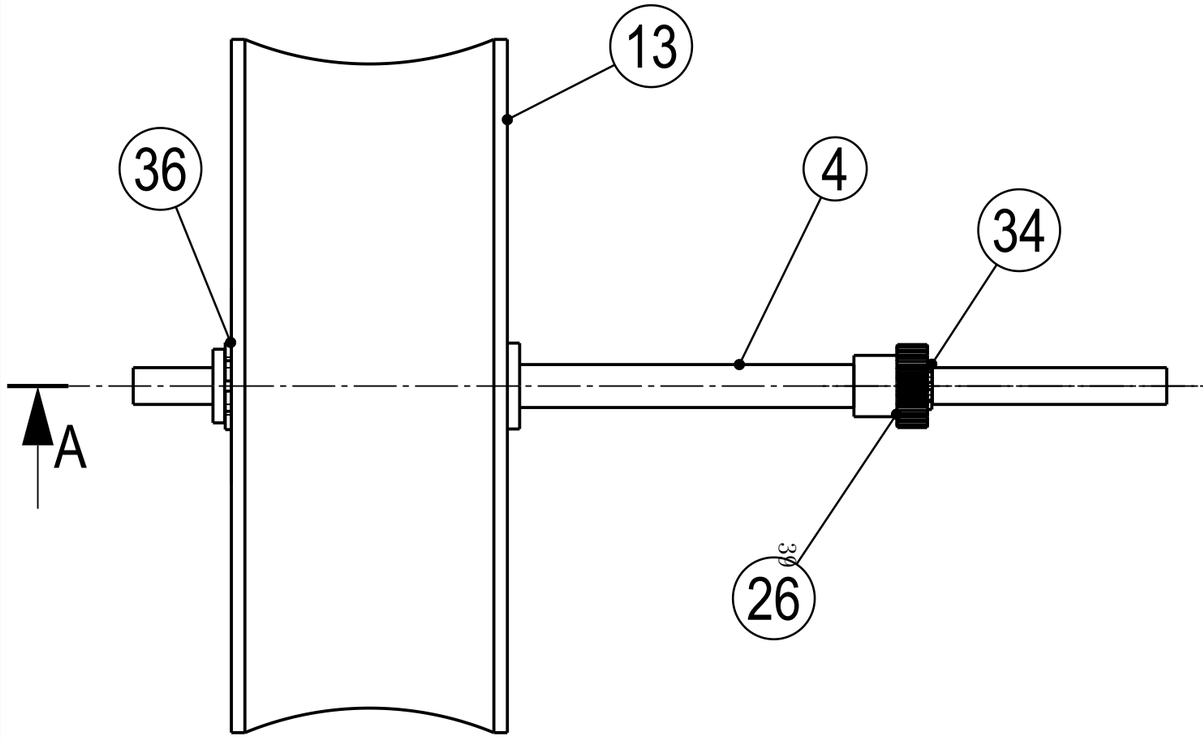
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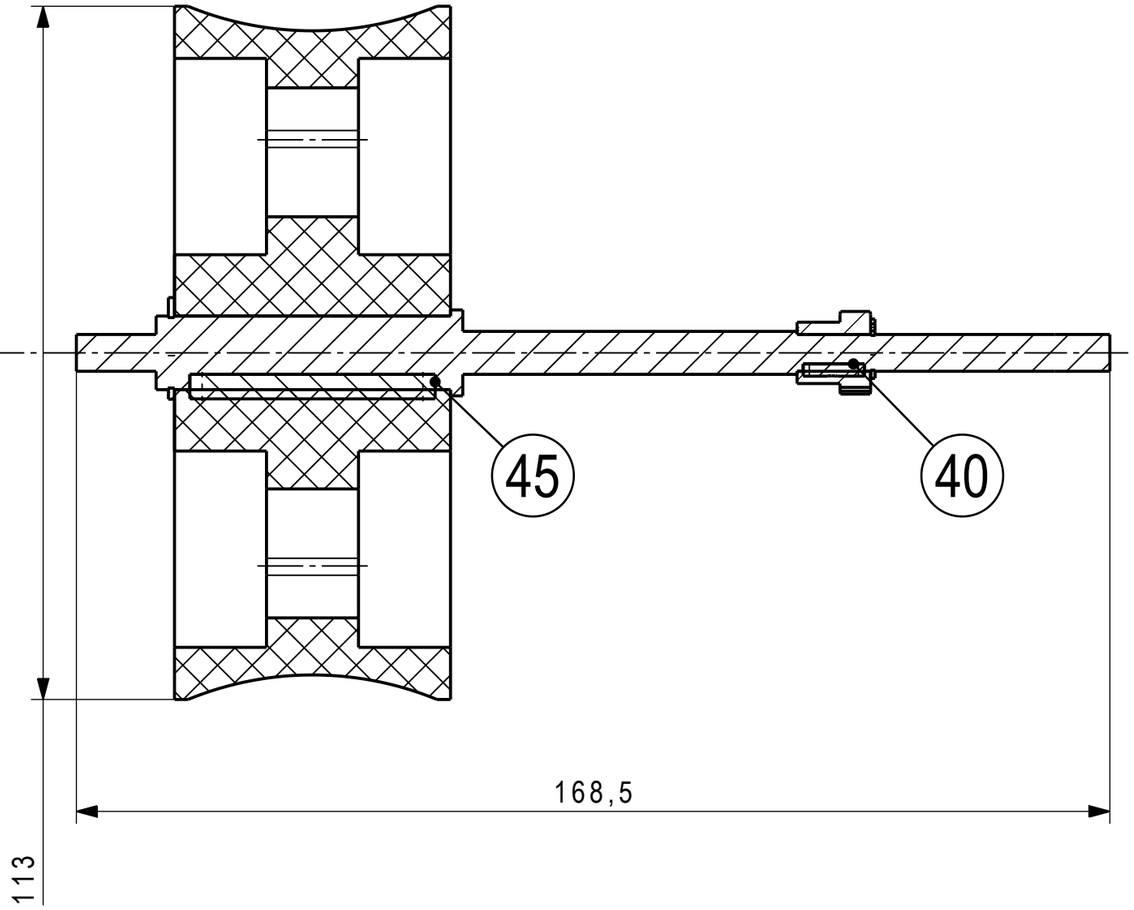


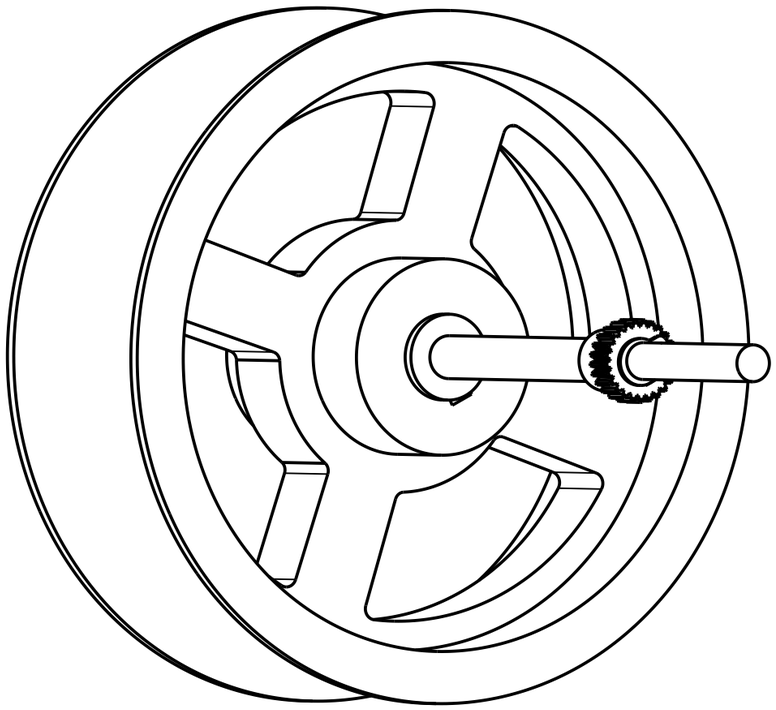
Isometric view

Part number	Quantity	Parts list	Material
		Shaft 3	Steel
25	1	Gear G0.5-45	Stainless steel
40	1	Key KK2-10	Alloy steel
60	1	Washer M6	Steel

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		Approved for prod.			
No separate parts list ☐		Order no.			
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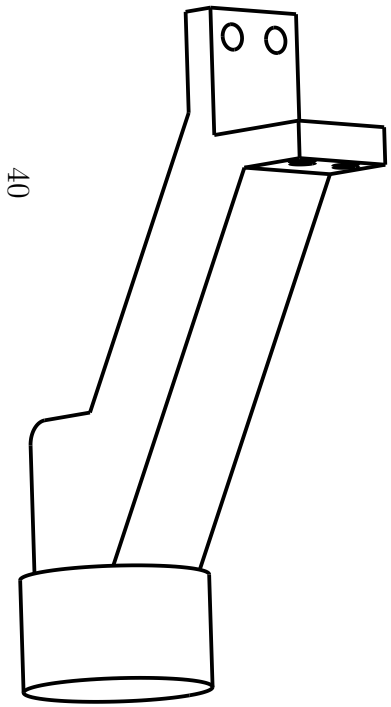


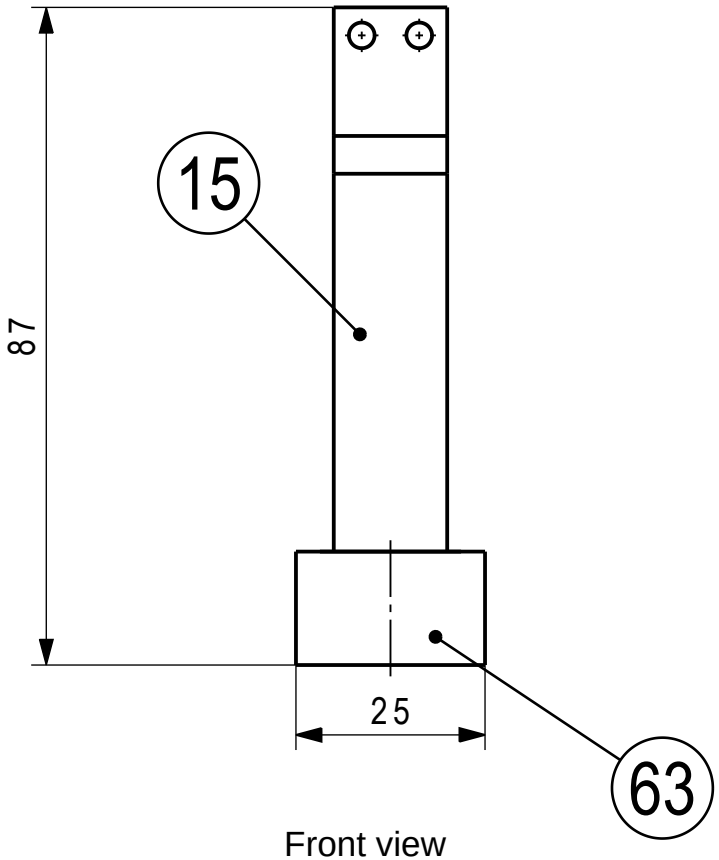




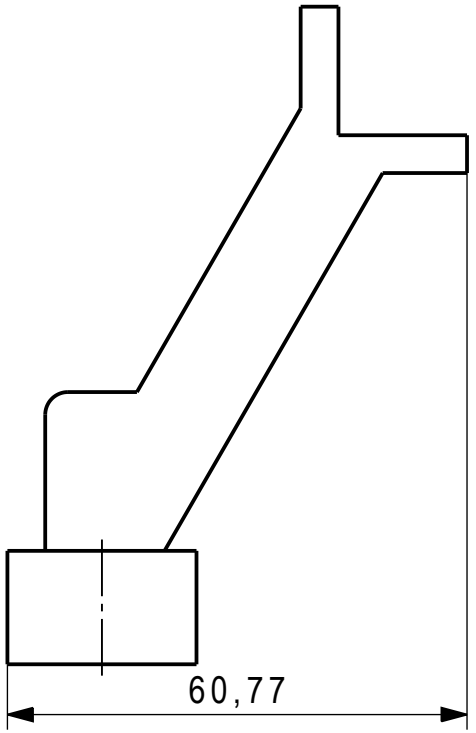
Part number	Quantity	Parts list	Material
		Shaft 4	Steel
13	1	Launch wheels	
26	1	Gear G0.5-25	Stainless steel
34	1		Stainless steel
36	1		Stainless steel
40	1	Key KK2-10	Stainless steel
45	1	Key KESF4-40	Stainless steel

Rev.	Rev.	Drawn	01/06/2025	MT_07	Scale 1:1
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		Std. compliant			
		Approved for prod.			
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Sep. parts list, same no. ☐	Material	Origin		Format	No. of sheets / Sheet no.
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EPFL	Designation: Assembly E			Drawing no.: E	





Front view



Isometric view

Part number	Quantity	Parts list	Material
		Legs	
63	1	Anti-slip foot	Steel and rubber

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				Std. compliant	
				Approved for prod.	
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Sep. parts list, same no. ☐				Origin	
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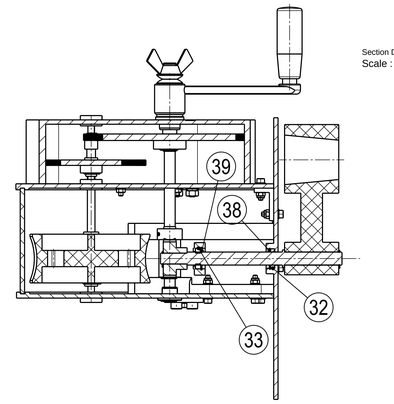
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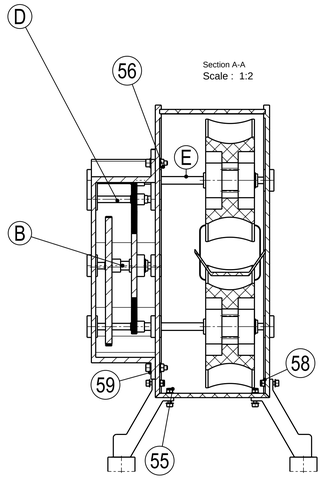
EPFL

Designation: Assembly F

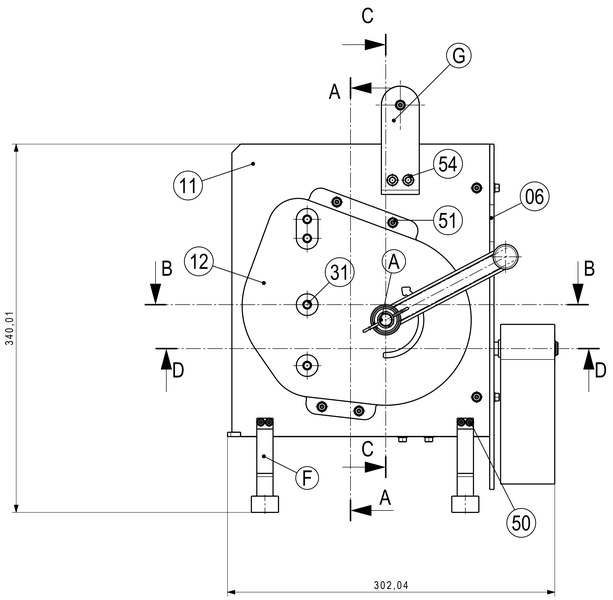
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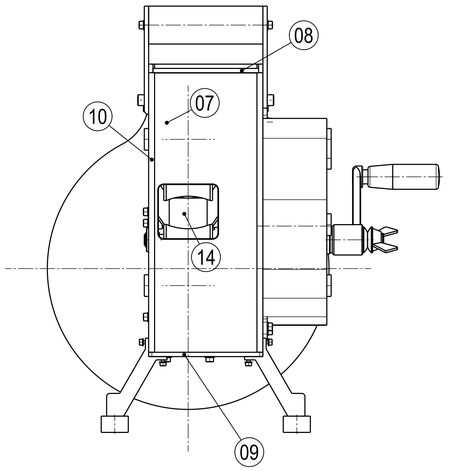


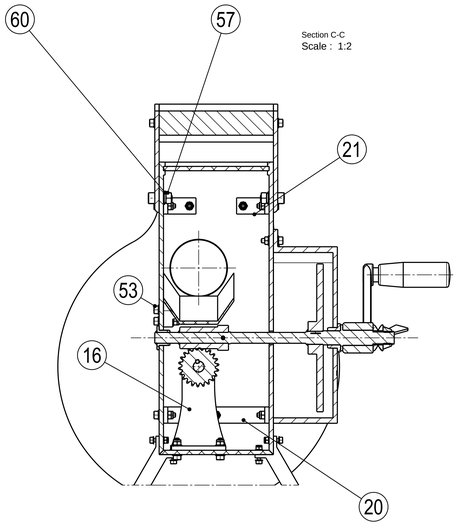
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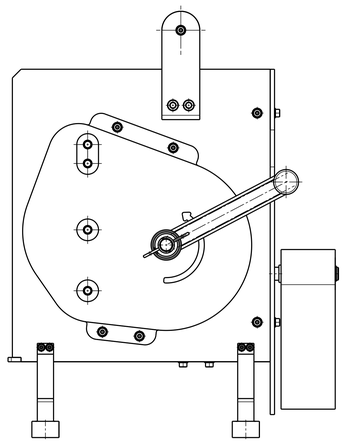
Section A-A
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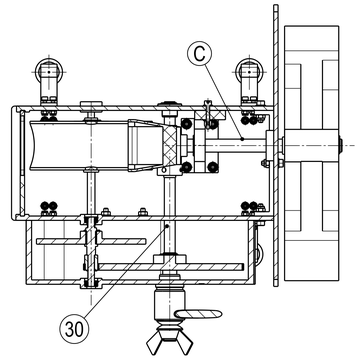






Section C-C
Scale : 1:2





Section B-B
Scale : 1:2

Part number	Quantity	Parts list	Material
		Assembly A	
B	1	Assembly B	
C	1	Assembly C	
D	1	Assembly D	
E	2	Assembly E	
F	4	Assembly F	
G	1	Assembly G	
6	1	Rear frame panel	Aluminium
7	1	Front frame panel	PC
8	2	Top frame panel	PC
9	2	Bottom frame panel	Aluminium
10	1	Right frame panel	Aluminium
11	1	Left frame panel	Aluminium
12	1	Gear housing	Aluminium
16	1	Shaft support	
20	1	Long bracket	Hardened steel
21	2	Short bracket	Hardened steel
30	1	Plain bearing SHFZ13-12	
32	1	Ball bearing	Stainless steel
33	1	Ball bearing	Stainless steel
38	4	Internal circlip CY-18S/B	Steel
39	1	Internal circlip CY-19S/B	Steel
50	16	Screw M3 1003801	Steel
51	11	Screw M4 1003941	Steel
53	2	Screw M4 1301705	Steel
54	7	Screw M6 1004301	Steel
55	15	Self-locking nut M4	Steel
56	4	Self-locking nut M3	Steel
57	7	Nut M6	Steel
58	16	Washer M3	Steel
59	15	Washer M4	Steel
60	7	Washer M6	Steel

Rev.		Rev.	Drawn: 01/06/2025	MT_07	Scale: 1:2
			Checked		
			Std. compliant		
			Approved for prod.		
No separate parts list			Order no.		
Seq. parts list, same no.		Material	Origin	Format	No. of sheets / Sheet no.
Seq. parts list, diff. no.		Mass	Replaces		
EPFL		Designation: Squash ball launcher		Drawing no.: 0	

1.3 Datasheets of purchased parts

The supplier datasheets of the purchased parts (gears, worm, bearings, circlips, keys, screws, nuts, washers, crank and feet; HPC Europe, Misumi, Bossard and McMaster-Carr) were attached to the original submission. They are not reproduced in this version; the references are listed in Table 3.

2 Transmission ratios

2.1 Ratio between $\mathbf{F_{out}}$ and $\mathbf{F_{in}}$

Let us calculate the ratio between the input force and the output force. A force F_{in} is applied to the crank of radius r_m. The resulting torque is:

$$C_0 = r_m \cdot F_{in}$$

This torque is transmitted to gear 22, so $C_{22} = C_0$. Writing the moment balance:

$$F_{22} \cdot r_{22} = r_m \cdot F_{in} \;\Rightarrow\; F_{22} = \frac{r_m}{r_{22}} \cdot F_{in}$$

Since the tangential forces are equal between gears 22 and 23:

$$F_{23} = \frac{C_{23}}{r_{23}} \quad \text{with} \quad C_{23} = r_{23} \cdot \frac{r_m}{r_{22}} \cdot F_{in}$$

Gears 23 and 24 are on the same shaft, so $C_{24} = C_{23}$, hence:

$$F_{24} = \frac{C_{24}}{r_{24}} = \frac{r_{23}}{r_{24}} \cdot \frac{r_m}{r_{22}} \cdot F_{in}$$

This force is then transmitted to gear 26:

$$C_{26} = r_{26} \cdot F_{24} = r_{26} \cdot \frac{r_{23}}{r_{24}} \cdot \frac{r_m}{r_{22}} \cdot F_{in}$$

This torque is finally transmitted to the launch wheels:

$$C_r = C_{26} = F_{out} \cdot r_r \;\Rightarrow\; F_{out} = \frac{r_{26}}{r_r} \cdot \frac{r_{23}}{r_{24}} \cdot \frac{r_m}{r_{22}} \cdot F_{in}$$

Replacing the radius ratios by tooth-count ratios $\left(\frac{r_{26}}{r_{24}} = \frac{Z_{26}}{Z_{24}},\ \frac{r_{23}}{r_{22}} = \frac{Z_{23}}{Z_{22}}\right)$ gives:

$$F_{out} = \frac{r_m}{r_r} \cdot \frac{Z_{26}}{Z_{24}} \cdot \frac{Z_{23}}{Z_{22}} \cdot F_{in}$$

With the numerical values:

$$F_{out} = \frac{0.125}{0.0525} \cdot \frac{25}{200} \cdot \frac{17}{180} \cdot F_{in} \approx \frac{1}{34} \cdot F_{in}$$

2.2 Ratio between $\boldsymbol{\omega}_{\mathbf{out}}$ and $\boldsymbol{\omega}_{\mathbf{in}}$

The wheels that launch the balls have a radius $r = 52.5$ mm. According to the specifications, the ball exit speed must be between 80 and 130 km/h. We choose a target speed of 100 km/h, i.e.:

$$v_{out} = 100\ \text{km/h} = \frac{100}{3.6} \approx 27.78\ \text{m/s}$$

The relation between linear and angular speed gives:

$$\omega_{out} = \frac{v_{out}}{r} = \frac{27.78}{0.0525} \approx 529\ \text{rad} \cdot \text{s}^{-1}$$

We assume that the crank turns at a realistic speed of one turn per second, i.e. $\omega_{in} = 2\pi$ rad $\cdot$ s^{-1}. The required step-up ratio is therefore:

$$i = \frac{\omega_{out}}{\omega_{in}} \approx \frac{529}{2\pi} \approx 84$$

3 Gear selection

To reach the required speed ratio, of the order of 80, we chose to use two successive gear pairs.

The first pair consists of a 180-tooth gear and a 17-tooth pinion, which gives a transmission ratio of:

$$\frac{Z_1}{Z_2} = \frac{180}{17} \approx 10.6$$

We allowed ourselves this ratio based on the following document, with the agreement of Prof. Soubielle, available on Moodle in the "Transmissions" course:

TABLE 8–7 Number of Pinion Teeth to Ensure No Interference

For a pinion meshing with a rack		For a 20°, full-depth pinion meshing with a gear		
Tooth form	Minimum number of teeth	Number of pinion teeth	Maximum number of gear teeth	Maximum ratio
14½°, involute, full-depth	32	17	1309	77.00
20°, involute, full-depth	18	16	101	6.31
25°, involute, full-depth	12	15	45	3.00
		14	26	1.85
		13	16	1.23

Figure 1: Justification of the 180/17 ratio

The second pair consists of a large 200-tooth gear and a 25-tooth pinion, i.e. a ratio of:

$$\frac{Z_3}{Z_5} = \frac{200}{25} = 8$$

This pair was chosen with a ratio of 8, following the advice of the professors, who recommended not to exceed a ratio of 8 between two gears. The product of the two ratios gives:

$$\frac{Z_3}{Z_5} \cdot \frac{Z_1}{Z_2} = \frac{200}{25} \cdot \frac{180}{17} \approx 85$$

This result validates the condition from section 2.2: the transmission ratio of the system must be at least 80.

To reverse the direction of rotation between the 200-tooth gear and the 25-tooth gear, so that the launch wheels turn the right way, we added a 45-tooth idler pinion. It does not change the overall ratio, but:

- it gives the driven wheel the correct direction of rotation;
- it moves the output pinion sideways, which leaves more room for the launch wheels.

4 Required input force

To estimate the force the user must apply on the crank to set the whole mechanism in motion, we first need the total moment of inertia of the system, referred to the crank axis.

The moments of inertia of the parts cannot simply be added, since they do not all turn at the same angular speed. We therefore use the reduction formula for the moment of inertia:

$$J_{eq} = \left(\frac{\omega}{\omega_e}\right)^2 \cdot J$$

where J_{eq} is the equivalent moment of inertia referred to the input (crank) axis, J is the moment of inertia of the part about its own axis, ω is the angular speed of that part, and ω_e is the input angular speed (here $\omega_e = 2\pi$ rad/s).

Applying this formula to each rotating part (gears, launch wheels, shafts, etc.) gives its contribution to the total moment of inertia seen from the input. The total moment of inertia referred to the crank is then:

$$J_{total} = \sum_i J_{eq,i}$$

Adding up the values from Table 7, we obtain $J_{total} \approx 5.10$ kg · m². This value is then used to calculate the required input force with the angular momentum theorem:

$$\vec{r} \times \vec{F} = J_{total} \cdot \frac{d\vec{\omega}}{dt}$$

Since these vectors are collinear, this simplifies to:

$$r \cdot F = J_{total} \cdot \frac{d\omega}{dt} \;\Rightarrow\; \frac{d\omega}{dt} = \frac{r \cdot F}{J_{total}}$$

Integrating over time, with the system initially at rest:

$$\omega(t) = \frac{r \cdot F}{J_{total}} \cdot t + \omega(t=0) = \frac{r \cdot F}{J_{total}} \cdot t$$

To find the force needed to bring the system to the desired speed, we set a target angular speed $\omega = 2\pi$ rad/s and a time to reach it $\Delta t = 11$ s (this choice is explained in section 5). With a crank radius $r = 0.125$ m and a total inertia $J_{total} = 5.10$ kg · m²:

$$F_{min} = \frac{\omega \cdot J_{total}}{r \cdot \Delta t} = \frac{2\pi \cdot 5.10}{0.125 \cdot 11} \approx 23.5 \text{ N}$$

For the sizing and safety calculations, we assume an input force much larger than this value, to keep a wide safety margin.

Note: this calculation assumes there are no mechanical losses while the machine is operated, which is of course not the case. The input force is recalculated later taking the efficiency of the machine into account.

5 Ball storage mechanism

5.1 Principle and dimensions

The storage mechanism is based on a vertical half-disc of radius $r_{disc} = 0.125$ m with five cylindrical pockets, each holding one squash ball. The disc turns about a central axis and is driven by the gear meshing with the worm, until each pocket reaches an opening at the top that leads to the chute guiding the balls to the launch mechanism.

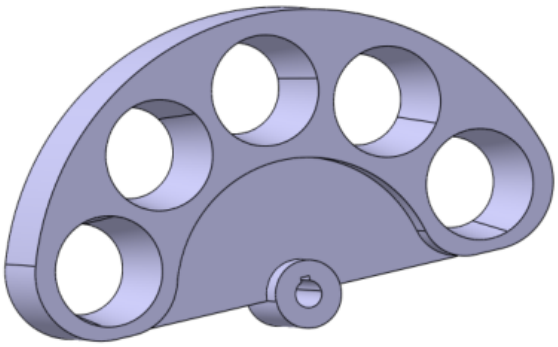


Figure 2: Ball magazine

Our first design used a full disc with a ball every fifth of a turn. With this layout, the first two balls reached the launch zone before the wheels were up to speed.

To solve this, we placed the five balls in a half-disc, with the starting position of the disc just after the launch opening. The mechanism therefore reaches full speed before the first balls arrive. The ratio between the input angular speed ω_{in} and the angular speed of the disc is $\frac{1}{22}$, since the worm has one start and the worm wheel has 22 teeth. It therefore takes about 11 seconds for the first ball to slide into the chute.

The pockets are machined with a 7.5° taper towards the axis, which limits friction between the pocket wall, the frame and the balls. The large radius is $r_{max} = 50.75$ mm and the small radius $r_{min} = 42$ mm. This taper re-centres the balls and lets them slide towards the opening when a pocket reaches the top. The ball then falls by gravity into an inclined chute, which brings it to the launch mechanism.

5.2 Ball track

A chute guides the balls from the magazine to the launch wheels. It is made of POM-H to keep it light, and is screwed to the frame.

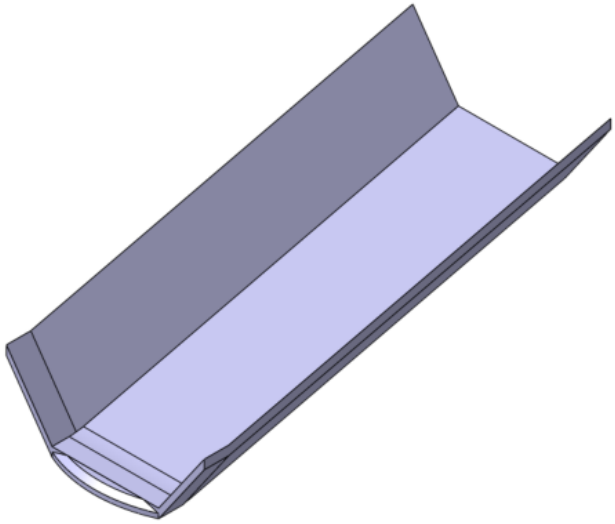


Figure 3: Ball track

5.3 Worm gear efficiency

Several measures improve the efficiency of the worm gear. A high-grade low-viscosity synthetic oil designed for worm gears significantly reduces friction between the hardened steel worm and the bronze wheel. The meshing surfaces are also superfinished and polished, which lowers roughness and sliding losses. Together, these measures give an overall efficiency of about 80 %. Since the system runs at moderate speed, heating stays low and the temperature stable, which preserves the lubrication over time without overheating or degrading the mechanical properties.

6 Efficiency

To calculate the efficiency of the system, we consider the energy lost in:

1. the worm gear,
2. friction between the gears,
3. the ball bearings,
4. the plain bearings.

Section 5.3 showed that the efficiency of the worm gear is about 0.8. For the gears, we use the course value of 0.98 per gear pair; with two pairs, $\eta_{gears} = 0.98^2 \approx 0.96$. As shown in section 10.2.2, the efficiency of the ball bearings is $\eta_{bearings} \approx 0.98$, and section 10.3.2 gives the efficiency of the plain bearings, $\eta_{plain} \approx 0.93$. The overall efficiency is the product of these:

$$\eta = 0.80 \cdot 0.96 \cdot 0.98 \cdot 0.93 \approx 0.70$$

7 Ball launch

In this section we calculate the exit speed of the ball, v_{out}, under the following assumptions:

- the ball arrives between the wheels with negligible speed;
- both wheels turn at the same speed;
- mechanical energy is conserved: there are no losses at contact;
- $\omega_{in} = 2\pi$ rad $\cdot$ s^{-1}.

Conservation of mechanical energy between the moment the wheels spin alone and the moment the ball has passed between them gives:

$$2 \cdot \tfrac{1}{2} \cdot J_r \cdot \omega_r^2 = 2 \cdot \tfrac{1}{2} \cdot J_r \cdot \omega_{out}^2 + \tfrac{1}{2} \cdot m_b \cdot v_{out}^2$$

where J_r is the moment of inertia of one wheel, ω_r the initial angular speed of the wheels, ω_{out} their angular speed after the launch, m_b the mass of the ball and v_{out} its exit speed. Expressing ω_{out} as a function of v_{out}, with r_r the wheel radius:

$$\omega_{out} = \frac{v_{out}}{r_r} \;\Rightarrow\; J_r \cdot \omega_r^2 = J_r \cdot \left(\frac{v_{out}}{r_r}\right)^2 + \tfrac{1}{2} \cdot m_b \cdot v_{out}^2$$

Factorising by v_{out}^2 and isolating v_{out}:

$$J_r \cdot \omega_r^2 = \left(\frac{J_r}{r_r^2} + \tfrac{1}{2} m_b\right) v_{out}^2 \;\Rightarrow\; v_{out} = \sqrt{\frac{J_r \cdot \omega_r^2}{\frac{J_r}{r_r^2} + \frac{1}{2} m_b}} = \omega_r \cdot r_r \cdot \sqrt{\frac{J_r}{J_r + \frac{1}{2} m_b r_r^2}}$$

Numerical application, with $J_r = 3.6 \cdot 10^{-4}$ kg · m² and $m_b = 0.025$ kg:

$$v_{out} = \omega_r \cdot r_r \cdot \sqrt{\frac{3.6 \cdot 10^{-4}}{3.6 \cdot 10^{-4} + \frac{1}{2} \cdot 0.025 \cdot 0.0525^2}} \approx 0.978 \cdot \omega_r \cdot r_r$$

The exit speed of the ball therefore depends directly on the angular speed and radius of the launch wheels, with a proportionality factor $K \approx 0.978$ set by the wheel inertia and the ball mass.

8 Safety checks

8.1 Gear module

An important question when choosing the gears was their module. In this section we calculate the minimum module of each gear pair, using the following condition from the "Transmissions" course:

$$2.34 \sqrt{\frac{F_T}{k \cdot \sigma_{pe}}} \leq m$$

Figure 4 – Condition on the gear module

With $F_T = \frac{C}{r}$ and $r = \frac{m \cdot Z}{2}$, so $F_T = \frac{2C}{mZ}$:

$$2.34 \cdot \sqrt{\frac{2C}{k \cdot \sigma_{pe} \cdot mZ}} \leq m \;\Rightarrow\; m \geq 2.22 \cdot \sqrt[3]{\frac{C}{k \cdot \sigma_{pe} \cdot Z}}$$

We calculate the minimum modules with a form factor $k = 10$ and an input force $F_{in} = 70$ N, to include a sufficient safety margin, and $\sigma_{pe} = 210$ MPa, a typical value for common steel used in the course.

For the first gear pair, we apply the formula to the large gear (22): transmitted torque $C = F_{in} \cdot r_m \approx 8.75$ N · m, number of teeth $Z = 180$:

$$m \geq 2.22 \cdot \sqrt[3]{\frac{8.75}{10 \cdot 210 \cdot 10^6 \cdot 180}} \;\Rightarrow\; m \geq 0.64 \text{ mm}$$

For mechanical safety, standardisation and manufacturing reasons, we set the module of this pair to $m = 0.75$ mm.

For the second gear pair, we apply the formula to the large gear (24), with $k = 10$, $\sigma_{pe} = 210$ MPa, $C = 0.86$ N · m and $Z = 200$:

$$m \geq 2.22 \cdot \sqrt[3]{\frac{0.86}{10 \cdot 210 \cdot 10^6 \cdot 200}} \;\Rightarrow\; m \geq 0.29 \text{ mm}$$

We set the module slightly above: $m = 0.5$ mm.

Finally, for the worm and its wheel, with $k = 10$, $\sigma_{pe} = 210$ MPa, $C = 14.4$ N · m and $Z = 22$:

$$m \geq 2.22 \cdot \sqrt[3]{\frac{14.4}{10 \cdot 210 \cdot 10^6 \cdot 22}} \;\Rightarrow\; m \geq 1.49 \text{ mm}$$

We set the module slightly above: $m = 1.5$ mm.

8.2 Shaft sizing

Here we size the transmission shafts so that they withstand the torsional loads, using the torsion criteria from the course "Torsion, Shear, Efficiency". The maximum shear stress in a solid cylindrical shaft under a torque M_t is:

$$\tau_{max} = \frac{M_t \cdot R}{I_0} \qquad \text{with} \qquad I_0 = \frac{\pi \cdot R^4}{2}$$

where M_t is the applied torque (N · m), R the shaft radius (m) and I_0 the polar second moment of area of a solid cylinder. With $R = \frac{d}{2}$:

$$\tau_{max} = \frac{2 \cdot M_t}{\pi \cdot R^3} = \frac{16 \cdot M_t}{\pi \cdot d^3}$$

For the shaft to be strong enough, $\tau_{max} \leq R_{pg}$, hence:

$$\frac{16 \cdot M_t}{\pi \cdot d_{min}^3} \leq R_{pg} \;\Rightarrow\; d_{min} \geq \left(\frac{16 \cdot M_t}{\pi \cdot R_{pg}} \right)^{1/3}$$

To choose R_{pg}, we use the following table from the course, which gives the safety factor s:

s	Loads on the structure	Stresses in the structure	Material behaviour	Remarks
$1 < s < 2$	regular and known	known	tested and known	constant operation without shocks
$2 < s < 3$	regular and fairly well known	fairly well known	tested and fairly well known	normal operation, with light shocks and moderate overloads
$3 < s < 4$	moderately known, or poorly known or uncertain	moderately known, or poorly known or uncertain	not tested, or known	

Figure 5 – Choice of the safety factor depending on operating conditions (indicative values)

Since the gears carry moderately known loads, with possible shocks or uncertainties, we use a safety factor $s = 4$. For the worm, which runs more smoothly under well-defined loads, $s = 2$ is enough. The resulting allowable shear stresses are:

$$R_{pg,\,gears} = \frac{210 \cdot 10^6}{4} = 52.5 \cdot 10^6 \text{ Pa} \qquad R_{pg,\,worm} = \frac{210 \cdot 10^6}{2} = 105 \cdot 10^6 \text{ Pa}$$

Substituting into the formula for d_{min}: for the gears, $d_{min} \geq 0.00459 \cdot M_t^{1/3}$; for the worm, $d_{min} \geq 0.00365 \cdot M_t^{1/3}$. Numerical values:

Shaft	M_t (N · m)	d_{min} (mm)	d_{actual} (mm)
1	8.75	9.58	10
2	0.859	4.36	6
3	0.19	2.63	6
4	0.11	2.20	6
5	14.4	9.17	10

Table 4 – Minimum shaft diameters as a function of the applied torque

Some shafts are locally larger for assembly or interface reasons, but the checks are always made on the smallest diameters. This guarantees the torsional strength of the whole system.

8.3 Sizing of the shoulders in the frame parts

To hold the ball bearings axially and prevent any movement or early damage, shoulders were integrated into the frame parts that receive the bearings. Three criteria guided their design:

- the minimum axial thickness of the shoulder, e: $e \geq 0.2 \cdot D$, where D is the outer diameter of the bearing;
- the minimum radial height of the shoulder, h: $h \geq 0.3 \cdot b$, where b is the bearing width;
- the remaining thickness behind the groove: $e_{min} \geq 0.3 \cdot D_{ext}$, where D_{ext} is the outer diameter of the bearing.

These conditions are met on all frame parts, to make the machine safe to use and as strong as possible.

8.4 Choice of keys

In our mechanism, torque is transmitted between the shafts and the gears by keys. Here we determine the minimum length L a key needs to withstand the stresses caused by the transmitted torque. The strength condition uses the allowable contact pressure R_{pg} between the key and the shaft.

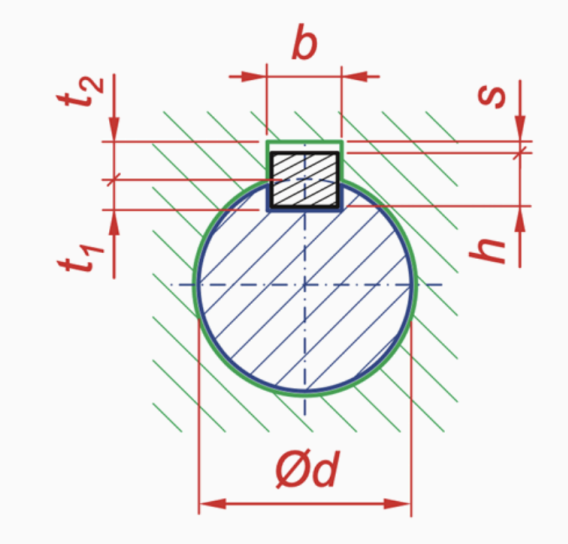


Figure 6: Key dimensions

The effective lateral contact area is:

$$A_c = \frac{\pi \cdot b^2}{4} + (L - b) \cdot b$$

where b is the standard key width, L its total length and A_c the area loaded in pressure. The allowable contact force is $F_c = R_{pg} \cdot A_c$, and the torque transmitted by the key is:

$$C = \frac{d}{2} \cdot F_c = \frac{d}{2} \cdot R_{pg} \cdot \left(\frac{\pi \cdot b^2}{4} + (L - b) \cdot b \right)$$

Isolating L gives the minimum length:

$$L = \frac{2 \cdot C}{d \cdot R_{pg} \cdot b} - \frac{\pi \cdot b}{4} + b$$

This expression gives L from the transmitted torque, the shaft diameter, the standard key dimensions and the allowable stress of the material. We have 9 keys on different shafts; for each, we apply the formula to find the minimum length that transmits the torque without exceeding the allowable contact pressure. For example, for key 1 ($C = 8.75$ N · m, $d = 14$ mm, $b = 5$ mm):

$$L_{1,min} = \frac{2 \cdot 8.75}{14 \cdot 210 \cdot 5} - \frac{\pi \cdot 5 \cdot 10^{-3}}{4} + 5 \cdot 10^{-3} = 2.26 \text{ mm}$$

The other minimum lengths are calculated with the same formula:

Key	C (N · m)	d (mm)	b (mm)	L_{min} (mm)	L_{actual} (mm)
1	8.75	12	5	2.26	25
2	8.75	12	4	2.67	12
3	0.859	6	2	1.13	12
4	0.859	10	3	0.72	10
5	0.19	6	2	0.59	10
6	0.11	6	2	0.51	10
7	0.11	12	4	0.21	40
8	14.4	10	3	5.84	20
9	14.4	12	4	4.01	45

Table 5 – Minimum required length of each key

We went beyond the minimum lengths so that the keys run along the full length of the gears and of the other parts they drive.

8.5 Choice of the pin

A pin is used to drive the worm. To check its shear strength, we verify that the maximum stress τ stays below the allowable shear stress R_{pg}. The stress is:

$$\tau = \frac{2 \cdot C}{\pi R \cdot d \cdot L}$$

where $C = 8.75$ N · m is the transmitted torque, $R = 5$ mm the shaft radius, $d = 3$ mm the pin diameter and $L = 20$ mm the contact length between the pin and the shaft. The strength condition is:

$$\tau \leq \frac{R_{pe}}{s}$$

where $R_{pe} = 230$ MPa is the yield strength of the pin material (stainless steel) and $s = 3$ the chosen safety factor, which leaves enough margin for mechanical uncertainties. We obtain:

$$\tau = \frac{2 \cdot 8.75}{\pi \cdot (5 \cdot 10^{-3}) \cdot 3 \cdot 10^{-3} \cdot 20 \cdot 10^{-3}} \approx 18.6 \text{ MPa}$$

$$R_{pg} = \frac{R_e}{s} = \frac{230 \text{ MPa}}{3} = 76.7 \text{ MPa} \qquad \Rightarrow \qquad \tau = 18.6 \text{ MPa} < R_{pg} = 76.7 \text{ MPa}$$

The shear stress in the pin stays below the allowable limit, so the sizing is validated.

9 Choice of parts

9.1 Launch wheels

The launch wheels are made of POM-H, a light, stiff and easily machined material. Their radius is 52.5 mm, slightly less than our initial target of 55 mm, to leave enough space for the ball to pass (40 mm diameter). The wheels are 45 mm thick, slightly more than the ball, for good guiding without jamming.

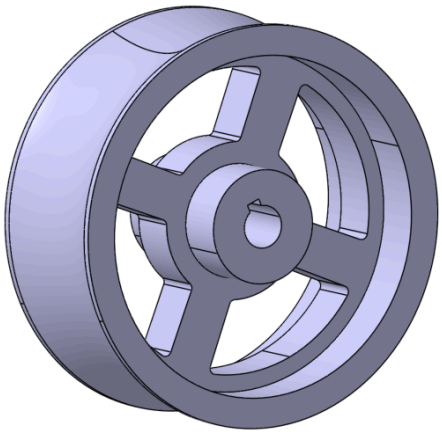


Figure 7: Launch wheel modelled in CATIA

Each wheel is mounted on its shaft with a key, which gives a rigid connection without slip. To reduce their inertia, symmetrical lightening holes were machined. The ball is in direct contact with the POM-H, without any coating.

Although the POM-H wheel is plastic and mounted on a steel shaft, it withstands the stresses thanks to the low torque and the large contact area. Let us check this with a calculation, knowing that the allowable contact pressure given by manufacturers is several MPa. The contact pressure between the bore of the wheel and the steel shaft is:

$$p = \frac{F}{A} \qquad \text{with} \qquad F = \frac{2C}{d} \quad \text{and} \quad A = \pi \cdot d \cdot L$$

with $d = 12$ mm the shaft diameter, $L = 40$ mm the contact length and $C = 0.11$ N · m the transmitted torque. The equivalent tangential force, the contact area and the mean pressure are:

$$F = \frac{2 \times 0.11}{0.012} = 18.33 \text{ N} \qquad A = \pi \cdot 12 \cdot 40 = 1507 \text{ mm}^2 \qquad p = \frac{18.33}{1507} \approx 0.012 \text{ MPa} = 12 \text{ kPa}$$

This value is far below the recommended limit pressure for POM-H.

9.2 Choice of the crank

We chose a crank with a 12.5 cm radius, a good compromise: not so large that the motion becomes slow, not so small that the torque becomes insufficient. The model was found online, which avoided machining it.

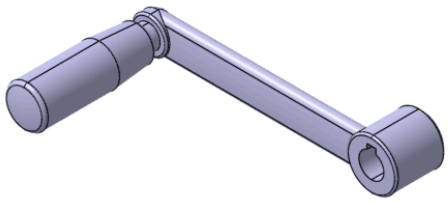


Figure 8: Selected crank

We had considered a second shaft end for left-handed users, but it would have complicated the overall assembly. We kept a one-sided configuration, which is simpler and more compact. The crank is fixed to the shaft with a key to transmit the rotation, and held on its axis by a wing nut, a simple and reliable solution. Since it is made of a light material, the crank has a very low inertia, which makes it easier to spin up the system.

9.3 Frame

The frame has the shape shown below, optimised to fit all the mechanical components efficiently. This configuration gives good stability during use. To reduce the overall size while increasing stiffness, a circular housing screwed to the frame was added to enclose the gears. It also supports the rotating shafts better, improving their stability and guidance.

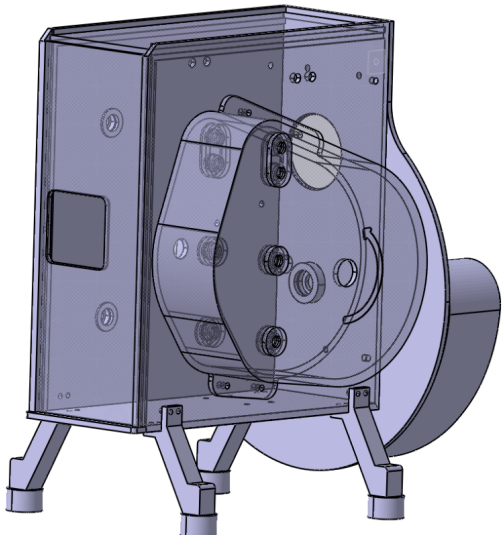


Figure 9: Machined frame

9.3.1 Materials

To stay under the maximum mass of 15 kg while keeping good structural stiffness, several materials were chosen for their properties:

- **Main frame walls:** aluminium, a light alloy ($\rho \approx 2.66$ g $\cdot$ cm^{-3}). The thickness is between 3 mm and 5 mm depending on the load on each area.
- **Top and front windows:** polycarbonate. These two panels carry no load and do not need to be as strong as the others.
- **Feet:** anti-slip rubber, which absorbs vibrations and keeps the machine stable.

Four faces of the frame are made of aluminium, for a structure that is stiff, light and easy to machine. This material withstands the torsion and bending loads from the shafts and fastening points well. Aluminium also allows precise machining of holes, bores and threads, which is essential for a reliable assembly. Its low density helps limit the total mass of the machine, an important requirement. Finally, aluminium resists corrosion well, which extends the life of the frame without extra treatment.

9.3.2 Ergonomics and features

- **A side crank**, on the left during use, lets the operator drive the machine comfortably.
- **A carrying handle** on the top makes the machine easy to move, and to hold during use.
- **A front cover that can be removed without tools** gives quick access to the launch mechanism for maintenance and disassembly.

9.3.3 Compactness and mass

The components are arranged to minimise the overall size while keeping critical areas easy to reach. The final mass of the machine (without balls) is estimated at about 12 kg, a comfortable margin below the specification. This low weight makes the machine easy to carry without compromising its stability during use.

9.4 Legs

The legs are inclined at 30° to give stability and keep the frame off the ground while it turns. Each leg has an anti-slip foot that improves grip and absorbs vibrations. The legs are screwed to the bottom panel of the frame with M3 screws, washers and self-locking nuts, for a reliable assembly that does not loosen.

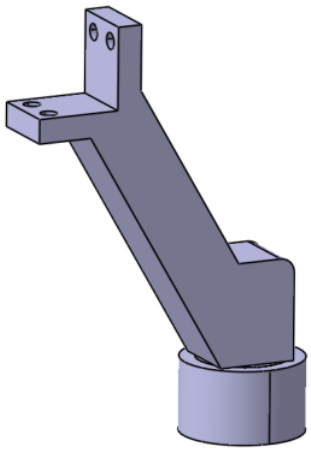


Figure 10: Frame leg

10 Fasteners and guiding systems

10.1 Screw dimensions

First, recall the basic sizing formula for the minimum cross-section of a screw:

$$A_{min} = \frac{F}{\sigma_{adm}}$$

where F is the force on the screw and σ_{adm} the allowable stress, i.e. the yield strength of the material divided by a safety factor. For each screw, we apply this formula to find the minimum cross-section, using the following table:

Nominal diameter $d = D$	Pitch P	Tensile stress area A_s (mm²)	Tap drill D_B
M4	0.5	9.79	3.5
M5	0.5	16.1	4.5
M6	0.75	22.0	5.25
M8	1	39.2	7
M10	1.25	61.2	8.75
M12	1.5	88.1	10.5
M16	1.5	167	14.5
M20	1.5	272	18.5

Figure 11 – Reference table for screw sizing (excerpt)

10.1.1 Fastening the gear housing

The housing is fixed to the frame with 4 screws, which must withstand the forces transmitted by the gears and by the user. The applied forces are:

- total tangential force: $F_t = 1263$ N;
- radial force from the gears: $F_r = 1193 \cdot \tan(\beta) \approx 434$ N;
- total resultant force on the assembly: $F_{res,tot} = F_t + F_r = 1263 + 434 = 1697$ N;
- force per screw (shared by 4 screws): $F_{screw} = \frac{1697}{4} \approx 424$ N.

The screws are property class 8.8, so $R_e = 640$ MPa and $\sigma_{adm} = \frac{R_e}{s} = \frac{640}{2} = 320$ MPa. The minimum cross-section is therefore:

$$A_{min} = \frac{F_{screw}}{\sigma_{adm}} = \frac{424}{320} \approx 1.325\ \text{mm}^2$$

We chose M4 screws, with a stress area of $A_s \approx 8.78$ mm², far above the required value. A 4.5 mm clearance hole is provided in each part for the screws, which are secured on the other side of the frame with nuts.

10.1.2 Fastening the rear shaft

The shaft carrying the worm wheel is supported by a bracket fixed to the bottom panel of the frame with four screws. We determine the minimum screw diameter, starting with the forces they carry.

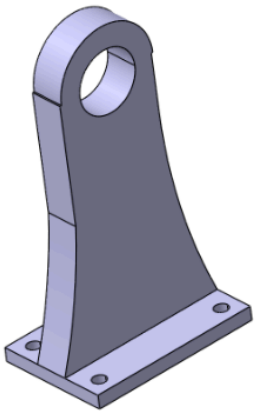


Figure 12: Shaft support

The total force F on the screws is the sum of the tangential force transmitted by the shaft through the gear, F_t, and of the weight of the supported parts, P:

$$F_t = \frac{2 \cdot C}{d} = \frac{2 \cdot 14.4}{12 \cdot 10^{-3}} = 2400 \text{ N} \qquad P = m \cdot g = 15 \cdot 9.81 = 147 \text{ N} \qquad F = 2400 + 147 = 2547 \text{ N}$$

The force per screw ($n = 4$) and the minimum cross-section are:

$$F_i = \frac{2547}{4} = 636.75 \text{ N} \qquad A_{min} = \frac{F_i}{\sigma_{pe}} = \frac{636.75}{640} \approx 0.995 \text{ mm}^2$$

With a safety factor $s = 3$: $A_{min} = 3 \cdot 0.995 = 2.985$ mm². We therefore chose M4 screws, whose stress area is larger than the calculated value, which makes the assembly safe.

10.1.3 Fastening the handle

We size the screws holding the handle used to carry the machine or to lean on during use, with the following assumptions: total (exaggerated) mass $m = 20$ kg, $n = 4$ screws, property class 8.8 steel ($R_e = 640$ MPa) and safety factor $s = 3$:

$$F = m \cdot g = 196.2 \text{ N} \qquad F_{screw} = \frac{F}{n} = 49.05 \text{ N} \qquad \sigma_{adm} = \frac{640}{3} \approx 213.3 \text{ MPa} \qquad A_{min} = \frac{49.05}{213.3} \approx 0.230 \text{ mm}^2$$

We chose M6 screws.

Note: the handle is placed on the axis passing through the centre of mass of the machine, to keep it stable when carried.

10.1.4 Fastening the chute

The forces from the squash balls on the chute are very small, a few newtons at most, even with a dynamic impact. The chute screws therefore only carry moderate loads, mainly in shear or to hold position. We chose M4 screws.

10.1.5 Fastening the legs and the lower side panels

In total, 16 screws fix the frame to the legs. To size them, we consider a conservative scenario in which the user leans fully on the frame, applying a uniformly distributed load. The applied mass is taken as $m = 70$ kg, i.e. a total force of $F_{total} = 70 \cdot 9.81 = 686.7$ N, shared equally by the 16 screws:

$$F_{screw} = \frac{686.7}{16} \approx 42.92 \text{ N} \qquad \sigma_{adm} = \frac{640}{3} \approx 213.3 \text{ MPa} \qquad A_{min} = \frac{42.92}{213.3} \approx 0.201 \text{ mm}^2$$

We chose M3 screws.

Note: we often use screws in plain (untapped) holes, secured with nuts and washers. We looked for partially threaded screws, but could not find the dimensions we needed.

10.2 Ball bearings

The system uses 12 ball bearings: 10 for the transmission shafts and 2 for the shaft driven by the worm. They provide precise guidance, reduce friction losses and carry radial and axial loads. This type of bearing was chosen for its compactness, good efficiency and ease of integration in a manual machine running at moderate speed.

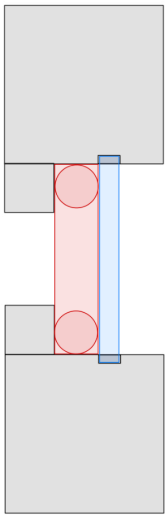


Figure 13: Bearing mounting

10.2.1 Sizing and safety

Here we determine the minimum requirements for the ball bearings by calculating the required dynamic load rating C and static load rating C_0. These values set quality and safety conditions for choosing compatible commercial bearings. We assume that each shaft is supported by two equally spaced bearings and that the load is evenly shared between the eight bearings of the system.

- Target total operating time: 200 h
- Rotation speed: $\omega = 1$ turn/s $= 60$ rpm
- Total number of turns: $L = \omega \cdot t = 1 \cdot 200 \cdot 3600 = 720\,000$ turns $= 0.72 \times 10^6$ turns
- Maximum load on the system (very high value, assuming the user leans on the machine): $P_{total} = 70$ kg $= 686.7$ N
- Load on one bearing (symmetric sharing): $P = \frac{686.7}{8} = 85.84$ N

Formulas used: the dynamic life in millions of turns, $L_{10} = \left(\frac{C}{P}\right)^3 \Rightarrow C = P \cdot L_{10}^{1/3}$, and the static safety factor, $S_0 = \frac{C_0}{P} \Rightarrow C_0 = S_0 \cdot P$. This gives:

$$C = 85.84 \cdot (0.72)^{1/3} \approx 85.84 \cdot 0.894 = 76.7 \text{ N} \qquad C_0 = 3 \cdot 85.84 = 257.5 \text{ N} \quad (S_0 = 3)$$

The bearings must therefore have a dynamic load rating $C \geq 77$ N and a static load rating $C_0 \geq 258$ N. We chose ball bearings whose ratings are higher than these values.

10.2.2 Corresponding losses

The friction losses in the ball bearings are estimated with:

$$P_{lost} = \frac{1}{2} \cdot \mu \cdot P \cdot d \cdot \omega$$

where μ is the friction coefficient of the bearings, P the radial load on one bearing (N), d the mean shaft diameter (m) and ω the angular speed (rad/s), with $\mu = 0.0015$ and $P = 49$ N.

Each shaft is supported by two ball bearings, which guide it radially and reduce friction. Note that the two bearings of a shaft do not necessarily have the same bore. The shafts turning at high angular speed (those of the wheels) are supported by three bearings, which improves stability and prevents vibrations.

Shaft	d_1 (mm)	d_2 (mm)	ω (rad/s)	P_{lost} (W)
2	6	10	66.5	0.08
3	6	6	295.7	0.26
4	6	12	532.2	0.70
5	6	6	0.3	$2.6 \cdot 10^{-4}$

Table 6 – Power lost to friction in the ball bearings

Multiplying the individual efficiencies gives $\eta_{bearings} \approx 0.98$.

10.3 Plain bearings

10.3.1 Strength check of the plain bearings

In a plain bearing, the pressure-velocity product must stay below a limit value $(p \cdot V)_{max}$, which defines the allowable operating range. The contact pressure is:

$$p = \frac{F}{d \cdot L}$$

where F is the radial force on the bearing (N), d the shaft diameter (m) and L the contact length (m). With a transmitted torque C (N · m), the force is $F = \frac{2C}{d}$, and the sliding speed is $V = \frac{d}{2} \cdot \omega$, where ω is the angular speed (rad/s). The pressure-velocity product becomes:

$$p \cdot V = \frac{F}{d \cdot L} \cdot \frac{d}{2} \cdot \omega = \frac{F \cdot \omega}{2 \cdot L} = \frac{2 \cdot C \cdot \omega}{2 \cdot d \cdot L} = \frac{C \cdot \omega}{d \cdot L}$$

This value must stay below the limit given by the supplier, $(p \cdot V)_{max} = 0.42$ MPa · m/s. With $C = 8.75$ N · m and $\omega = 2\pi$ rad/s, and assuming the pressure is shared equally between the two plain bearings (so the product is divided by 2):

$$\text{Crank side } (L = 8\text{ mm},\ d = 14\text{ mm}): \quad p \cdot V = \frac{8.75 \cdot 2\pi}{2 \cdot 14 \cdot 10^{-3} \cdot 8 \cdot 10^{-3}} \approx 0.34\ \text{MPa} \cdot \text{m/s}$$

$$\text{Other side } (L = 10\text{ mm},\ d = 10\text{ mm}): \quad p \cdot V = \frac{8.75 \cdot 2\pi}{2 \cdot 10 \cdot 10^{-3} \cdot 10 \cdot 10^{-3}} \approx 0.28\ \text{MPa} \cdot \text{m/s}$$

Both values are below the limit, so the operation of the plain bearings is acceptable.

10.3.2 Estimated energy losses

In a plain bearing, friction between the shaft and the inner surface of the bushing creates a parasitic resisting moment M_a that opposes the driving torque C. The efficiency is:

$$\eta = \frac{C - M_a}{C} = 1 - \frac{M_a}{C}$$

The driving torque is applied at a 12.5 cm radius through the crank, i.e. an equivalent diameter $d_R = 25$ mm, so $C = F_{tan} \cdot \frac{d_R}{2}$. The friction moment is generated at the contact between the shaft and the bearing, at a radius $\frac{d_a}{2}$: $M_a = F_{tan} \cdot \mu \cdot \frac{d_a}{2}$, where $\mu = 0.08$ is the friction coefficient given by the supplier and d_a the shaft diameter (bore of the bearing). Combining the two expressions:

$$\eta = 1 - \mu \cdot \frac{d_a}{d_R} \qquad \eta_1 = 1 - 0.08 \cdot \frac{14}{25} \approx 0.96\ \text{(crank side)} \qquad \eta_2 = 1 - 0.08 \cdot \frac{10}{25} \approx 0.97\ \text{(other side)}$$

The efficiency of the plain bearings is therefore $\eta_{plain} = \eta_1 \cdot \eta_2 \approx 0.93$.

11 Critical review of the project

Although our machine fulfils its purpose overall, it has some limitations and possible improvements, detailed here.

11.1 Compliance with the specifications

The vast majority of the functional and dimensional requirements are met. One small deviation remains: ambidextrous use. The machine was meant to be usable by both right- and left-handed people, but the crank is currently only on the left side. This was chosen for mechanical simplicity: a left-handed user turns with their dominant hand, while a right-handed user can lean firmly on the structure. An interchangeable handle was considered, but it would have made the assembly impossible.

11.2 Risk of jamming

Mechanical measures keep the rotating parts in position and moving smoothly: circlips, pins, shoulders and keys transmit the loads, while the shafts are guided by ball bearings and plain bearings.

The main identified risk concerns the ball feed. A badly inserted ball, or two balls arriving at the same time, could block the mechanism. To prevent this:

- the storage pocket is inclined at 12° to help the balls descend by gravity;
- the time for a ball to fall through the chute is estimated at 0.4 s, far less than the interval between two shots;
- the feed system creates enough spacing between balls, so that one is fully launched before the next arrives.

11.3 User safety

Operator safety was built in from the first design stages:

- a fixed, closed housing fully encloses the gears, preventing any contact with rotating parts;
- the user naturally stands on the opposite side from the line of fire, at a safe distance;
- even if the guidance fails, a ball cannot enter the gear compartment, which protects the system.

These precautions eliminate the risk of injury or critical malfunction during use.

11.4 Machinability

All parts were designed to be machined on 3-axis CNC machine tools, as required by the project, with particular attention to simple geometries. The only relatively demanding part is the ball-guiding housing, which requires a lot of material removal. This operation is nevertheless feasible on a conventional 3-axis milling machine, without advanced processes.

12 Useful calculations

12.1 Summary tables

The following table summarises the basic data of our gears and launch wheels:

Part no.	**22**	**23**	**24**	**25**	**26**	**27**	**13**
Number of teeth	180	17	200	45	25	22	–
Diameter (mm)	135	12.75	100	22.5	33	16.5	105
Mass (kg)	0.67	0.01	0.30	0.02	0.007	0.18	0.26
Moment of inertia ($\times 10^{-5}$ kg · m^2)	1516	0.27	383	1.2	0.17	29.2	360
Ratio ω/ω_e	1	10.6	10.6	47.1	85	0.05	85

Table 7 – Basic data of the gears and launch wheels

The next tables are based on Newtonian mechanics: the tangential forces are equal between two gears in direct contact, and the torques are equal for gears fixed on the same shaft.

Gear no.	**22**	**23**	**24**	**25**	**26**	**27**
Torque (N · m)	8.75	0.86	0.86	0.19	0.11	14.4
Tangential force (N)	134.8	134.8	17.2	17.2	17.2	889

Table 8 – Tangential forces and torques for $F_{in} = 70$ N

Shaft no.	**01**	**02**	**03**	**04**	**05**
Torque (N · m)	8.75	0.86	0.19	0.11	14.4

Table 9 – Shaft torques for $F_{in} = 70$ N

The following table gives the angular and tangential speeds of each gear and of the launch wheels, using the basic rules of gear trains: two gears in contact have the same tangential speed at the contact point, and gears fixed on the same shaft turn at the same angular speed.

Part no.	**22**	**23**	**24**	**25**	**26**	**27**	**13**
Angular speed (rad · s^{-1})	6.4	66.5	66.5	295.7	532.2	0.28	532.2
Tangential speed (m · s^{-1})	0.42	0.42	3.32	3.32	3.32	0.005	27.9

Table 10 – Angular and tangential speeds with $\omega_{in} = 2\pi$ rad · s^{-1}

12.2 Required force for a target exit speed

The efficiency of the machine is about 0.7. It does not directly affect the exit speed, but rather the force needed to reach that speed in a given time, or the time needed with a given force. Here the acceleration time is set to 11 seconds, so reaching a target speed in that time requires a higher input force than a simple energy balance gives. The force is corrected by the efficiency:

$$F_{actual} = \frac{F_{ideal}}{\eta}$$

To express this force in terms of the mechanical parameters of the system, we use the relations shown earlier:

$$F = \frac{\omega_{in} \cdot J}{r_m \cdot \Delta t} \qquad \text{where} \qquad \omega_{in} = \frac{\omega_{out}}{85}, \qquad \omega_{out} = \frac{v_{target}}{K \cdot r}$$

with $J = 5.1$ kg · m² the equivalent inertia, $r_r = 0.0525$ m the radius of the launch wheels, $r_m = 0.125$ m the crank radius and $K = 0.978$ the rotation-to-translation conversion factor.

Case 1 — exit speed v = 80 km/h:

$$v = \frac{80}{3.6} = 22.22 \text{ m/s}, \quad \omega_{out} = \frac{22.22}{0.978 \cdot 0.0525} \approx 434 \text{ rad/s}, \quad \omega_{in} = \frac{434}{85} \approx 5.1 \text{ rad/s}$$

$$F_{ideal} = \frac{5.1 \cdot 5.1}{0.125 \cdot 11} \approx 18.9 \text{ N}, \qquad F_{actual} = \frac{18.9}{0.7} \approx 27.0 \text{ N}$$

Case 2 — exit speed v = 130 km/h:

$$v = \frac{130}{3.6} = 36.11 \text{ m/s}, \quad \omega_{out} = \frac{36.11}{0.978 \cdot 0.0525} \approx 706 \text{ rad/s}, \quad \omega_{in} = \frac{706}{85} \approx 8.3 \text{ rad/s}$$

$$F_{ideal} = \frac{8.3 \cdot 5.1}{0.125 \cdot 11} \approx 30.7 \text{ N}, \qquad F_{actual} = \frac{30.7}{0.7} \approx 43.8 \text{ N}$$

To reach the target speed in 11 seconds, the user must therefore apply an input force of about 27 N for 80 km/h and 43.8 N for 130 km/h, taking into account the overall efficiency of the machine and the energy transferred to the ball as it passes between the wheels.

12.3 Firing rate

We estimate the time between two successive shots, taking into account the time for a ball to descend the chute and the time for the next ball to move into position. The disc turns at $\frac{1}{22}$ turn per second, and the pockets are spaced $36.5^\circ = \frac{36.5}{360} \approx 0.1014$ turn apart. The time to cover this angle is:

$$t_{rot} = 22 \times 0.1014 \approx 2.23 \text{ s}$$

The time between two shots is dominated by t_{rot}, so $t_{shot} \approx 2.2$ s.

12.4 Kinematics and centre distances

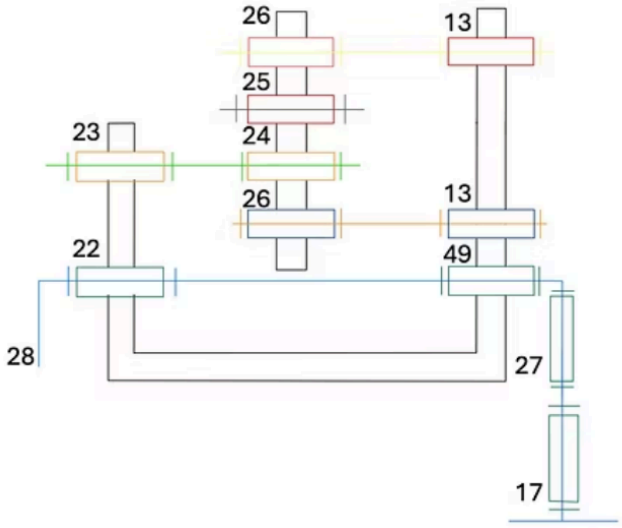


Figure 14: Kinematic chain of the mechanism

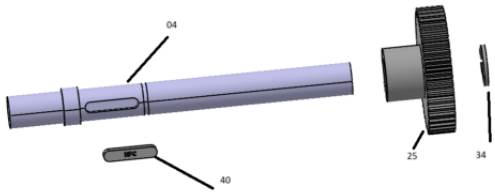
The figure shows the kinematic chain of the launch system. The motion is generated by hand with the crank (28), fixed to gear (22), which forms the first transmission stage.

Gear (22) drives (23), mounted on the same shaft as (24). Gear (24) meshes with (25), itself linked to (26), which transmits the rotation to the central shaft. On this same shaft, a second gear (26) drives gear (49), fixed to the worm (27).

The worm provides a large reduction ratio and transmits the motion to the vertical shaft supporting the two launch wheels (13). These wheels propel the ball coming from the magazine (17), located on the vertical axis.

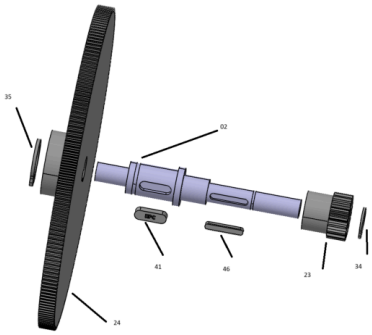
Each connection between two components of the diagram is a direct coupling, a key or a rigid joint, which keeps everything synchronised. The diagram shows the hierarchical structure of the mechanism: manual generation of the motion, transmission through successive gears, reduction by the worm gear, and finally launching of the balls by friction.

13 Assembly



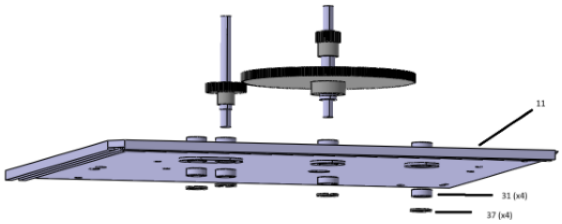
1

- Place the gear over the key and secure it with the circlip.



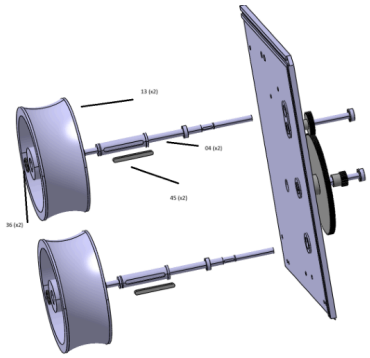
2

- Place the gears over their keys and secure them with the circlips.



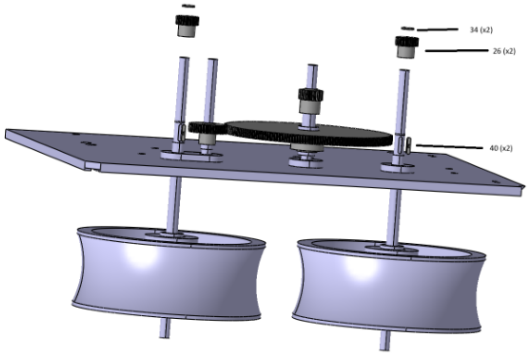
3

- Fit the bearings and internal circlips in the panel.
- Insert the two shafts assembled previously into the panel.



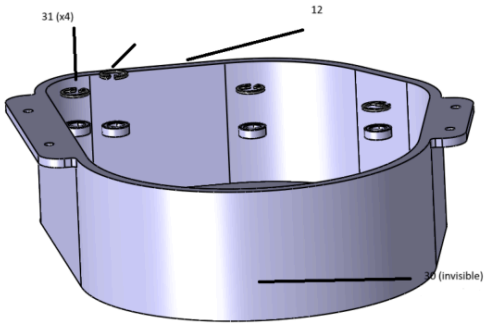
4

- Place and secure the launch wheels on the shaft.
- Insert the shaft into the panel assembled previously.



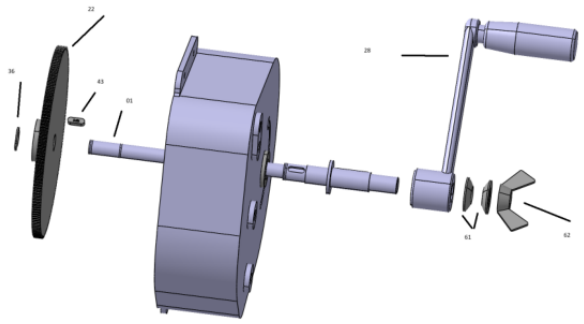
5

- Fit the indicated gears on the launch-wheel shaft.



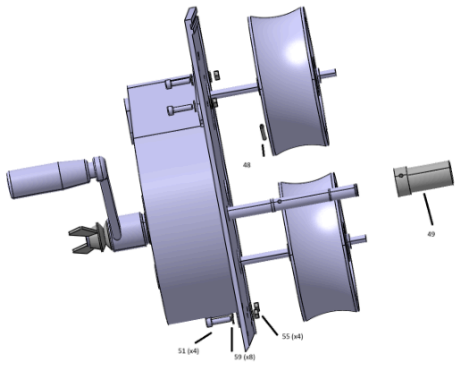
6

- Take the gear housing and fit the ball bearings with their circlips.



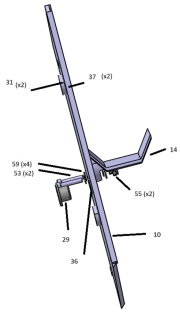
7

- Start by inserting the shaft into the housing as shown.
- Fit the crank and its fasteners.
- On the other side, fit the gear, the key and its circlip.



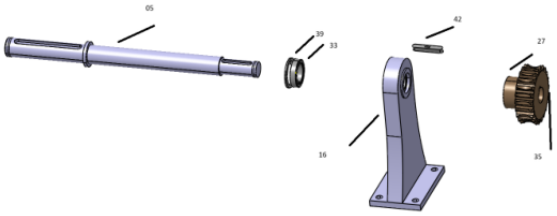
8

- Join the two previous assemblies.
- Fasten them with the indicated screws, washers and nuts.
- Add the worm and its pin.



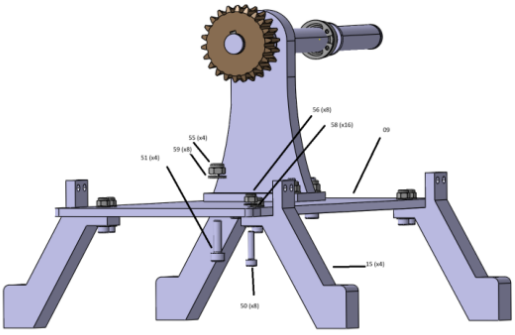
9

- Prepare the right panel of the frame.
- Fit the ball bearings and plain bearings.
- Fit the specified screws with their washers and nuts.



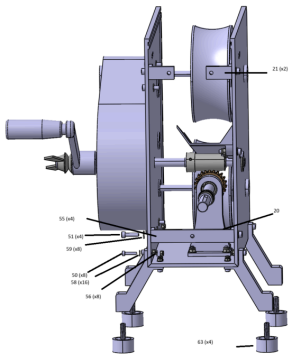
10

- Start by inserting the bearing into the support.
- Then insert the shaft to mount it.



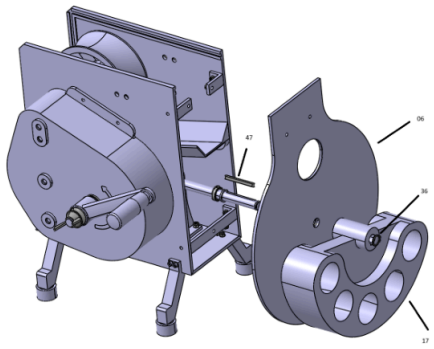
11

- Join this assembly with the bottom panel of the frame.
- Screw them together as indicated.
- Screw the legs to the bottom panel.



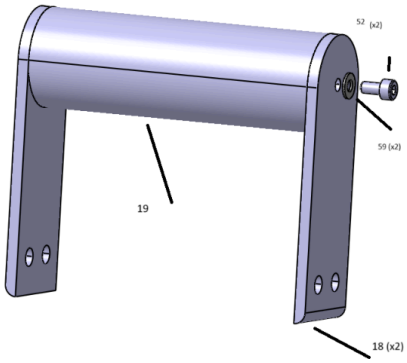
12

- Join the last two assemblies.
- Fasten them with the brackets.
- Screw as indicated.



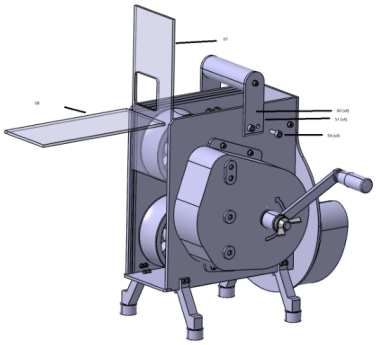
13

- Add the rear panel of the mechanism.
- Add the key after fitting the circular panel.



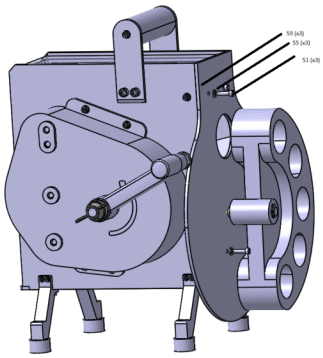
14

- Assemble the carrying handle of the frame.
- Tighten the indicated screws.



15

- Tighten the screws before adding the frame panels, for easy access to the nuts.
- Fit the panels in the desired order.



16

- Tighten the last screws, and shift the top panel of the frame if needed.

14 Operating instructions

1. Place the machine on a flat, stable surface.
2. Stand behind the machine, hold the handle with your right hand and the crank with your left hand.
3. Turn the crank until the first pocket of the storage half-disc is just before the exit opening, on the right side.
4. Load the balls into the pockets of the storage half-disc.
5. Turn the crank in the direction of the arrow, applying a force between 27 and 44 N for 11 s, to reach launch speed.
6. Once the acceleration time has elapsed, keep turning at a constant speed without accelerating further.
7. Release the crank to stop firing; the system comes to rest on its own.

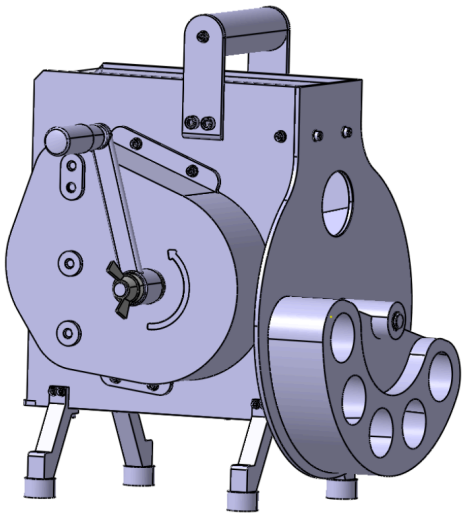


Figure 15: Direction of rotation